The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for the main idea, the purpose, or the focus of the text, speech, or conversation.
- The topic can be used to organize, summarize, or evaluate the text, speech, or conversation.
- The topic can be related to other topics by finding similarities, differences, causes, effects, or implications.

Engineering Chemistry Lab

Engineering chemistry lab is a practical course that complements the theoretical concepts of engineering chemistry. It aims to develop the skills and knowledge of students in performing various experiments related to chemical engineering, such as polymerization, corrosion, water analysis, fuel analysis, spectroscopy, electrochemistry, etc. It also introduces the students to the scientific method, experimental design, chemical instrumentation, data collection and analysis, and preparation of laboratory reports.

The syllabus of engineering chemistry lab may vary depending on the institution, branch, and curriculum. However, some of the common topics and experiments that are covered in engineering chemistry lab are:

- High polymers and plastics: This topic covers the introduction, mechanism, methods, and properties of polymerization and polymers. The experiments may include the preparation and characterization of polymers such as nylon, bakelite, PVC, etc.
- Water and its treatment: This topic covers the sources, impurities, hardness, and methods of water treatment and analysis. The experiments may include the estimation of hardness, alkalinity, chloride, dissolved oxygen, etc. in water samples.
- Fuels and combustion: This topic covers the classification, characteristics, and analysis of solid, liquid, and gaseous fuels. The experiments may include the determination of calorific value, flash point, fire point, viscosity, etc. of fuels.
- Corrosion and its control: This topic covers the causes, types, effects, and prevention of corrosion of metals and alloys. The experiments may include the study of corrosion rate, corrosion inhibitors, electrochemical corrosion, etc.
- Phase rule and alloys: This topic covers the introduction, terminology, and applications of phase rule and phase diagrams. It also covers the types, properties, and preparation of alloys. The experiments may include the

construction of phase diagrams, preparation and analysis of alloys, etc.

- Instrumental methods of analysis: This topic covers the principles, instrumentation, and applications of various spectroscopic and electrochemical techniques for chemical analysis. The experiments may include the use of UV-Vis, IR, NMR, AAS, potentiometer, conductometer, etc. for the analysis of unknown samples.
- Nanochemistry: This topic covers the introduction, synthesis, characterization, and applications of nanomaterials and nanotechnology. The experiments may include the preparation and characterization of nanoparticles, nanocomposites, nanocatalysts, etc.

The engineering chemistry lab course requires the students to follow the safety rules, instructions, and procedures given by the instructor and the lab manual. The students are expected to perform the experiments in groups, record the observations and results, and submit the lab reports on time. The students are also evaluated based on their performance, understanding, and presentation of the experiments.

Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To familiarize the students with the methods and techniques of knowledge representation, reasoning, planning, search, and machine learning in AI.
- To develop the skills and abilities to design, implement, and evaluate AI systems and agents using various tools and languages.
- To expose the students to the current trends and challenges in AI research and practice.

1. To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- Analytical instruments can provide qualitative or quantitative information about the composition, structure, or function of samples, such as elements, molecules, cells, tissues, organs, etc.
- Analytical instruments can be used for various purposes, such as research, development, quality control, diagnosis, forensics, environmental monitoring, etc.
- Analytical instruments can have different features, such as sensitivity, selectivity, accuracy, precision, resolution, speed, throughput, etc.
- Analytical instruments can have different components, such as sources, detectors, transducers, amplifiers, filters, processors, displays, etc.

- Analytical instruments can have different modes of operation, such as continuous, discrete, batch, online, offline, etc.
- Analytical instruments can have different challenges, such as interference, noise, drift, calibration, validation, maintenance, etc.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- Chloride content is the amount of chloride ions (Cl-) present in water. Chloride ions are derived from various sources such as seawater intrusion, industrial effluents, agricultural runoff, and domestic sewage. Chloride content is an important parameter for assessing the quality of water for various purposes such as drinking, irrigation, and industrial use. Chloride content can be determined by titrating a known volume of water sample with a standard solution of silver nitrate (AgNO3) using potassium chromate (K2CrO4) as an indicator. The end point of the titration is marked by the appearance of a reddish-brown precipitate of silver chromate (Ag2CrO4).
- Hardness of water is the measure of the concentration of calcium (Ca2+) and magnesium (Mg2+) ions in water. These ions are mainly derived from the dissolution of carbonate and bicarbonate minerals such as limestone, dolomite, and gypsum. Hardness of water affects the performance of soap and detergents, causes scaling and corrosion of pipes and boilers, and affects the taste and quality of beverages such as tea and coffee. Hardness of water can be classified into temporary hardness and permanent hardness. Temporary hardness is caused by the presence of bicarbonate ions (HCO3-) and can be removed by boiling the water. Permanent hardness is caused by the presence of other ions such as sulfate (SO42-), chloride (Cl-), and nitrate (NO3-) and cannot be removed by boiling. Hardness of water can be determined by titrating a known volume of water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T (EBT) as an indicator. The end point of the titration is marked by the change of color of the indicator from wine red to blue.
- Alkalinity of water is the measure of the capacity of water to neutralize acids. Alkalinity of water is mainly due to the presence of carbonate (CO32-), bicarbonate (HCO3-), and hydroxide (OH-) ions in water. These ions are derived from the dissolution of carbonate and bicarbonate minerals, the hydrolysis of salts, and the photosynthesis of aquatic plants. Alkalinity of water affects the pH and buffering capacity of water, the solubility of metals and minerals, and the biological activity of aquatic organisms. Alkalinity of water can be determined by titrating a known volume of water sample with a standard solution of sulfuric acid (H2SO4) using phenolphthalein and methyl orange as indicators. The end point of the titration with phenolphthalein is marked by the change of color of the indicator from pink to colorless, indicating the consumption of hydroxide and carbonate ions. The end point of the titration with methyl orange is

marked by the change of color of the indicator from yellow to orange, indicating the consumption of bicarbonate ions. The difference between the two end points gives the phenolphthalein alkalinity (P-alkalinity), which represents the concentration of hydroxide and carbonate ions. The total amount of acid used gives the total alkalinity (T-alkalinity), which represents the concentration of hydroxide, carbonate, and bicarbonate ions.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic:

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- pH is a measure of how acidic or basic a liquid is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. The pH scale ranges from 0 to 14, where 0 is the most acidic, 7 is neutral, and 14 is the most basic. Some examples of liquids with different pH values are:
  - Lemon juice: pH 2
  - Vinegar: pH 3
  - Pure water: pH 7
  - Baking soda solution: pH 9
  - Ammonia solution: pH 11
  - Bleach: pH 13
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is caused by the imbalance of intermolecular forces between the liquid and the surrounding air or other substances. Surface tension is expressed in units of force per unit length, such as newtons per meter (N/m) or dynes per centimeter (dyn/cm). Some factors that affect the surface tension of a liquid are:
  - Temperature: As the temperature increases, the surface tension decreases, because the molecules have more kinetic energy and can overcome the intermolecular forces more easily.
  - Impurities: Adding impurities such as salt, sugar, or detergent to a liquid can lower the surface tension, because they disrupt the uniformity of the intermolecular forces at the surface.
  - Shape: The shape of the container or the object that touches the liquid can also influence the surface tension, because it can create different curvatures and pressures at the surface.
- Viscosity is a measure of how resistant a liquid is to flow. It is related to the internal friction between the molecules of the liquid as they move past each other. Viscosity is expressed in units of force times time per unit area, such as pascal-seconds ($Pa \cdot s$) or poise (P). Some factors that affect the viscosity of a liquid are:

- Temperature: As the temperature increases, the viscosity decreases, because the molecules have more kinetic energy and can slide past each other more easily.
- Molecular size and shape: Larger and more complex molecules tend to have higher viscosity, because they have more intermolecular forces and interactions that hinder their movement.
- Interactions between molecules: Different types of intermolecular forces, such as hydrogen bonding, dipole-dipole, or van der Waals, can affect the viscosity of a liquid, depending on their strength and directionality.

4. To enable the students to understand about the preparation of different resins.

- Resins are natural or synthetic organic substances that are usually solid or semi-solid, and have a high molecular weight.
- Resins are widely used in various industries, such as paints, adhesives, plastics, rubber, pharmaceuticals, cosmetics, etc.
- Resins can be classified into two main types: thermoplastic and thermosetting.
- Thermoplastic resins are resins that can be melted and reshaped by heating, such as polyethylene, polypropylene, polystyrene, etc.
- Thermosetting resins are resins that undergo irreversible chemical reactions when heated, and form cross-linked networks that cannot be remelted, such as epoxy, phenolic, polyester, etc.
- Resins can be prepared by different methods, depending on their type and properties. Some of the common methods are:
  - Polymerization: This is the process of joining small molecules (monomers) to form large molecules (polymers) by covalent bonds. For example, ethylene can be polymerized to form polyethylene by using a catalyst and high pressure and temperature.
  - Condensation: This is the process of combining two or more molecules (usually with the elimination of a small molecule, such as water) to form a larger molecule. For example, phenol and formaldehyde can be condensed to form phenolic resin by using an acid or a base as a catalyst and heating.
  - Copolymerization: This is the process of polymerizing two or more different monomers to form a copolymer, which has properties that are different from those of the individual monomers. For example, styrene and acrylonitrile can be copolymerized to form styrene-acrylonitrile (SAN) resin by using a free radical initiator and heating.
  - Modification: This is the process of changing the properties of a resin by adding or reacting with another substance. For example, epoxy resin can be modified by adding a curing agent, such as an amine or an anhydride, which reacts with the epoxy groups and forms a cross-linked network.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

- Adipic acid is a dicarboxylic acid that is used as a raw material for the production of nylon-6,6, a synthetic polymer widely used in various applications.
- Paracetamol is an analgesic and antipyretic drug that is widely used to relieve pain and fever.
- Conventional route of synthesis of adipic acid involves the oxidation of cyclohexane or a mixture of cyclohexanol and cyclohexanone with nitric acid, which produces large amounts of nitrous oxide, a greenhouse gas and an ozone depleter.
- Conventional route of synthesis of paracetamol involves the acetylation of phenol with acetic anhydride, which produces large amounts of acetic acid, a corrosive and toxic by-product.
- Green route of synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in a microemulsion, which produces water as the only by-product. The microemulsion is composed of water, cyclohexene, hydrogen peroxide, sodium tungstate as a catalyst, and alkyldimethylbenzylammonium chloride as a surfactant. The microemulsion can be recycled and reused, and the adipic acid can be easily separated and purified .
- Green route of synthesis of paracetamol involves the reaction of 4-aminophenol with acetic acid in water, which produces water as the only by-product. The reaction can be catalyzed by a green catalyst such as iron oxide nanoparticles, which can be easily recovered and reused.
- The advantages of green route of synthesis are:
  - It reduces the environmental impact of the chemical processes by minimizing the use of hazardous reagents, solvents, and catalysts, and by avoiding the generation of harmful by-products and wastes.
  - It improves the efficiency and selectivity of the chemical reactions by using milder conditions, renewable resources, and biodegradable materials.
  - It lowers the cost and energy consumption of the chemical processes by simplifying the reaction steps, reducing the separation and purification steps, and recycling and reusing the reaction media.

LIST OF EXPERIMENTS

- An experiment is a procedure that tests a hypothesis or a question by collecting and analyzing data under controlled conditions.
- Experiments can be classified into different types based on their design, purpose, or outcome.
- Some common types of experiments are:

- **Randomized controlled trial (RCT)**: An experiment that randomly assigns participants to either a treatment group or a control group, and compares the outcomes of the two groups. RCTs are often used to test the effectiveness or safety of a new intervention, such as a drug, a vaccine, or a behavioral program.
- **Factorial design**: An experiment that manipulates two or more independent variables (factors) simultaneously, and measures their effects on one or more dependent variables (outcomes). Factorial designs can test the main effects of each factor, as well as the interactions between them.
- **Quasi-experiment**: An experiment that lacks random assignment or a control group, but still attempts to measure the causal effect of a treatment or an intervention. Quasi-experiments are often used when it is not feasible or ethical to conduct a true experiment, such as in natural or social settings.
- **Natural experiment**: An experiment that exploits a naturally occurring event or variation as a source of randomization or comparison. Natural experiments are often used to study the effects of policies, events, or environmental changes that are beyond the control of the researcher.
- **Observational study**: A study that does not manipulate any variables, but only observes and measures the existing relationships between them. Observational studies can be descriptive, exploratory, or inferential, depending on the research question and the statistical methods used.

1. **Calibration of Analytical Equipment and apparatus**

- Calibration is the process of evaluating and adjusting the precision and accuracy of measurement equipment.
- Calibration ensures that the equipment and apparatus produce valid and reliable results that are consistent with the standards and specifications.
- Calibration is essential for maintaining the quality and safety of analytical products and processes, as well as complying with the regulatory requirements and guidelines.
- Calibration involves comparing the output of the equipment or apparatus with a known reference value, such as a certified standard or a traceable measurement.
- Calibration can be performed at different levels of accuracy and frequency, depending on the type, purpose, and usage of the equipment or apparatus .
- Calibration can be classified into two types: internal and external calibration.
  - Internal calibration is performed by the user or operator of the equipment or apparatus, using the built-in features or functions of the device.

- External calibration is performed by a third-party service provider or a qualified laboratory, using independent standards or instruments.

- Calibration can be documented by a calibration certificate or a calibration report, which records the details of the calibration procedure, the reference values, the measurement results, the uncertainty estimates, and the calibration date and interval .

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is the measure of the concentration of calcium and magnesium ions in water.
- EDTA (Ethylenediaminetetraacetic acid) is a chelating agent that can form stable complexes with metal ions such as Ca^{2+} and Mg^{2+}.
- EDTA titration is a method of determining the hardness of water by titrating a water sample with a standard solution of EDTA in the presence of a suitable indicator.
- The indicator used is Eriochrome Black T (EBT), which is a dye that changes color from wine red to blue when it binds to metal ions.
- The procedure for EDTA titration is as follows :
  - Take a sample volume of 20 ml (V ml) of water to be tested.
  - Dilute the sample to 40 ml by adding 20 ml of distilled water in an Erlenmeyer flask.
  - Add 1 ml of ammonia buffer to adjust the pH to 10 ± 0.1. This is to prevent the precipitation of metal hydroxides and to ensure the formation of EDTA-metal complexes.
  - Add 1 or 2 drops of EBT indicator to the sample. The sample will turn wine red due to the presence of Ca^{2+} and Mg^{2+} ions.
  - Titrate the sample with a standard solution of EDTA of known concentration (C mg/ml) from a burette with vigorous shaking until the color changes from wine red to blue. This indicates the end point of the titration.
  - Record the volume of EDTA used (V1 ml).
- The calculation of hardness of water by EDTA titration is based on the following equation :
  - $C * V1 = V * H$
  - where C is the concentration of EDTA in mg/ml, V1 is the volume of EDTA used in ml, V is the volume of water sample in ml, and H is the hardness of water in mg/L as $CaCO_3$.
  - The hardness of water can be expressed as total hardness or calcium hardness, depending on whether the EDTA titrates both Ca^{2+} and Mg^{2+} ions or only Ca^{2+} ions.

  - The total hardness is obtained when the pH of the sample is 10 ± 0.1, and the calcium hardness is obtained when the pH of the sample is 12 ± 0.1.
  - The hardness of water can be classified as soft, moderately hard, hard, or very hard, depending on the value of H in mg/L as $CaCO_3$, as shown in the table below:

Hardness of water	H (mg/L as $CaCO_3$)
Soft	< 75
Moderately hard	75 - 150
Hard	150 - 300
Very hard	> 300

3. Determination of Alkalinity of water sample.

- Alkalinity is the measure of the water's ability to neutralize acids. It is mainly caused by the presence of bicarbonates, carbonates, and hydroxides in water.
- Alkalinity is important for water quality because it affects the pH, corrosion, and disinfection processes. High alkalinity can cause scaling and reduce the effectiveness of chlorine. Low alkalinity can cause corrosion and pH fluctuations.
- Alkalinity can be determined by titrating a water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a suitable indicator, such as phenolphthalein or methyl orange.
- The procedure for determining alkalinity is as follows:
  - Take a 100 mL aliquot of the water sample in a conical flask and add a few drops of phenolphthalein indicator. If the sample turns pink, it indicates the presence of hydroxides or carbonates. Record the initial pH of the sample using a pH meter.
  - Titrate the sample with the standard acid solution until the pink color disappears. Record the volume of acid used as V_1. This is the phenolphthalein alkalinity (P-alkalinity) of the sample, which represents the amount of hydroxides and half of the carbonates in the sample.
  - Add a few drops of methyl orange indicator to the same sample and continue the titration until the color changes from yellow to orange. Record the volume of acid used as V_2. This is the total alkalinity (T-alkalinity) of the sample, which represents the amount of hydroxides, carbonates, and bicarbonates in the sample.
  - Calculate the alkalinity of the sample using the following formulas:

    - P-alkalinity (mg/L as $CaCO_3$) = (V1 x N x 50,000) / mL of sample
    - T-alkalinity (mg/L as $CaCO_3$) = (V2 x N x 50,000) / mL of sample
    - N is the normality of the standard acid solution
    - 50,000 is the equivalent weight of $CaCO_3$ in mg/L
    - mL of sample is the volume of the water sample taken for titration
  - Based on the values of P-alkalinity and T-alkalinity, the types and concentrations of the alkaline species in the sample can be determined using the following table:

P-alkalinity	T-alkalinity	Alkaline species present	Concentrations (mg/L as $CaCO_3$)
0	> 0	Only bicarbonates	Bicarbonates = T-alkalinity
> 0	= 2 x P	Only carbonates and hydroxides	Carbonates = Hydroxides = P-alkalinity
> 0	$>$ 2 x P	Bicarbonates, carbonates, and hydroxides	Bicarbonates = T-alkalinity - 2 x P-alkalinity Carbonates = Hydroxides = P-alkalinity
> 0	$<$ 2 x P	Impossible condition	N/A

- Report the results of the alkalinity determination, including the initial pH, the volumes of acid used, the P-alkalinity, the T-alkalinity, the types and concentrations of the alkaline species, and any observations or sources of error.

4. **Determination of pH by titrimetric method**

- Titration is a laboratory technique that involves adding a known amount of a solution (called the titrant) to another solution (called the analyte) until a chemical reaction is complete.
- The point at which the reaction is complete is called the endpoint, and it is usually indicated by a color change of an indicator or a change in the electrical properties of the solution.
- The pH is a measure of the acidity or basicity of a solution, and it is related to the concentration of hydrogen ions (H^+) or hydronium ions (H_3O^+) in the solution.
- The pH can be determined by titrating the solution with a standard acid or base solution of known concentration and measuring the volume of the titrant required to reach the endpoint.
- The pH can also be calculated from the titration data using the following formula:

pH = -log[H+]

where [H+] is the concentration of hydrogen ions in the solution at the endpoint.

- The titration method can be used to determine the pH of various types of solutions, such as strong acids, strong bases, weak acids, weak bases, buffer solutions, and polyprotic acids.
- The titration method can also be used to determine the concentration of an unknown acid or base solution by using a standard solution of known pH and concentration as the titrant.
- The titration method requires the use of appropriate equipment, such as a burette, a pipette, a titration flask, a pH meter, an indicator, and a stirrer.
- The titration method also requires the use of proper procedures, such as rinsing the equipment, measuring the volumes accurately, adding the titrant slowly and carefully, observing the color change or the pH change, and recording the data.

5. Determination of surface tension of given liquid.

- Surface tension is a property of liquids that causes them to behave as if they have a thin elastic skin on their surface.
- Surface tension is caused by the cohesive forces between the molecules of the liquid, which tend to minimize the surface area of the liquid.
- Surface tension can be measured by various methods, such as the capillary rise method, the drop weight method, the ring method, and the du Noüy method.
- The capillary rise method is based on the principle that a liquid will rise or fall in a narrow tube (capillary) depending on the relative strength of the adhesive forces between the liquid and the tube, and the cohesive forces within the liquid.
- The drop weight method is based on the principle that the weight of a drop of liquid that detaches from a vertical capillary tube is proportional to the surface tension of the liquid.
- The ring method is based on the principle that a horizontal ring of wire (usually platinum) immersed in a liquid will experience a force due to the surface tension of the liquid, which tends to contract the ring.
- The du Noüy method is based on the principle that a platinum ring attached to a balance is dipped in a liquid and then pulled out slowly, and the force required to detach the ring from the liquid surface is measured by the balance. This force is proportional to the surface tension of the liquid.

6. Determination of Viscosity of a given liquid by viscometer.

- Viscosity is a measure of the resistance of a fluid to flow or deform under an applied force.

- A viscometer is a device that measures the viscosity of a fluid by observing its flow through a narrow tube or a capillary.
- The flow rate of a fluid through a capillary depends on the pressure difference across the capillary, the length and radius of the capillary, and the viscosity of the fluid.
- The Hagen-Poiseuille equation relates these variables as follows:

$$Q = \frac{\pi r^4 \Delta P}{8\eta L}$$

where Q is the volumetric flow rate, r is the radius of the capillary, ΔP is the pressure difference, η is the viscosity, and L is the length of the capillary.

- To determine the viscosity of a given liquid by viscometer, the following steps are followed:
  - Fill the viscometer with the liquid to be tested and immerse it in a water bath at a constant temperature.
  - Allow the liquid to flow through the capillary under the influence of gravity and measure the time taken for a fixed volume of liquid to flow between two marked points on the viscometer.
  - Repeat the experiment for different volumes of liquid and calculate the average time for each volume.
  - Plot a graph of volume versus time and find the slope of the best-fit line. This slope is equal to the flow rate Q.
  - Using the known values of r, L, and ΔP, calculate the viscosity η from the Hagen-Poiseuille equation.
  - Compare the experimental value of viscosity with the theoretical value and calculate the percentage error.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator.

- Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron(II) sulfate and ammonium sulfate, with the formula (NH4)2Fe(SO4)2 · 6H2O.
- It is commonly used as a standard solution for titrating oxidizing agents, such as potassium permanganate or potassium dichromate, in redox titrations.
- The strength of a ferrous ammonium sulfate solution can be determined by titrating it with a standard solution of potassium permanganate, using an external indicator such as starch or potassium ferricyanide.
- The reaction between ferrous ammonium sulfate and potassium permanganate is:

(NH4)2Fe(SO4)2 + KMnO4 + H2SO4 -> K2SO4 + (NH4)2SO4 + Fe2(SO4)3 + MnSO4 + H2O

- The reaction between ferrous ammonium sulfate and potassium dichromate is:

$(NH_4)_2Fe(SO_4)_2 + K_2Cr_2O_7 + H_2SO_4 \rightarrow K_2SO_4 + (NH_4)_2SO_4 + Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + H_2O$

- The end point of the titration is detected by the change of color of the external indicator. Starch forms a blue-black complex with iodine, which is produced by the reaction of potassium permanganate and potassium iodide. Potassium ferricyanide forms a blue precipitate of Prussian blue with ferrous ions, which are present in excess at the end point.

- The procedure for the titration is as follows:
  - Prepare a standard solution of potassium permanganate or potassium dichromate of known concentration and a solution of potassium iodide (if using starch as an indicator).
  - Pipette a known volume of the ferrous ammonium sulfate solution into a conical flask and add some dilute sulfuric acid.
  - Add a few drops of the external indicator (starch or potassium ferricyanide) to the flask.
  - Fill a burette with the standard solution of potassium permanganate or potassium dichromate and note the initial reading.
  - Titrate the ferrous ammonium sulfate solution with the standard solution until the end point is reached, indicated by the color change of the external indicator.
  - Note the final reading of the burette and calculate the volume of the standard solution used.
  - Repeat the titration for at least two more times and take the average volume of the standard solution used.
  - Calculate the strength of the ferrous ammonium sulfate solution using the following formula:

Strength of ferrous ammonium sulfate = (Volume of standard solution x Normality of standard solution x Equivalent weight of ferrous ammonium sulfate) / (Volume of ferrous ammonium sulfate solution x 1000)

- The equivalent weight of ferrous ammonium sulfate is 196 g/mol.

8. Determination of the strength of Potassium dichromate using internal indicator.

- Potassium dichromate is a primary standard substance that can be used to determine the strength of a reducing agent such as ferrous ammonium sulphate.

- The reaction between potassium dichromate and ferrous ammonium sulphate is as follows:

$$K_2Cr_2O_7{+}6FeSO_4{+}7H_2SO_4 \rightarrow Cr_2(SO_4)_3{+}3K_2SO_4{+}6Fe_2(SO_4)_3{+}7H_2O$$

- The end point of this titration can be detected by the change in colour of the solution from green to reddish-brown, due to the formation of ferric ions.
- This colour change is the internal indicator of the titration, and no external indicator is required.
- The strength of potassium dichromate can be calculated by using the following formula:

$$\text{Strength of } \mathrm{K_2Cr_2O_7} = \frac{6 \times \text{Normality of } \mathrm{FeSO_4} \times \text{Volume of } \mathrm{FeSO_4}}{\text{Volume of } \mathrm{K_2Cr_2O_7}}$$

- The procedure for this experiment is as follows:
  - Prepare a standard solution of ferrous ammonium sulphate by dissolving 3.92 g of the salt in 250 mL of distilled water and adding 10 mL of dilute sulphuric acid.
  - Transfer 20 mL of this solution to a conical flask and add 10 mL of dilute sulphuric acid.
  - Pipette out 10 mL of potassium dichromate solution of unknown strength into a burette and titrate it against the ferrous ammonium sulphate solution in the conical flask.
  - Note the initial and final burette readings and calculate the volume of potassium dichromate used.
  - Repeat the titration for two more times and take the average volume of potassium dichromate used.
  - Calculate the strength of potassium dichromate using the formula given above.

9. **Determination of available chlorine in bleaching powder**

- Bleaching powder is a compound of calcium, chlorine and oxygen with the formula Ca(OCl)Cl. It is used as a disinfectant and a bleaching agent.
- Available chlorine is the amount of chlorine that can be liberated from bleaching powder when it reacts with dilute acids. It is expressed as a percentage of the mass of bleaching powder.
- The reaction of bleaching powder with dilute acid is as follows:

Ca(OCl)Cl + H2SO4 -> CaSO4 + H2O + Cl2

- The available chlorine can be determined by titrating the chlorine gas evolved with a standard solution of sodium thiosulfate (Na2S2O3) using starch as an indicator. The reaction is as follows:

$Cl_2 + 2Na_2S_2O_3 \rightarrow 2NaCl + Na_2S_4O_6$

- The procedure for the determination of available chlorine in bleaching powder is as follows:
  - Weigh about 0.5 g of bleaching powder and transfer it to a 250 mL conical flask.
  - Add about 100 mL of distilled water and shake well to dissolve the bleaching powder.
  - Add 10 mL of dilute sulfuric acid and stopper the flask immediately.
  - Allow the flask to stand for 10 minutes and then titrate the liberated chlorine gas with 0.1 N sodium thiosulfate solution using starch as an indicator.
  - The end point is the disappearance of the blue color of the starch-iodine complex.
  - Repeat the titration with a blank solution of water and acid.
  - Calculate the percentage of available chlorine in the bleaching powder using the formula:

% available chlorine = (B - S) x N x 35.5 x 100 / W

where B is the volume of sodium thiosulfate solution used for the blank titration, S is the volume of sodium thiosulfate solution used for the sample titration, N is the normality of sodium thiosulfate solution, W is the weight of bleaching powder taken, and 35.5 is the equivalent weight of chlorine.

10. Determination of chloride content in water sample.

- Chloride is one of the major anions found in natural waters. It originates from various sources, such as seawater intrusion, industrial effluents, agricultural runoff, and domestic sewage.

- Chloride can affect the taste, corrosion, and biological activity of water. Therefore, it is important to monitor and control its concentration in drinking water, wastewater, and environmental samples.

- The most common method for determining chloride content in water sample is the **argentometric method**. It is based on the precipitation of chloride ions as silver chloride by adding a known excess of silver nitrate solution and titrating the excess silver ions with a standard solution of sodium or potassium thiocyanate using ferric ammonium sulfate as an indicator.

- The procedure for the argentometric method is as follows:
  - Pipette a 50 mL aliquot of the water sample into a 250 mL conical flask and add 2 mL of 6 M nitric acid and a few drops of potassium chromate indicator.

  - Titrate the sample with 0.02 M silver nitrate solution until a faint reddish-brown color appears, indicating the end point. Record the volume of silver nitrate solution used as V1.

- Add 2 mL of 10% ferric ammonium sulfate solution and continue the titration with 0.1 M sodium or potassium thiocyanate solution until a permanent brownish-red color appears, indicating the end point. Record the volume of thiocyanate solution used as V2.
- Repeat the procedure with a blank sample of distilled water and record the volumes of silver nitrate and thiocyanate solutions used as V3 and V4, respectively.
- Calculate the chloride content in the water sample using the following formula:
  * Chloride (mg/L) = (V1 - V3) x 0.02 x 35.45 x 1000 / 50 - (V2 - V4) x 0.1 x 35.45 x 1000 / 50
  * Where 35.45 is the molar mass of chloride ion in g/mol.

11. Preparation of Phenol formaldehyde (PF) resin

- Phenol formaldehyde resin is a type of thermosetting polymer that is obtained by the reaction of phenol and formaldehyde.
- Phenol and formaldehyde can react in different ratios to produce different types of PF resins, such as novolacs, resoles, and co-condensed resins.
- The reaction can be catalyzed by either acid or base, depending on the desired properties and applications of the resin.
- The general steps for the preparation of PF resin are:
  - Weigh the appropriate amount of phenol and formaldehyde, and mix them in a reaction vessel.
  - Add the catalyst, either acid or base, and heat the mixture to initiate the polymerization reaction.
  - Monitor the reaction parameters, such as temperature, pH, viscosity, and water content, and adjust them as needed to control the degree of polymerization and the molecular weight distribution of the resin.
  - Stop the reaction when the desired properties are achieved, and isolate the resin by filtration, precipitation, or solvent extraction.
  - Dry and cure the resin, either by heating or by adding a hardener, to form a cross-linked network that gives the resin its strength and stability.
- Some examples of PF resins and their preparation methods are:
  - Novolacs: These are linear PF resins that are prepared by using an excess of phenol and an acid catalyst, such as hydrochloric acid or sulfuric acid. The reaction ratio of phenol to formaldehyde is usually between 1.5:1 and 2:1. The reaction temperature is typically between

60°C and 100°C, and the reaction time is between 2 and 4 hours. The novolac resin is soluble in organic solvents and can be molded into various shapes by adding a hardener, such as hexamethylenetetramine, and applying heat and pressure.
  - Resoles: These are branched PF resins that are prepared by using an excess of formaldehyde and a base catalyst, such as ammonia or sodium hydroxide. The reaction ratio of phenol to formaldehyde is usually between 1:1.5 and 1:3. The reaction temperature is typically between 80°C and 120°C, and the reaction time is between 10 and 60 minutes. The resole resin is self-curing and can be hardened by heating alone, without the need of a hardener.
  - Co-condensed resins: These are PF resins that are prepared by using a mixture of phenol and another phenolic compound, such as urea, melamine, or cresol, and either an acid or a base catalyst. The reaction ratio of phenol to formaldehyde and the other phenolic compound varies depending on the desired properties and applications of the resin. The reaction temperature and time are also adjusted accordingly. The co-condensed resin can have improved properties, such as higher thermal stability, lower water absorption, and better adhesion, than the pure PF resin.

12. Preparation of Urea formaldehyde (UF) resin

- Urea formaldehyde (UF) resin is a thermosetting polymer that is widely used as an adhesive and a coating material for wood products, such as plywood, particleboard, and fiberboard.
- UF resin is formed by the condensation reaction of urea and formaldehyde in the presence of a catalyst, such as ammonia or an acid .
- The reaction can be carried out in two or three steps, depending on the desired properties and applications of the resin .
- In the two-step process, urea and formaldehyde are first mixed in a molar ratio of 2-3:1 at pH 6-11 and heated to at least 80°C to form methylolureas, which are intermediate compounds with one or more hydroxymethyl groups attached to the nitrogen atom of urea .
- Then, the pH of the mixture is lowered to 0.5-3.5 by adding a mineral or organic acid, such as sulfuric acid or formic acid, and the reaction is continued to form a viscous resin with a high degree of polymerization .
- In the three-step process, urea and formaldehyde are first mixed in a molar ratio of 1.5-2:1 at pH 8-9 and heated to 50-70°C to form monomethylolurea, which is the simplest methylolurea with one hydroxymethyl group .
- Then, ammonia is added to the mixture to increase the pH to 9-10 and the reaction is continued to form dimethylolurea, which has two hydroxymethyl groups .
- Finally, the pH of the mixture is lowered to 4.5-5.5 by adding an acid and the reaction is completed to form a resin with a low degree of polymerization and a low formaldehyde content .

- The properties and applications of UF resin depend on the molar ratio of urea and formaldehyde, the pH and temperature of the reaction, the type and amount of catalyst, and the degree of polymerization .
- UF resin has high tensile strength, hardness, and chemical resistance, but low thermal stability and moisture resistance .
- UF resin is mainly used as an adhesive for wood products, such as plywood, particleboard, and fiberboard, because it has good bonding strength and low cost .
- UF resin can also be used as a coating material for paper, textiles, and electrical appliances, because it has good surface finish and electrical insulation .
- However, UF resin has some drawbacks, such as the emission of formaldehyde during curing and service, which can cause health and environmental problems .
- Therefore, various methods have been developed to reduce the formaldehyde content and emission of UF resin, such as using formaldehyde scavengers, modifying the resin structure, and applying post-treatment techniques .

13. Preparation of Adipic acid / Paracetamol

- Adipic acid is the most important dicarboxylic acid, mainly used as a precursor for the production of nylon.
- Adipic acid can be prepared from a mixture of cyclohexanone and cyclohexanol called KA oil, which is oxidized with nitric acid to give adipic acid, via a multistep pathway .
- The oxidation of KA oil involves the following steps :
  - Formation of nitrosonium ion (NO+) from nitric acid and sulfuric acid.
  - Nitrosation of cyclohexanone and cyclohexanol to form nitrosocyclohexanone and nitrosocyclohexanol, respectively.
  - Rearrangement of nitrosocyclohexanone and nitrosocyclohexanol to form 6-nitrosocaproic acid and 6-nitroso-1-hydroxycaproic acid, respectively.
  - Oxidation of 6-nitrosocaproic acid and 6-nitroso-1-hydroxycaproic acid to form 6-nitrocaproic acid and 6-nitro-1-hydroxycaproic acid, respectively.
  - Hydrolysis of 6-nitrocaproic acid and 6-nitro-1-hydroxycaproic acid to form adipic acid and nitrous acid, respectively.
  - Recycle of nitrous acid to nitric acid by reaction with oxygen and water.
- Paracetamol is an effective antipyretic and analgesic, widely used for the relief of pain and fever.
- Paracetamol can be prepared from p-aminophenol by acetylation with acetic anhydride in the presence of concentrated sulfuric acid as a catalyst .

- The acetylation of p-aminophenol involves the following steps :
  - Formation of acylium ion (CH3CO+) from acetic anhydride and sulfuric acid.
  - Electrophilic substitution of p-aminophenol with acylium ion to form N-acetyl-p-aminophenol (paracetamol) and acetic acid.
  - Purification of paracetamol by recrystallization from water.

14. Determination of Cell Conductance of a Solution

- Cell conductance of a solution is the measure of how well the solution can conduct electric current. It is the reciprocal of the resistance of the solution.
- Cell conductance of a solution depends on the concentration, charge, and mobility of the ions present in the solution.
- To measure the cell conductance of a solution, a special device called a conductance cell is used. A conductance cell consists of two electrodes immersed in the solution, connected to a Wheatstone bridge circuit that measures the resistance of the solution.
- The conductance cell has a constant value, called the cell constant, which is the ratio of the distance between the electrodes to the area of cross-section of the electrodes. The cell constant is determined by using a solution of known specific conductance, such as potassium chloride (KCl).
- The specific conductance of a solution is the conductance of the solution per unit length and per unit cross-sectional area. It is also called the conductivity of the solution. The specific conductance of a solution is inversely proportional to the resistance of the solution and directly proportional to the cell constant.
- The formula for specific conductance of a solution is:

  ```
  Specific conductance = (Cell constant x Conductance) /
  Resistance
  ```

- The unit of specific conductance is siemens per meter (S/m) or ohm inverse meter (Ω^-1 m^-1).
- The specific conductance of a solution varies with temperature, concentration, and nature of the solute. Generally, the specific conductance of a solution increases with increasing temperature and concentration, and decreases with increasing viscosity and molecular size of the solute.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of determination of rate constant of hydrolysis of esters.

15. Determination of Rate constant of hydrolysis of esters

- Esters are organic compounds that have the general formula RCOOR', where R and R' are alkyl or aryl groups.
- Esters can undergo hydrolysis, which is a chemical reaction with water, to produce carboxylic acids and alcohols.
- The hydrolysis of esters can be catalyzed by acids or bases, or can occur spontaneously under certain conditions.
- The rate of hydrolysis of esters depends on several factors, such as the nature of the ester, the temperature, the pH, and the concentration of the reactants and products.
- The rate of hydrolysis of esters can be expressed by the following rate law:

```
rate = k[ester][water]
```

where k is the rate constant, [ester] and water are the molar concentrations of the ester and water, respectively.

- The rate constant k can be determined experimentally by measuring the rate of hydrolysis of a known concentration of ester at a constant temperature and pH.
- One method to measure the rate of hydrolysis of esters is to use a conductometric titration, which is a technique that measures the electrical conductivity of a solution as a function of the volume of titrant added.
- In a conductometric titration, the ester is hydrolyzed by a known concentration of a strong base, such as sodium hydroxide, and the resulting carboxylic acid is titrated by a known concentration of a strong acid, such as hydrochloric acid.
- The conductivity of the solution changes as the titration proceeds, due to the formation and consumption of different ions.
- The conductivity data can be plotted as a graph of conductivity versus volume of titrant, and the shape of the graph can be used to determine the rate of hydrolysis of the ester.
- The rate of hydrolysis of the ester can be calculated by using the following formula:

```
rate = (Vb x Cb x Ca) / (Vt x Ct x t)
```

where Vb is the volume of base used for hydrolysis, Cb is the concentration of base, Ca is the concentration of acid, Vt is the total volume of the solution, Ct is the concentration of ester, and t is the time taken for hydrolysis.

- The rate constant k can be obtained by dividing the rate by the concentration of ester and water, assuming that the concentration of water is constant and much larger than the concentration of ester.

16. Element detection and identification of functional groups in organic compounds.

- Organic compounds are composed of carbon, hydrogen, and other elements such as oxygen, nitrogen, sulfur, phosphorus, halogens, etc.
- The detection and identification of these elements and functional groups in organic compounds are important for determining their structure and properties.
- There are various methods for detecting and identifying elements and functional groups in organic compounds, such as:
  - Qualitative analysis: This involves testing the presence or absence of an element or a functional group by observing a characteristic reaction, color change, precipitate formation, etc. For example, carbon and hydrogen can be detected by heating the organic compound with copper oxide and observing the formation of carbon dioxide and water. Oxygen can be detected by heating the organic compound with sodium metal and observing the evolution of hydrogen gas. Nitrogen can be detected by the Lassaigne's test, which involves heating the organic compound with sodium metal, dissolving the residue in water, and adding ferrous sulfate and ferric chloride solutions. A blue or green color indicates the presence of nitrogen. Sulfur can be detected by the same test, but by adding sodium nitroprusside solution instead of ferric chloride. A violet color indicates the presence of sulfur. Halogens can be detected by the Beilstein test, which involves heating the organic compound with a copper wire and observing the color of the flame. A green flame indicates the presence of halogens. Functional groups can be detected by various chemical tests that involve specific reagents and indicators. For example, alcohols can be detected by the Lucas test, which involves adding Lucas reagent (a mixture of zinc chloride and concentrated hydrochloric acid) to the organic compound and observing the formation of a turbid solution. Aldehydes and ketones can be detected by the Fehling's test, which involves adding Fehling's solution (a mixture of copper sulfate, sodium hydroxide, and potassium sodium tartrate) to the organic compound and heating it. A red precipitate of copper oxide indicates the presence of aldehydes or ketones. Carboxylic acids can be detected by the litmus test, which involves adding a drop of litmus solution (a natural indicator) to the organic compound and observing the color change. A red color indicates the presence of carboxylic acids.
  - Quantitative analysis: This involves measuring the amount or percentage of an element or a functional group in an organic compound by using various techniques, such as:
    - Gravimetric analysis: This involves isolating and weighing the element or the functional group as a pure compound or a precipitate. For example, carbon and hydrogen can be determined by combustion analysis, which involves burning the organic com-

pound in excess oxygen and collecting and weighing the carbon dioxide and water formed. Oxygen can be determined by difference, which involves subtracting the percentage of carbon and hydrogen from 100. Nitrogen can be determined by the Dumas method, which involves heating the organic compound in a sealed tube with excess copper oxide and collecting and measuring the nitrogen gas evolved. Sulfur can be determined by the Carius method, which involves heating the organic compound in a sealed tube with fuming nitric acid and collecting and weighing the barium sulfate precipitate formed. Halogens can be determined by the same method, but by collecting and weighing the silver halide precipitate formed. Functional groups can be determined by various methods that involve converting them into measurable compounds or precipitates. For example, alcohols can be determined by the acetylation method, which involves reacting the organic compound with acetic anhydride and measuring the acetic acid formed. Aldehydes and ketones can be determined by the oxime method, which involves reacting the organic compound with hydroxylamine and measuring the oxime formed. Carboxylic acids can be determined by the neutralization method, which involves titrating the organic compound with a standard base and measuring the volume of the base used.

* Volumetric analysis: This involves measuring the volume of a solution of a known concentration that reacts with the element or the functional group in an organic compound. For example, nitrogen can be determined by the Kjeldahl method, which involves digesting the organic compound with concentrated sulfuric acid and a catalyst, distilling the ammonia formed, and titrating it with a standard acid. Halogens can be determined by the Volhard method, which involves precipitating the halides with silver nitrate, filtering and washing the precipitate, dissolving it in ammonium thiocyanate, and titrating the excess thiocyanate with a standard iron(III) solution. Functional groups can be determined by various methods that involve titrating them with a standard solution of a

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- This note indicates that the instructor has the authority and flexibility to select the experiments that are most suitable for the course objectives and the available resources.
- The instructor can choose any 10 experiments from the list of experiments given in the syllabus or the lab manual, depending on the relevance, difficulty, and duration of each experiment.
- The instructor can also modify or replace any two of the experiments from

the list, if they find them to be outdated, impractical, or redundant.
- The instructor should inform the students about the chosen experiments and any changes made to them before the start of the lab sessions, so that the students can prepare accordingly.
- The instructor should also provide the rationale and the learning outcomes for each experiment, as well as the expected results and the evaluation criteria.

Course Outcomes:

- By the end of this course, you will be able to:
  - Identify and explain the basic concepts and principles of artificial intelligence, such as search, knowledge representation, reasoning, planning, learning, and natural language processing.
  - Apply various artificial intelligence techniques and algorithms to solve problems, such as heuristic search, constraint satisfaction, logic, inference, probabilistic models, decision making, neural networks, and machine learning.
  - Evaluate the strengths and limitations of different artificial intelligence approaches and methods, and compare their performance and applicability to different domains and scenarios.
  - Design and implement artificial intelligence systems and applications using appropriate tools and frameworks, such as Python, TensorFlow, PyTorch, scikit-learn, NLTK, etc.
  - Communicate and present artificial intelligence concepts, methods, results, and ethical issues effectively using oral, written, and visual means.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply appropriate methods and techniques to solve various artificial intelligence problems, such as game playing, constraint satisfaction, logical inference, probabilistic reasoning, machine learning, natural language understanding, image analysis, and robot control.
- Evaluate the strengths and limitations of different artificial intelligence approaches and paradigms, such as symbolic, connectionist, evolutionary, and hybrid systems.
- Analyze the ethical, social, and legal implications of artificial intelligence applications and systems, such as autonomy, responsibility, privacy, bias, and human-AI interaction.
- Design and implement simple artificial intelligence programs and projects using relevant tools and frameworks, such as Python, TensorFlow, Py-

Torch, OpenAI Gym, and ROS.

Course Outcomes Bloom's

- Course outcomes are brief statements that describe what students will be expected to learn by the end of the course.
- Bloom's taxonomy is a model of cognitive skills used to classify educational learning objectives and is a helpful tool for the development of learning outcomes.
- Bloom's taxonomy consists of six levels of thinking: remember, understand, apply, analyze, evaluate, and create.
- The levels of Bloom's taxonomy are arranged in a hierarchical order, meaning that each level requires the mastery of the previous one.
- Bloom's taxonomy provides instructors with a focus for designing their course materials, activities, and assessments that align with the learning outcomes .
- Bloom's taxonomy also helps students to monitor their own learning progress and to develop higher-order thinking skills.

Level

- A level is a measure of the amount or degree of something, such as height, quantity, quality, intensity, etc.
- A level can be expressed in different units, such as meters, liters, decibels, degrees, etc.
- A level can be compared to a standard or a reference point, such as sea level, ground level, boiling point, etc.
- A level can be used to classify or rank something, such as difficulty level, skill level, education level, etc.
- A level can be changed or adjusted, such as raising or lowering the water level, increasing or decreasing the sound level, etc.

CO-1 Get an understanding of the use of different analytical instruments. K3

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.

- Different analytical instruments have different principles, applications, advantages, and limitations.

- Some examples of analytical instruments are:

  - Spectrophotometers: measure the intensity of light absorbed or emitted by a sample at different wavelengths.
  - Chromatographs: separate the components of a mixture based on their affinity to a stationary phase and a mobile phase.

- Mass spectrometers: identify the molecular mass and structure of compounds by ionizing them and measuring their mass-to-charge ratio.
- Microscopes: magnify the image of small objects or structures using visible light, electrons, or other forms of radiation.
- Electrochemical analyzers: measure the electrical potential, current, or charge of a sample in relation to a reference electrode.
- Thermal analyzers: measure the changes in physical or chemical properties of a sample as a function of temperature or time.
- Nuclear magnetic resonance (NMR) spectrometers: measure the resonance frequency of atomic nuclei in a magnetic field, which reflects their chemical environment and interactions.
- X-ray diffraction (XRD) instruments: measure the diffraction pattern of X-rays by a crystalline sample, which reveals its atomic structure and symmetry.
- Atomic absorption (AA) spectrometers: measure the absorption of light by free atoms in a flame or a furnace, which reflects their concentration and identity.
- Inductively coupled plasma (ICP) instruments: measure the emission of light by atoms or ions in a plasma, which reflects their concentration and identity.

- To get an understanding of the use of different analytical instruments, one should:
  - Know the basic principles and components of each instrument and how they affect the performance and accuracy of the measurements.
  - Know the types of samples and parameters that can be analyzed by each instrument and the sample preparation and handling methods required.
  - Know the advantages and limitations of each instrument and the sources of errors and interferences that can affect the results.
  - Know the calibration and validation methods for each instrument and the quality control and assurance procedures to ensure the reliability and reproducibility of the measurements.
  - Know the interpretation and presentation of the data obtained by each instrument and the statistical methods to evaluate the significance and uncertainty of the results.
  - Know the applications and relevance of each instrument in different fields of science, engineering, medicine, industry, and environment.

CO2 Molecular Properties and Surface Tension

- Carbon dioxide (CO2) is a linear molecule with a bond length of 116.3 pm and a bond angle of 180 degrees.
- CO2 is a non-polar molecule with a dipole moment of zero and a symmetrical distribution of electrons.

- CO2 has a molar mass of 44.0095 g/mol and a critical point at 304.25 K and 7.38 MPa.
- CO2 is a gas at standard temperature and pressure, but can be liquefied or solidified under certain conditions.
- The density and specific weight of CO2 vary with temperature and pressure, and can be calculated using online tools or tables.
- The surface tension of CO2 is the energy required to increase the surface area of the liquid phase due to intermolecular forces.
- The surface tension of CO2 depends on the temperature, pressure, and the presence of other substances in the solution .
- The surface tension of CO2 can be measured using various methods, such as the pendant drop method, the Wilhelmy plate method, or the capillary rise method.
- The surface tension of CO2 is important for applications such as carbon capture and storage, enhanced oil recovery, and supercritical fluid extraction .

Viscosity

- Viscosity is the property of a fluid that measures its resistance to flow.
- Viscosity depends on the internal friction between the molecules of the fluid, which is affected by factors such as temperature, pressure, and composition.
- Viscosity is expressed by the formula: ` = F / (A * dv/dy)`, where is the viscosity, `F` is the force applied, `A` is the area of contact, and `dv/dy` is the velocity gradient of the fluid.
- The SI unit of viscosity is the pascal-second ($\mathrm{Pa \cdot s}$), which is equivalent to one newton-second per square meter ($\mathrm{N \cdot s/m^2}$). Other common units are the poise (P), which is equal to 0.1 $\mathrm{Pa \cdot s}$, and the centipoise (cP), which is equal to 0.001 $\mathrm{Pa \cdot s}$.
- Viscosity can be classified into two types: dynamic viscosity and kinematic viscosity. Dynamic viscosity is the ratio of the shear stress to the shear rate of a fluid, while kinematic viscosity is the ratio of the dynamic viscosity to the density of the fluid.
- Examples of fluids with high viscosity are honey, molasses, and tar, while examples of fluids with low viscosity are water, gasoline, and air.

K3

- K3 is a type of synthetic cannabinoid that mimics the effects of THC, the main psychoactive compound in cannabis.
- K3 is also known as Spice, K2, or synthetic marijuana, and is sold as a herbal incense or potpourri.
- K3 is sprayed with various chemicals that bind to the same receptors in the brain as THC, but with much higher potency and unpredictability.
- K3 can cause a range of adverse effects, such as paranoia, anxiety, hallu-

cinations, psychosis, seizures, vomiting, and cardiac problems.

- K3 is illegal in many countries, but some users may seek it out as a cheaper or more accessible alternative to cannabis, or to evade drug tests.
- K3 is not detected by standard urine tests, but can be identified by specialized tests that look for specific metabolites.
- K3 is not safe to use and can have serious health consequences, especially when mixed with other substances or taken in high doses.

CO-3 Measure the hardness and alkalinity of the water. K3

- Hardness and alkalinity are two parameters that indicate the water quality and the suitability of water for various purposes.
- Hardness measures the total amount of divalent salts, mainly calcium and magnesium, that can form insoluble deposits or scale in pipes, boilers, and appliances.
- Alkalinity measures the total amount of bases, mainly bicarbonate and carbonate, that can neutralize acids and buffer the pH of water.
- Hardness and alkalinity are related through common ions formed in aquatic systems. The counter-ions associated with the bicarbonate and carbonate fraction of alkalinity are the principal cations responsible for hardness.
- Hardness and alkalinity are influenced by the geology of the area where the water is located and the dissolution of carbon dioxide from the atmosphere. The ions responsible for hardness and alkalinity originate from the dissolution of geological minerals into rain and groundwater.
- Hardness and alkalinity can be measured by titration methods using standard solutions of acids and indicators. The end point of the titration is determined by a color change of the indicator or a pH meter.
- Hardness can be classified into two types: temporary and permanent. Temporary hardness is caused by the presence of bicarbonate ions and can be removed by boiling the water. Permanent hardness is caused by the presence of other ions such as sulfate, chloride, and nitrate and can be removed by ion exchange or other methods.
- Alkalinity can be classified into three types: carbonate, bicarbonate, and hydroxide. Carbonate alkalinity is present when the pH of the water is above 8.3 and can be neutralized by adding a strong acid. Bicarbonate alkalinity is present when the pH of the water is between 4.5 and 8.3 and can be neutralized by adding a weak acid. Hydroxide alkalinity is present when the pH of the water is above 10.3 and can be neutralized by adding a strong acid.
- Hardness and alkalinity are important for various applications of water such as drinking, irrigation, industrial, and aquatic life. Hardness and alkalinity affect the taste, corrosiveness, scaling potential, and buffering capacity of water .

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic compound with a hydroxyl group (-OH) attached to a benzene ring. Phenol has the chemical formula C_6H_5OH and is also known as carbolic acid. Phenol is a weak acid and can form phenoxide ions by losing a proton from the hydroxyl group.

There are several methods to prepare phenol from different starting materials. Some of the common methods are:

- From haloarenes: Haloarenes are aromatic compounds with a halogen atom (-X) attached to a benzene ring. For example, chlorobenzene is a haloarene with a chlorine atom (-Cl) attached to a benzene ring. Haloarenes can be converted to phenol by heating them with sodium hydroxide (NaOH) at high temperature and pressure. This is called the Dow process. The reaction is as follows:

  $C_6H_5X + NaOH \rightarrow C_6H_5ONa + H_2O + NaX$

  $C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

  The sodium phenoxide (C_6H_5ONa) formed in the first step is hydrolyzed to phenol in the second step. The halide ion (X^-) is removed as a sodium salt (NaX).

- From benzene: Benzene is an aromatic compound with a six-carbon ring and three double bonds. Benzene can be converted to phenol by first sulfonating it with concentrated sulfuric acid (H_2SO_4) to form benzene sulfonic acid ($C_6H_5SO_3H$). This is called the sulfonation reaction. The reaction is as follows:

  $C_6H_6 + H_2SO_4 \rightarrow C_6H_5SO_3H + H_2O$

  The benzene sulfonic acid can then be fused with sodium hydroxide at high temperature to form sodium phenoxide, which can be hydrolyzed to phenol as in the previous method. This is called the desulfonation reaction. The reaction is as follows:

  $C_6H_5SO_3H + NaOH \rightarrow C_6H_5ONa + NaHSO_4$

  $C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

- From cumene: Cumene is an alkylbenzene with an isopropyl group (-$CH(CH_3)_2$) attached to a benzene ring. Cumene can be oxidized by air to form cumene hydroperoxide ($C_6H_5C(CH_3)_2OOH$), which can then be cleaved by acid to form phenol and acetone (CH_3COCH_3). This is called the cumene process. The reaction is as follows:

  $C_6H_5CH(CH_3)_2 + O_2 \rightarrow C_6H_5C(CH_3)_2OOH$

  $C_6H_5C(CH_3)_2OOH + H_2O \rightarrow C_6H_5OH + CH_3COCH_3$

- From diazonium salts: Diazonium salts are compounds with a diazonium group ($-N_2^+$) attached to an aromatic ring. For example, benzenediazonium chloride ($C_6H_5N_2^+Cl^-$) is a diazonium salt with a chlorine ion ($-Cl$) attached to the diazonium group. Diazonium salts can be hydrolyzed to phenol by heating them with water. This is called the Sandmeyer reaction. The reaction is as follows:

  $C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + N_2 + HCl$

  The nitrogen gas (N_2) and the halide ion (Cl^-) are removed as by-products.

Formaldehyde

- Formaldehyde is a naturally occurring organic compound with the formula CH_2O and structure H−CHO.
- Formaldehyde is a colorless, highly toxic, and flammable gas at room temperature.
- Formaldehyde is an intermediate in the oxidation (or combustion) of methane, as well as of other carbon compounds, e.g. in forest fires, automobile exhaust, and tobacco smoke.
- Formaldehyde is also produced in the atmosphere by the action of sunlight and oxygen on atmospheric methane and other hydrocarbons, and it becomes part of smog.
- Formaldehyde is used widely by industry to manufacture building materials and numerous household products, such as fertilizer, paper, plywood, resins, antiseptics, medicines, and cosmetics .
- Formaldehyde is also used as a food preservative and in embalming fluids.
- Formaldehyde can cause irritation of the skin, eyes, nose, and throat, as well as respiratory and neurological effects .
- Formaldehyde is classified as a human carcinogen by the International Agency for Research on Cancer and the US National Toxicology Program, based on evidence from studies of workers exposed to high levels of formaldehyde and from animal experiments .
- Formaldehyde exposure can be reduced by using products that contain low or no formaldehyde, ventilating indoor spaces, avoiding smoking, and following safety precautions when handling formaldehyde-containing products .

Urea formaldehyde resin

- Urea formaldehyde resin is a synthetic polymer made by reacting urea and formaldehyde in an aqueous solution.
- Urea formaldehyde resin is a thermosetting resin, meaning that it can be cured by heat and pressure to form a hard and durable material.
- Urea formaldehyde resin is widely used as an adhesive for wood products, such as particleboard, plywood, and fiberboard, because of its high strength, water resistance, and low cost .

- Urea formaldehyde resin is also used as a coating for paper, textiles, and electrical appliances, as well as a molding compound for buttons, knobs, and electrical components .
- Urea formaldehyde resin can release formaldehyde gas during its production, processing, and use, which can cause health problems for workers and consumers .
- Urea formaldehyde resin can also emit formaldehyde gas over time from products that contain it, especially when exposed to heat and humidity .
- Urea formaldehyde resin is regulated by various agencies and standards to limit its formaldehyde emissions and ensure its safe use .
- Urea formaldehyde resin can be replaced by other resins that have lower or no formaldehyde emissions, such as phenol formaldehyde, melamine formaldehyde, or soy-based resins .

Adipic acid

- Adipic acid is an organic compound with the formula C6H10O4 and structure HOOC−(CH2)4−COOH.
- Adipic acid is a white, crystalline solid that is slightly soluble in water and has a sour taste.
- Adipic acid is mainly produced by oxidation of cyclohexane or cyclohexanol with nitric acid, or by fermentation of glucose or other sugars .
- Adipic acid is the most widely used dicarboxylic acid in the chemical industry, accounting for about 60% of the global market.
- Adipic acid is mainly used as a precursor for the production of nylon 6,6, a synthetic polymer that is used for making fibers, fabrics, carpets, plastics, and other products .
- Adipic acid is also used as a monomer for making polyurethane, a versatile material that is used for making foams, coatings, adhesives, and elastomers .
- Adipic acid is also used as an additive for food, beverages, and pharmaceuticals, as it can act as a flavoring agent, an acidulant, a buffering agent, and a gelling agent .
- Adipic acid is generally considered to be safe and biode

K3

- K3 is a type of surface mount technology (SMT) component package that is used for integrated circuits (ICs).
- K3 stands for **Kilby 3**, named after Jack Kilby, the inventor of the integrated circuit.
- K3 is a **quad flat no-lead (QFN)** package, which means it has four flat sides with no leads or pins protruding from the edges.
- K3 has a **thermal pad** at the bottom of the package, which is used to dissipate heat from the IC and improve its performance and reliability.
- K3 has a **pitch** of 0.5 mm, which means the distance between the center

of adjacent pads on the same side is 0.5 mm.

- K3 has a **body size** of 3 mm x 3 mm, which means the length and width of the package are both 3 mm.
- K3 has a **pad size** of 0.25 mm x 0.5 mm, which means the length and width of each pad are 0.25 mm and 0.5 mm respectively.
- K3 has a **pad count** of 16, which means it has 16 pads on the four sides of the package, with 4 pads on each side.
- K3 is a **low-profile** package, which means it has a height of less than 1 mm.
- K3 is a **lead-free** package, which means it does not contain any lead or other hazardous substances that could harm the environment or human health.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction, K3, is a measure of how fast the reaction proceeds at a given temperature and concentration of reactants.
- The rate constant of a reaction can be estimated by using the Arrhenius equation, which relates K3 to the activation energy, Ea, and the frequency factor, A, of the reaction.
- The Arrhenius equation is:

K3 = A * exp(-Ea / RT)

where R is the universal gas constant and T is the absolute temperature.

- To estimate K3, one needs to know the values of A and Ea, which can be obtained from experimental data or theoretical calculations.
- Alternatively, one can use the linearized form of the Arrhenius equation, which is:

ln(K3) = ln(A) - Ea / RT

- By plotting ln(K3) versus 1/T for different temperatures, one can obtain a straight line with slope -Ea / R and intercept ln(A).
- The values of Ea and A can then be calculated from the slope and intercept of the line, and K3 can be estimated for any temperature.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.

- The topic can be used to guide the reader's or listener's attention and interest, such as by using hooks, transitions, examples, or questions.
- The topic can be used to communicate the purpose and goal of the text, such as by using a thesis statement, a claim, an argument, or a conclusion.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

- History
- Science
- Literature
- Art
- Music
- Sports
- Politics
- Philosophy
- Economics
- Geography
- Culture
- Language
- Technology
- Health
- Education

Please choose a topic or type your own.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have not specified the topic you want to write about. Please enter the topic name after the colon:

The topic is:

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.
- The topic can be used to guide the research and analysis of the text, such as by finding relevant sources, evaluating the evidence, and drawing conclusions.

- The topic can be used to communicate the purpose and message of the text, such as by using a thesis statement, a hook, a transition, or a call to action.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

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- History
- Science
- Mathematics
- Literature
- Art
- Music
- Sports
- Technology
- Philosophy
- Politics

Please choose a topic or type your own.

Page 29 of 40

- The topic of this page is **the structure and function of the human eye**.
- The human eye is a complex organ that allows us to see the world around us.
- The eye consists of several parts, each with a specific role in vision.

The main parts of the eye are:

- The **cornea**: a transparent layer that covers the front of the eye and refracts (bends) light rays entering the eye.
- The **iris**: a circular muscle that controls the size of the pupil, the opening in the center of the iris that regulates the amount of light entering the eye.
- The **lens**: a flexible structure behind the pupil that changes shape to focus light rays onto the retina, the light-sensitive layer at the back of the eye.
- The **retina**: a thin layer of cells that converts light into electrical signals that are sent to the brain via the optic nerve.
- The **optic nerve**: a bundle of nerve fibers that carries visual information from the retina to the brain.
- The **vitreous humor**: a clear gel that fills the space between the lens and the retina and helps maintain the shape of the eye.

The function of the eye is to:

- Capture light rays from the environment and focus them onto the retina.
- Convert light into electrical signals that can be processed by the brain.
- Transmit visual information to the brain for interpretation and recognition.

The process of vision involves:

- Refraction of light by the cornea and the lens to form an inverted image on the retina.
- Phototransduction by the photoreceptor cells (rods and cones) in the retina that convert light into electrical signals.
- Transmission of electrical signals by the bipolar cells and the ganglion cells in the retina to the optic nerve.
- Integration of visual information by the brain in the visual cortex and other areas that enable us to perceive color, shape, depth, motion, and other aspects of vision.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

- ## History of Artificial Intelligence
- ## Basics of Programming
- ## Current Events and News
- ## Science and Technology
- ## Arts and Culture
-

- ### Health and Wellness

- ### Sports and Games

- ### Travel and Tourism

- ### Hobbies and Interests

- ### Personal Development

Please choose a topic or type your own.

ENGINEERING PHYSICS LAB KCS

- Engineering physics is an interdisciplinary field that combines physics and engineering to solve practical problems and develop new technologies.
- Engineering physics lab is a course that provides hands-on experience and skills in conducting experiments, analyzing data, and reporting results related to engineering physics topics.
- KCS is an acronym for Kennedy College of Sciences, a division of the University of Massachusetts Lowell that offers various programs in science and engineering, including engineering physics.
- Some of the objectives of the engineering physics lab course are:
  - To familiarize students with the basic principles and methods of experimental physics and engineering.
  - To enhance students' understanding of the theoretical concepts and models of physics and engineering by applying them to real-world situations.
  - To develop students' ability to design, perform, and troubleshoot experiments using various instruments and equipment.
  - To improve students' communication and teamwork skills by preparing and presenting lab reports and collaborating with peers.
- Some of the topics covered in the engineering physics lab course are:

  - Mechanics: kinematics, dynamics, conservation laws, rotational motion, oscillations, waves, fluids, etc.
  - Thermodynamics: temperature, heat, work, energy, entropy, heat engines, etc.
  - Electricity and magnetism: electric fields, electric potential, capacitance, resistance, Ohm's law, Kirchhoff's laws, magnetic fields, Faraday's law, induction, etc.
  - Optics: reflection, refraction, lenses, mirrors, interference, diffraction, polarization, etc.
  - Modern physics: atomic structure, nuclear physics, radioactivity, quantum mechanics, etc.

- Some of the experiments performed in the engineering physics lab course are:

  - Measurement of g using a simple pendulum
  - Verification of Hooke's law using a spring
  - Determination of Young's modulus using a cantilever beam
  - Measurement of viscosity using a falling ball viscometer
  - Calibration of a thermocouple and a thermistor
  - Measurement of specific heat capacity using a calorimeter
  - Verification of Ohm's law and Kirchhoff's laws using a circuit board
  - Measurement of the magnetic field of a solenoid using a Hall probe
  - Measurement of the focal length of a convex lens using a spherometer
  - Verification of Malus' law using a polarimeter
  - Measurement of the wavelength of a laser using a diffraction grating
  - Measurement of the half-life of a radioactive source using a Geiger counter
  - Measurement of the Planck's constant using a photoelectric effect apparatus

Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To develop the skills and techniques for designing, implementing, and evaluating AI systems and agents.
- To explore the ethical, social, and legal implications of AI and its impact on human society.
- To provide a foundation for further study and research in AI and related fields.

1. To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- Analytical instruments can be used for various purposes, such as quality control, research and development, environmental monitoring, forensic analysis, clinical diagnosis, etc.
- Analytical instruments can provide qualitative or quantitative information about the samples, such as their identity, composition, structure, concentration, purity, etc.
- Analytical instruments can have different components, such as a source, a detector, a signal processor, a display, a sample holder, etc.
- Analytical instruments can have different characteristics, such as sensitivity, selectivity, accuracy, precision, resolution, range, etc.
- Analytical instruments can have different advantages and limitations, such as speed, cost, reliability, complexity, safety, etc.
- Analytical instruments can follow different standards and protocols, such as calibration, validation, verification, maintenance, etc.

Analysis of Chloride Content, Hardness, and Alkalinity

- Chloride content is a measure of the amount of chloride ions (Cl-) present in a water sample. Chloride ions are derived from sources such as salt, rocks, soil, industrial effluents, and sewage. Chloride content affects the taste, corrosion, and disinfection of water.
- Hardness is a measure of the amount of calcium (Ca2+) and magnesium (Mg2+) ions present in a water sample. These ions are derived from sources such as limestone, dolomite, gypsum, and fertilizers. Hardness affects the lathering, scaling, and softening of water.
- Alkalinity is a measure of the ability of a water sample to neutralize acids. It is mainly due to the presence of bicarbonate (HCO3-), carbonate (CO32-), and hydroxide (OH-) ions. Alkalinity affects the pH, buffering, and coagulation of water.

Methods of Analysis

- Chloride content can be determined by titrating a water sample with silver nitrate (AgNO3) solution using potassium chromate (K2CrO4) as an

indicator. The end point is marked by the formation of a reddish-brown precipitate of silver chromate (Ag_2CrO_4).
- Hardness can be determined by titrating a water sample with ethylenediaminetetraacetic acid (EDTA) solution using eriochrome black T (EBT) as an indicator. The end point is marked by a color change from wine red to blue.
- Alkalinity can be determined by titrating a water sample with sulfuric acid (H_2SO_4) solution using phenolphthalein and methyl orange as indicators. The end point is marked by a color change from pink to colorless with phenolphthalein and from yellow to orange with methyl orange.

Applications of Analysis

- Chloride content is used to assess the suitability of water for drinking, irrigation, and industrial purposes. It is also used to monitor the salinity and pollution of water sources.
- Hardness is used to assess the quality of water for domestic, industrial, and agricultural purposes. It is also used to design and operate water treatment processes such as softening, reverse osmosis, and ion exchange.
- Alkalinity is used to assess the acid-base balance and buffering capacity of water. It is also used to control the pH and coagulation of water during treatment and distribution.

Water

Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states. It is one of the most plentiful and essential of compounds. A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.

Some of the main properties and characteristics of water are:

- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen. Water molecules cling to each other because of a force called hydrogen bonding. It's the reason why water can do amazing things.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid. Liquid water is the most common form of water on Earth. Gas water is also called water vapor or steam. Solid water is also called ice or snow.
- Water is the liquid that makes life on Earth possible. As water cycles from the air to the land to the sea and back again, water shapes our planet — and nearly every aspect of our lives. All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H_2O. It is a transparent, tasteless, odorless, and nearly colorless chemical substance,

and it is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

- Water has a high specific heat capacity, meaning it can absorb and release a lot of heat without changing its temperature much. This helps regulate the climate and the body temperature of living organisms.
- Water has a high surface tension, meaning it can form droplets and bubbles and resist external forces. This allows water to flow, stick, and form shapes.
- Water has a high latent heat of vaporization, meaning it takes a lot of energy to change water from liquid to gas. This helps cool down the environment and the body when water evaporates.
- Water is a universal solvent, meaning it can dissolve many other substances. This makes water a medium for chemical reactions and transport of nutrients and wastes.
- Water is a polar molecule, meaning it has a slight positive charge on one end and a slight negative charge on the other. This makes water attract other polar molecules and ions, and repel nonpolar molecules. This also affects the solubility, conductivity, and reactivity of water.

Water is a vital resource for human activities, such as drinking, cooking, washing, agriculture, industry, and energy production. Water is also a source of recreation, beauty, and inspiration. However, water is also a limited and threatened resource, as many regions face water scarcity, pollution, and degradation. Therefore, water conservation, management, and protection are important challenges for the present and future generations.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of how acidic or basic a solution is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. pH = -log[H+]
- The pH scale ranges from 0 to 14, with 7 being neutral, lower than 7 being acidic, and higher than 7 being basic. The pH of pure water is 7 at 25°C.
- pH can be measured using indicators, such as litmus paper, phenolphthalein, or universal indicator, that change color depending on the acidity or basicity of the solution. pH can also be measured using a pH meter, which is an electronic device that measures the voltage difference between a reference electrode and a pH-sensitive electrode.
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is defined as the force per unit length that must be applied to the surface to break it. Surface tension = F/L

- Surface tension causes liquids to form spherical droplets, to minimize their surface area. It also allows some objects, such as insects or needles, to float on the surface of water, if they do not break the surface.
- Surface tension can be measured using a force gauge, which measures the force required to pull a ring or a plate out of the liquid surface. Surface tension can also be measured using a stalagmometer, which measures the number of drops of liquid that fall from a capillary tube under gravity.
- Viscosity is a measure of how resistant a fluid is to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity = /
- The shear stress is the force per unit area that is applied parallel to the surface of the fluid. The shear rate is the rate of change of velocity per unit distance in the direction perpendicular to the surface of the fluid.
- Viscosity depends on the temperature and the composition of the fluid. Generally, viscosity decreases as temperature increases, and increases as the fluid becomes more complex or concentrated.
- Viscosity can be measured using a viscometer, which is a device that measures the time it takes for a fluid to flow through a tube or a gap under a fixed pressure difference. Viscosity can also be measured using a rheometer, which is a device that measures the torque required to rotate a cylinder or a plate in a fluid.

A Liquid

- A liquid is one of the four states of matter, along with solid, gas, and plasma.
- A liquid is a fluid that has a definite volume but no fixed shape. It can flow and take the shape of its container.
- A liquid is composed of molecules that are loosely bound by intermolecular forces, such as hydrogen bonds or van der Waals forces.
- A liquid can undergo phase transitions to a solid (freezing) or a gas (evaporation or boiling) depending on the temperature and pressure.
- A liquid has some properties that are different from solids and gases, such as viscosity, surface tension, capillarity, and vapor pressure.

Preparation of Different Resins

Resins are organic compounds that are solid or semi-solid at room temperature and have a high molecular weight. They are usually derived from natural sources, such as plants, animals, or microorganisms, or synthesized from simple monomers, such as phenol, formaldehyde, or acrylic acid. Resins have various applications in industries, such as coatings, adhesives, plastics, rubber, and cosmetics.

Some of the common methods of preparing different resins are:

- **Polymerization**: This is the process of joining monomers together to form long chains of repeating units, called polymers. Polymerization can be initiated by heat, light, catalysts, or chemical agents. For example, phenol and formaldehyde can be polymerized to form phenol-formaldehyde resin, which is widely used as a thermosetting plastic and an adhesive.
- **Condensation**: This is the process of removing water or other small molecules from two or more reactants to form a larger molecule. Condensation can also be catalyzed by heat, acid, or base. For example, glycerol and phthalic anhydride can be condensed to form glyptal resin, which is used as a coating and a plasticizer.
- **Esterification**: This is the process of forming an ester bond between an alcohol and a carboxylic acid. Esterification can be catalyzed by acid or base. For example, glycerol and rosin (a natural resin obtained from pine trees) can be esterified to form glycerol ester of rosin, which is used as a tackifier and a stabilizer.
- **Hydrolysis**: This is the process of breaking down a large molecule into smaller molecules by adding water. Hydrolysis can be catalyzed by acid or base. For example, urea and formaldehyde can be hydrolyzed to form urea-formaldehyde resin, which is used as a fertilizer and an adhesive.

Synthesis of Adipic Acid

- Adipic acid is a dicarboxylic acid with the formula C6H10O4. It is mainly used for the production of nylon, polyurethanes, and other polymers.

- The conventional method for the synthesis of adipic acid is the oxidation of a mixture of cyclohexanone and cyclohexanol (KA oil) with nitric acid . This process involves several steps and produces nitrous oxide (N2O) as a by-product, which is a greenhouse gas and an ozone-depleting agent .

- The steps of the nitric acid oxidation of KA oil are as follows:

  - Cyclohexanol is converted to cyclohexanone by dehydrogenation, releasing nitrous acid (HNO2).
  - Cyclohexanone is converted to 6-hydroxyimino-6-nitrohexane (HINH) by nitrosation, releasing water.
  - HINH is converted to 6-oxo-6-nitrohexanoic acid (ONHA) by hydrolysis, releasing ammonia (NH3).
  - ONHA is converted to adipic acid by hydrogenation, releasing nitrous oxide (N2O).

- The overall reaction can be written as:

  C6H10O + 4 HNO3 $\rightarrow$ C6H10O4 + 4 NO + 2 H2O

- The nitric acid oxidation of KA oil is usually carried out in a batch reactor or a continuous stirred tank reactor (CSTR) at temperatures of 50-70

°C and pressures of 5-10 atm. The reaction is catalyzed by ammonium vanadate or other metal oxides .

- The yield of adipic acid from KA oil is about 80-85%, and the selectivity is about 90-95%. The main side products are glutaric acid, succinic acid, and nitrobenzene.
- The nitric acid oxidation of KA oil is an energy-intensive and environmentally unfriendly process, and alternative methods for the synthesis of adipic acid from renewable sources are being explored . Some of these methods include:
  - Oxidation of cyclohexene or cyclohexane with hydrogen peroxide or oxygen in the presence of various catalysts.
  - Oxidation of glucose or its derivatives with nitric acid, nitrous acid, or oxygen in the presence of various catalysts.
  - Fermentation of glucose or its derivatives by microorganisms that produce adipic acid or its precursors.
  - Biocatalytic conversion of cyclohexanol or cyclohexanone to adipic acid by enzymes or whole cells.

Acid and Paracetamol by Conventional and Green Route

- Paracetamol, also known as acetaminophen, is a common painkiller and antipyretic drug that is derived from phenol.
- The conventional route for the synthesis of paracetamol involves the following steps :
  - Nitration of phenol with nitric acid to form p-nitrophenol.
  - Reduction of p-nitrophenol with tin and hydrochloric acid to form p-aminophenol.
  - Acetylation of p-aminophenol with acetic anhydride to form paracetamol.
- The conventional route has some drawbacks, such as:
  - Low yield and selectivity of p-nitrophenol due to the formation of o-nitrophenol as a by-product.
  - Generation of toxic and corrosive wastes, such as nitric acid, tin chloride, and acetic acid.
  - Use of hazardous and expensive reagents, such as nitric acid, tin, and acetic anhydride.
- The green route for the synthesis of paracetamol involves the following steps :
  - Oxidation of phenol with hydrogen peroxide and a catalyst, such as iron(III) chloride or manganese dioxide, to form p-hydroquinone.
  - Amidation of p-hydroquinone with acetic acid and a catalyst, such as sodium acetate or sodium hydroxide, to form paracetamol.

- The green route has some advantages, such as:
  - High yield and selectivity of paracetamol due to the direct conversion of phenol to paracetamol without intermediate steps.
  - Reduction of environmental impact and cost due to the use of benign and cheap reagents, such as hydrogen peroxide, acetic acid, and sodium acetate.
  - Elimination of the need for solvents and heating due to the use of microwave irradiation or solvent-free conditions.

List of Experiments

- An experiment is a scientific procedure that tests a hypothesis or a prediction by manipulating one or more variables and observing the outcomes.
- Experiments are essential for the advancement of scientific knowledge and the development of new technologies and applications.
- There are many types of experiments, such as natural experiments, controlled experiments, randomized experiments, quasi-experiments, field experiments, laboratory experiments, etc.
- Some examples of famous experiments in history are:
  - The Michelson-Morley experiment, which disproved the existence of the luminiferous ether and paved the way for the theory of relativity.
  - The Rutherford gold foil experiment, which revealed the structure of the atom and the existence of the nucleus.
  - The Stanford prison experiment, which studied the effects of social roles and power on human behavior and ethics.
  - The Milgram experiment, which measured the willingness of people to obey authority figures even if it meant harming others.
  - The Pavlov's dog experiment, which demonstrated the phenomenon of classical conditioning and the association of stimuli and responses.
  - The double-slit experiment, which showed the wave-particle duality of light and matter and the role of observation in quantum mechanics.
  - The Asch conformity experiment, which tested the influence of social pressure and group norms on individual judgments and opinions.
  - The Skinner box experiment, which investigated the principles of operant conditioning and the effects of reinforcement and punishment on behavior.
  - The Watson and Rayner's Little Albert experiment, which induced a phobia in a child by pairing a loud noise with a white rat.
  - The Harlow's monkey experiment, which explored the importance of maternal attachment and social deprivation on the emotional development of primates.

Calibration of Analytical Equipment and Apparatus

- Calibration is the process of verifying the accuracy and precision of an analytical instrument or device by comparing its measurements with a known standard of reference.
- Calibration ensures that the instrument or device is functioning properly and producing reliable and consistent results within the specified limits of error.
- Calibration is essential for maintaining the quality and validity of analytical data and for complying with the regulatory and quality standards.
- Calibration should be performed periodically, before and after each use, or whenever there is a change in the operating conditions or performance of the instrument or device.
- Calibration should be documented and recorded in a calibration log or certificate that includes the date, time, operator, standard used, calibration method, results, and any corrective actions taken.
- Calibration methods vary depending on the type and purpose of the instrument or device, but generally involve the following steps:
  - Selecting an appropriate standard of reference that has a known or certified value and is traceable to a national or international standard.
  - Preparing the instrument or device for calibration by ensuring that it is clean, stable, and in good working condition.
  - Performing the calibration procedure by applying the standard to the instrument or device and measuring its response or output.
  - Comparing the measured response or output with the expected value of the standard and calculating the error or deviation.
  - Adjusting the instrument or device if the error or deviation exceeds the acceptable limits or specifications.
  - Repeating the calibration procedure until the instrument or device meets the required accuracy and precision criteria.
  - Confirming the calibration by testing the instrument or device with a different standard or sample and verifying that the results are within the expected range.
- Examples of analytical equipment and apparatus that require calibration include:
  - Balances and weighing devices that measure mass or weight of samples and reagents.
  - Volumetric devices that measure volume or deliver liquid samples and reagents, such as pipettes, burettes, flasks, and cylinders.
  - Spectrophotometers and colorimeters that measure the absorbance or transmittance of light by samples and reagents.
  - pH meters and electrodes that measure the acidity or alkalinity of solutions.
  - Thermometers and thermocouples that measure the temperature of

samples and reagents.
  - Pressure gauges and manometers that measure the pressure of gases and liquids.
  - Conductivity meters and electrodes that measure the electrical conductivity of solutions.
  - Chromatographs and detectors that separate and quantify the components of mixtures.

2. Determination of Hardness of water sample by EDTA method.

- Hardness of water is the measure of the concentration of calcium and magnesium ions in water.
- Hard water causes problems such as scaling, soap scum formation, corrosion and reduced efficiency of boilers and heaters.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that forms stable complexes with metal ions such as Ca2+ and Mg2+.
- EDTA method is a titrimetric method that involves titrating a water sample with a standard solution of EDTA using a suitable indicator.
- The indicator used is Eriochrome Black T (EBT), which is a metallochromic indicator that changes color depending on the metal ion present.
- EBT forms a wine-red complex with Ca2+ and Mg2+ in water, but when EDTA is added, it displaces EBT and forms a more stable colorless complex with the metal ions.
- The end point of the titration is marked by the change of color of the solution from wine-red to blue, indicating that all the metal ions have been complexed by EDTA.
- The hardness of water can be calculated from the volume of EDTA used and its concentration.

Procedure:

- Pipette out 50 mL of the water sample into a conical flask and add 2-3 drops of EBT indicator.
- Titrate the sample with standard EDTA solution from a burette until the color changes from wine-red to blue.
- Record the volume of EDTA used as V1.
- Repeat the titration with another 50 mL of the water sample and record the volume of EDTA used as V2.
- Take the average of V1 and V2 as V.
- Calculate the hardness of water using the formula:

Hardness of water (mg/L as CaCO3) = (V x M x 1000) / 50

where V is the average volume of EDTA used (mL), M is the molarity of EDTA solution (mol/L), and 50 is the volume of water sample taken (mL).

Determination of Alkalinity of Water Sample

- Alkalinity is the measure of the ability of water to neutralize acids.
- Alkalinity is mainly caused by the presence of bicarbonates, carbonates, and hydroxides in water.
- Alkalinity is important for aquatic life, corrosion control, and water treatment processes.
- Alkalinity can be determined by titrating a water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a suitable indicator, such as phenolphthalein or methyl orange.
- The endpoint of the titration is marked by a color change of the indicator from pink to colorless (phenolphthalein) or from yellow to orange (methyl orange).
- The amount of acid used to reach the endpoint is proportional to the alkalinity of the water sample.
- The alkalinity can be expressed in different units, such as milligrams per liter (mg/L) of calcium carbonate (CaCO3), milliequivalents per liter (meq/L), or degrees of hardness (dH).
- The alkalinity can be classified into different types, such as total alkalinity, phenolphthalein alkalinity, methyl orange alkalinity, carbonate alkalinity, bicarbonate alkalinity, and hydroxide alkalinity, depending on the pH range and the predominant species in the water sample.

Determination of pH by titrimetric method

- Titrimetric method is a technique of quantitative analysis that involves measuring the volume of a solution of known concentration (titrant) that is required to react completely with a solution of unknown concentration (analyte) .
- pH is a measure of the acidity or alkalinity of a solution, defined as the negative logarithm of the hydrogen ion concentration: pH = -log[H3O+] .
- Determination of pH by titrimetric method involves using a pH meter to monitor the change in pH of the analyte solution as the titrant is added .
- A pH meter consists of a glass electrode that responds to the hydrogen ion activity in the solution, and a reference electrode that provides a constant potential .
- The potential difference between the two electrodes is proportional to the pH of the solution, according to the Nernst equation .
- A plot of the pH of the solution versus the volume of the titrant added is called a titration curve .
- The titration curve can be used to determine the endpoint or equivalence point of the titration, where the analyte and the titrant are stoichiometri-

cally equivalent .

- The shape and features of the titration curve depend on the type and strength of the acid and base involved in the titration .
- For example, a strong acid-strong base titration curve has a steep rise near the equivalence point, where a small change in volume causes a large change in pH .
- A weak acid-strong base titration curve has a buffer region before the equivalence point, where the pH changes gradually, and a sharp rise at the equivalence point, where the pH jumps to a basic value .
- A weak base-strong acid titration curve has a similar shape, but inverted, with a buffer region after the equivalence point and a sharp drop at the equivalence point, where the pH jumps to an acidic value .
- A weak acid-weak base titration curve has a flat shape, with a small change in pH at the equivalence point, where the pH is close to 7 .
- The titration curve can also be used to calculate the pKa or pKb of the weak acid or base, by finding the pH at the half-equivalence point, where the concentration of the acid or base is equal to the concentration of its conjugate base or acid .
- The pH at the half-equivalence point is equal to the pKa or pKb of the weak acid or base, according to the Henderson-Hasselbalch equation .
- Determination of pH by titrimetric method is a useful technique for characterizing and quantifying acids and bases in solution, as well as for determining their dissociation constants .

5. Determination of surface tension of given liquid

- Surface tension is a property of a liquid surface that makes it act as if it were a stretched elastic membrane. It is caused by the attraction of the molecules in the surface layer by the bulk of the liquid, which tends to minimize the surface area .

- Surface tension can be expressed as the amount of force exerted in the surface perpendicular to a line of unit length. The SI unit of surface tension is N/m (newton per meter), and the cgs unit is dyn/cm (dyne per centimeter).

- Surface tension can be measured by various methods, such as the capillary rise method, the drop weight method, the ring method, and the bubble pressure method. Each method has its own advantages and limitations, depending on the accuracy, precision, and convenience required.

- In this experiment, we will use the ring method, also known as the Du Noüy method, to determine the surface tension of a given liquid. The ring method involves suspending a metal ring from a balance and dipping it into the liquid. The balance measures the force required to detach the ring from the liquid surface, which is proportional to the surface tension of the liquid.

- The apparatus required for this experiment are: a ring of known diameter and mass, a balance, a beaker, a thermometer, a stirrer, and the liquid to be tested.
- The procedure for this experiment is as follows:
  1. Fill the beaker with the liquid to be tested and place it on the balance. Note the initial reading of the balance.
  2. Attach the ring to the hook of the balance and lower it into the liquid until it is completely submerged. Stir the liquid gently to remove any air bubbles from the ring.
  3. Slowly raise the ring until it just touches the liquid surface. Note the reading of the balance. This is the weight of the ring in air, W_a.
  4. Continue to raise the ring until it breaks away from the liquid surface. Note the reading of the balance. This is the weight of the ring in liquid, W_l.
  5. Repeat steps 3 and 4 for at least three times and take the average values of W_a and W_l.
  6. Measure the temperature of the liquid and record it.
  7. Calculate the surface tension of the liquid using the formula:

     gamma = (W_a - W_l) * g / (4 * pi * r)

     where gamma is the surface tension, W_a and W_l are the average weights of the ring in air and liquid, g is the acceleration due to gravity, and r is the radius of the ring.
  8. Report the result with the appropriate units and significant figures.

Determination of Viscosity of a given liquid by viscometer

- Viscosity is a measure of the resistance of a fluid to flow or deform under an applied force.
- A viscometer is a device that measures the viscosity of a fluid by observing its flow through a narrow tube or a capillary.
- The flow rate of a fluid through a capillary depends on the pressure difference across the capillary, the length and radius of the capillary, and the viscosity of the fluid.
- The Hagen-Poiseuille equation relates these variables as follows:

$$Q = \frac{\pi r^4 \Delta P}{8 \eta L}$$

where Q is the volumetric flow rate, r is the radius of the capillary, ΔP is the pressure difference, η is the viscosity, and L is the length of the capillary.

- To determine the viscosity of a given liquid by viscometer, the following steps are followed:
  - Fill the viscometer with the liquid to be tested and immerse it in a water bath at a constant temperature.
  - Allow the liquid to flow through the capillary under the influence of gravity and measure the time taken for a fixed volume of liquid to flow out of the capillary.
  - Repeat the experiment for different volumes of liquid and calculate the average flow rate.
  - Measure the radius and length of the capillary using a micrometer and a meter scale.
  - Measure the pressure difference across the capillary using a manometer or a pressure gauge.
  - Substitute the values of Q, r, L, and ΔP in the Hagen-Poiseuille equation and solve for η.
  - Report the viscosity of the liquid with appropriate units and significant figures.

Determination of the strength of Ferrous ammonium sulfate using external indicator

- Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron(II) sulfate and ammonium sulfate, with the formula $(NH4)2Fe(SO4)2 \cdot 6H2O$.
- It is often used as a standard solution for titrations of reducing agents, such as oxalic acid, potassium permanganate, or potassium dichromate.
- The strength of a ferrous ammonium sulfate solution can be determined by titrating it with a standard solution of potassium permanganate, using an external indicator such as starch or potassium ferricyanide.
- The reaction between ferrous ammonium sulfate and potassium permanganate is:

(NH4)2Fe(SO4)2 + KMnO4 + H2SO4 -> K2SO4 + (NH4)2SO4 + Fe2(SO4)3 + MnSO4 + H2O

- The reaction is a redox reaction, in which ferrous ions are oxidized to ferric ions, and permanganate ions are reduced to manganese(II) ions.
- The end point of the titration is marked by the disappearance of the pink color of permanganate, or by the appearance of a blue color of starch-

iodine complex or a brown color of ferric ferrocyanide, depending on the external indicator used.

- The strength of the ferrous ammonium sulfate solution can be calculated by using the following formula:

Strength (g/L) = (V1 x N1 x M x 1000) / V2

Where,

V1 = volume of potassium permanganate solution used (mL)

N1 = normality of potassium permanganate solution

M = molar mass of ferrous ammonium sulfate (g/mol)

V2 = volume of ferrous ammonium sulfate solution taken (mL)

- The procedure for the titration is as follows:

1. Prepare a standard solution of potassium permanganate by dissolving a known mass of KMnO4 in distilled water and standardizing it against a primary standard, such as sodium oxalate.
2. Pipette out a known volume of ferrous ammonium sulfate solution into a conical flask and add a few drops of dilute sulfuric acid to acidify the solution.
3. Add a few drops of starch or potassium ferricyanide solution as an external indicator to the flask.
4. Titrate the ferrous ammonium sulfate solution with the standard potassium permanganate solution from a burette, swirling the flask constantly.
5. Note the burette reading when the end point is reached, indicated by the disappearance of the pink color of permanganate, or by the appearance of a blue color of starch-iodine complex or a brown color of ferric ferrocyanide.
6. Repeat the titration until concordant results are obtained.
7. Calculate the strength of the ferrous ammonium sulfate solution using the formula given above.

8. Determination of the strength of Potassium dichromate using internal indicator

- Potassium dichromate is a primary standard substance that can be used to determine the strength of a reducing agent, such as ferrous ammonium sulphate, by titration.
- The reaction between potassium dichromate and ferrous ammonium sulphate is as follows:

$$K_2Cr_2O_7+6FeSO_4(NH_4)_2SO_4+7H_2SO_4 \rightarrow Cr_2(SO_4)_3+3K_2SO_4+6(NH_4)_2SO_4+6Fe_2(SO_4)_3+7H_2O$$

- The end point of the titration can be detected by using an internal indicator, such as diphenylamine sulphonic acid, which changes color from colorless to blue when the solution becomes slightly acidic due to the formation of sulphuric acid.
- The procedure for the determination of the strength of potassium dichromate using internal indicator is as follows:
  - Weigh accurately about 0.5 g of potassium dichromate and dissolve it in 100 mL of distilled water in a volumetric flask. This is the standard solution of potassium dichromate.
  - Pipette out 10 mL of the standard solution into a conical flask and add 10 mL of dilute sulphuric acid and a few drops of diphenylamine sulphonic acid indicator.
  - Titrate the solution with the given ferrous ammonium sulphate solution from a burette until the color changes from colorless to blue. Note the burette reading. This is the first titration.
  - Repeat the titration two more times and take the average of the three readings. This is the second and third titration.
  - Calculate the strength of the ferrous ammonium sulphate solution using the following formula:

$$\text{Strength of ferrous ammonium sulphate solution} = \frac{\text{Weight of potassium dichromate} \times \text{Normality of potassiu}}{\text{Average volume of ferrous ammonium sulphate soluti}}$$

- The normality of potassium dichromate is 0.1 N, since one mole of potassium dichromate reacts with six moles of ferrous ions.

Determination of Available Chlorine in Bleaching Powder

- Bleaching powder is a white powder that contains calcium hypochlorite [Ca(OCl)2] and calcium chloride (CaCl2) as the main components.
- Bleaching powder is used as a disinfectant, bleaching agent, and oxidizing agent.
- Available chlorine is the amount of chlorine that can be liberated from bleaching powder when treated with dilute acids .
- Available chlorine is expressed as a percentage of the mass of bleaching powder.
- The principle of the determination of available chlorine is based on the reaction of bleaching powder with excess iodide in acidic medium to produce iodine, which is then titrated with sodium thiosulfate (hypo) solution .
- The reactions involved are as follows:

    - $Ca(OCl)_2 + 2H^+ + 2I^- \rightarrow Ca^{2+} + 2Cl^- + I_2 + H_2O$
    - $I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

- The procedure for the determination of available chlorine is as follows:

    - Weigh about 0.4 g of bleaching powder and transfer it to a 100 mL volumetric flask. Add distilled water and shake well. Filter the solution and collect the filtrate in a clean bottle. This is the stock solution of bleaching powder.

    - Pipette out 25 mL of the stock solution into a 250 mL Erlenmeyer flask. Add 10 mL of 10% potassium iodide (KI) solution and 10 mL of 1 M acetic acid (CH_3COOH) solution. Swirl the flask to mix the contents and allow it to stand for 5 minutes.

    - Titrate the liberated iodine with 0.1 N sodium thiosulfate ($Na_2S_2O_3$) solution using starch as an indicator. The end point is the disappearance of the blue color of the starch-iodine complex.

    - Repeat the titration with a blank solution containing 25 mL of distilled water, 10 mL of 10% KI solution, and 10 mL of 1 M CH_3COOH solution.

    - Calculate the percentage of available chlorine in the bleaching powder sample using the formula:

        * % available chlorine = (B - S) x N x 35.45 x 100 / W
        * where B is the volume of $Na_2S_2O_3$ solution used for the blank titration, S is the volume of $Na_2S_2O_3$ solution used for the sample titration, N is the normality of $Na_2S_2O_3$ solution, W is the weight of bleaching powder taken, and 35.45 is the equivalent weight of chlorine.

Determination of chloride content in water sample

- Chloride is one of the major anions found in water and wastewater. It originates from natural sources such as seawater, rocks and minerals, as well as from human activities such as industrial processes, agricultural runoff, road salt and domestic sewage.

- Chloride can affect the taste, corrosion, and biological activity of water. Therefore, it is important to monitor and control the chloride content in water for various purposes such as drinking, irrigation, and industrial use.

- One of the common methods for determining the chloride content in water is the Mohr method, which is based on the precipitation and titration of chloride with silver nitrate using potassium chromate as an indicator.

- The principle of the Mohr method is that when water containing chloride ions is titrated with silver nitrate solution, chloride ions are precipitated as white silver chloride. Potassium chromate is used as an indicator, which supplies chromate ions that react with excess silver ions to form a reddish-brown precipitate of silver chromate. The end point of the titration is marked by the appearance of this precipitate.
- The Mohr method can be used to determine the chloride content in water samples from various sources such as seawater, stream water, river water and estuary water. The pH of the sample solutions should be between 6.5 and 10 to avoid the interference of other anions such as carbonate, bicarbonate, phosphate, and sulfide.
- The procedure for the Mohr method is as follows:
  - Take 50 mL of sample (V mL) and dilute to 100 mL with distilled water in a volumetric flask.
  - If the sample is highly colored, add 3 mL of aluminum hydroxide and shake well, allow to settle, filter, wash and collect the filtrate.
  - Bring the sample pH to 7-8 by adding acid or alkali if necessary.
  - Add 1 mL of indicator (potassium chromate) to the sample solution in a conical flask.
  - Titrate the solution against standard silver nitrate solution (N/50 or 0.02 M) from a burette until a reddish-brown precipitate is obtained.
  - Record the volume of silver nitrate solution used (V1 mL).
  - Repeat the titration with a blank solution (distilled water) and record the volume of silver nitrate solution used (V2 mL).
  - Calculate the chloride content in the sample as follows:
    - Chloride content (mg/L) = (V1 - V2) x N x 35.45 x 1000 / V
    - Where N is the normality of silver nitrate solution, 35.45 is the equivalent weight of chloride, and V is the volume of sample taken in mL.

Preparation of Phenol Formaldehyde (PF) Resin

- Phenol formaldehyde resin is a type of thermosetting polymer that is obtained by the condensation reaction of phenol and formaldehyde.
- Phenol and formaldehyde are mixed in a reaction vessel in a molar ratio of 1:1 or higher, depending on the desired properties of the resin.
- The reaction can be catalyzed by either an acid or a base. Acid catalysts include hydrochloric acid and sulfuric acid, while base catalysts include ammonia and sodium hydroxide.

- The reaction proceeds through the formation of methylene bridges (-CH2-) between the phenol molecules, resulting in linear or branched chains of phenolic units.
- The reaction can be controlled by adjusting the temperature, pH, catalyst concentration, and reaction time. The higher the temperature and pH, the faster the reaction and the higher the degree of polymerization.
- The reaction can be stopped by adding water or an alcohol to the mixture, which terminates the chain growth and precipitates the resin.
- The resin can be filtered, washed, and dried to obtain the final product, which can be further processed into various forms, such as powders, granules, flakes, or preforms.
- The resin can be cured by heating it with a hardener, such as hexamethylenetetramine, under pressure. The curing process involves further cross-linking of the resin chains, forming a rigid and insoluble network.
- The cured resin has high mechanical strength, thermal stability, chemical resistance, and electrical insulation, making it suitable for various applications, such as adhesives, coatings, laminates, molding compounds, and wood products .

Preparation of Urea Formaldehyde (UF) Resin

- Urea formaldehyde (UF) resin is a thermosetting polymer that is widely used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- UF resin is synthesized by the condensation reaction of urea and formaldehyde in the presence of a base or an acid catalyst .
- The reaction can be carried out in one or more steps, depending on the desired properties and applications of the resin .
- The general steps for the preparation of UF resin are as follows:
  - Step 1: Preparation of methylolureas
    - Urea and formaldehyde are mixed in an aqueous solution in a molar ratio of 2-3:1 at pH 6-11.
    - The mixture is heated to at least 80°C to form methylolureas, which are intermediates with one or more hydroxymethyl groups attached to the nitrogen atoms of urea .
    - The reaction is exothermic and can be controlled by adjusting the pH, temperature, and concentration of the reactants .
    - The degree of methylation determines the reactivity and solubility of the resin .
    - The most common methylolureas are monomethylolurea (MMU), dimethylolurea (DMU), and trimethylolurea (TMU) .
    - MMU has a melting point of 110°C and is soluble in water.

        * DMU and TMU are solid at room temperature and are less soluble in water.
        * The reaction can be stopped by cooling the mixture or adding an acid to lower the pH .
        * The methylolureas can be isolated by filtration, washing, and drying.
    - Step 2: Preparation of UF resin
        * The methylolureas are further reacted with formaldehyde and urea in the presence of an acid catalyst, such as sulfuric acid, hydrochloric acid, or oxalic acid .
        * The pH is adjusted to 0.5-3.5 and the temperature is maintained at 60-100°C .
        * The reaction leads to the formation of methylene bridges (-CH2-) between the nitrogen atoms of different methylolureas, resulting in a linear or branched polymer chain .
        * The degree of polymerization and cross-linking depends on the pH, temperature, time, and molar ratio of the reactants .
        * The resin can be modified by adding other additives, such as ammonia, melamine, or phenol, to improve its properties .
        * The resin can be neutralized by adding a base, such as sodium hydroxide, to increase its stability and shelf life .
        * The resin can be concentrated by removing water under vacuum or by adding organic solvents, such as ethanol or acetone .
        * The resin can be stored in liquid or solid form until it is cured by heat and pressure .

Preparation of Adipic acid / Paracetamol

- Adipic acid is the most important dicarboxylic acid, mainly used as a precursor for the production of nylon.
- Paracetamol is an effective antipyretic and analgesic, widely used for the relief of fever, headaches, and other minor aches and pains.

Preparation of Adipic acid

- Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil.
- The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .
- The overall reaction is shown below:

Adipic acid synthesis

- The main steps are:
    - Formation of nitrosonium ion (NO+) from nitric acid and sulfuric acid.

  - Nitrosation of cyclohexanone and cyclohexanol to form nitrosocyclohexane and nitrosocyclohexanol, respectively.
  - Rearrangement of nitrosocyclohexane and nitrosocyclohexanol to form 6-nitroso-1-cyclohexanol and 6-nitroso-2-cyclohexanol, respectively.
  - Oxidation of 6-nitroso-1-cyclohexanol and 6-nitroso-2-cyclohexanol to form 6-nitrocyclohexanone and 6-nitrocyclohexanol, respectively.
  - Hydrolysis of 6-nitrocyclohexanone and 6-nitrocyclohexanol to form adipic acid and nitrous acid, respectively.
  - Recycle of nitrous acid to nitric acid by oxidation with air.

Preparation of Paracetamol

- Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulfuric acid as catalyst .
- The reaction is shown below:

Paracetamol synthesis

- The main steps are:
  - Formation of acylium ion (CH3CO+) from acetic anhydride and sulfuric acid.
  - Electrophilic substitution of p-aminophenol by acylium ion to form N-acetyl-p-aminophenol (paracetamol) and acetic acid.
  - Purification of paracetamol by recrystallization from water.

Determination of Cell Conductance of a Solution

- Cell conductance of a solution is the reciprocal of its resistance, i.e. G = 1/R, where G is the conductance and R is the resistance.
- Cell conductance is measured in Siemens (S) or ohms^{-1}.
- Cell conductance depends on the concentration, charge, and mobility of the ions in the solution, as well as the geometry of the cell used to measure it.
- A cell used to measure the conductance of a solution is called a conductance cell. It consists of two electrodes immersed in the solution, connected to a source of alternating current and a device to measure the resistance, such as a Wheatstone bridge.
- The electrodes of a conductance cell are usually made of platinum, which is inert and has a high conductivity. They are also coated with platinum black, which is a finely divided form of platinum that increases the surface area and minimizes polarization effects.
- The distance between the electrodes (l) and the cross-sectional area of the electrodes (a) determine the cell constant (C), which is a characteristic of the cell and has units of m^{-1}. C = l/a

- The cell constant can be determined by measuring the resistance of a cell filled with a solution of known conductivity, such as 0.0200 M KCl, and using the formula: C = kR, where k is the conductivity of the solution and R is the resistance of the cell.
- Once the cell constant is known, the conductivity of any solution can be calculated from the measured resistance by using the formula: k = C/R
- The conductivity of a solution is proportional to the concentration of the ions in the solution, but not linearly. The relation between conductivity and concentration is given by the Kohlrausch law: k = A√C, where A is a constant that depends on the nature of the electrolyte and C is the concentration of the solution.
- The conductivity of a solution also depends on the temperature, as the mobility of the ions increases with temperature. The relation between conductivity and temperature is given by the equation: k = k0(1 + T), where k0 is the conductivity at 0°C, is the temperature coefficient of conductivity, and T is the temperature in °C.

Determination of Rate constant of hydrolysis of esters

- Esters are organic compounds that have the general formula RCOOR', where R and R' are alkyl or aryl groups.
- Esters can be hydrolyzed by water in the presence of an acid or a base catalyst to form carboxylic acids and alcohols.
- The hydrolysis of esters is a first-order reaction, which means that the rate of the reaction depends only on the concentration of the ester.
- The rate constant of the hydrolysis of esters can be determined by measuring the concentration of the ester or the products at different time intervals and applying the integrated rate law for first-order reactions.
- The integrated rate law for first-order reactions is:

  ln[A] = -kt + ln[A]0

  where [A] is the concentration of the ester at time t, [A]0 is the initial concentration of the ester, k is the rate constant, and t is the time.
- To determine the rate constant of the hydrolysis of esters, the following steps can be followed:

  1. Prepare a solution of the ester and the catalyst in a suitable solvent, such as water or ethanol, and measure its initial concentration using a titration method or a spectrophotometer.
  2. Transfer a known volume of the solution to a flask and start a stopwatch.

3. At regular time intervals, take a small aliquot of the solution and quench the reaction by adding a large excess of a strong acid or a base, depending on the catalyst used.
4. Measure the concentration of the ester or the products in the quenched solution using the same method as in step 1.
5. Repeat steps 3 and 4 until the concentration of the ester or the products reaches a constant value, indicating that the reaction is complete.
6. Plot the natural logarithm of the concentration of the ester or the products versus time and obtain the slope of the line, which is equal to -k.
7. Calculate the rate constant of the hydrolysis of esters using the slope and the equation:

   k = -slope

Element detection and identification of functional groups in organic compounds

- Organic compounds are those that contain carbon atoms, usually bonded to hydrogen, oxygen, nitrogen, sulfur, or halogens.
- The structure and properties of organic compounds are largely determined by the functional groups, which are specific groups of atoms or bonds that have characteristic chemical behavior.
- Functional groups can be classified into different categories, such as alcohols, phenols, aldehydes, ketones, carboxylic acids, amines, ethers, esters, amides, halides, alkenes, alkynes, and aromatics.
- To detect and identify the elements and functional groups present in an organic compound, various qualitative and quantitative methods can be used, such as:
  - Physical tests: These involve observing the physical properties of the compound, such as color, odor, state, melting point, boiling point, solubility, etc.
  - Chemical tests: These involve reacting the compound with specific reagents and observing the changes, such as color change, precipitate formation, gas evolution, etc. Some common chemical tests are:
    * Lassaigne's test: This is a general test for the detection of nitrogen, sulfur, and halogens in an organic compound. It involves fusing the compound with sodium metal and extracting the sodium fusion extract, which is then tested for the presence of cyanide, sulfide, and halide ions.
    * Ignition test: This is a general test for the detection of carbon and hydrogen in an organic compound. It involves burning a

small amount of the compound in a flame and observing the nature of the flame and the residue. A yellow flame indicates the presence of carbon, while a blue flame indicates the presence of hydrogen. A black residue indicates the presence of carbon, while a white residue indicates the presence of metal oxides or halides.

* Alcoholic -OH group test: This is a specific test for the detection of alcohols in an organic compound. It involves reacting the compound with sodium metal, Lucas reagent, or ceric ammonium nitrate and observing the changes. A brisk effervescence of hydrogen gas indicates the presence of an alcohol. A turbidity or cloudiness in the solution indicates the presence of a primary, secondary, or tertiary alcohol, depending on the time taken for the reaction. A red coloration of the solution indicates the presence of a phenol.
* Carbonyl group test: This is a specific test for the detection of aldehydes and ketones in an organic compound. It involves reacting the compound with 2,4-dinitrophenylhydrazine, Tollens' reagent, Fehling's solution, or Benedict's solution and observing the changes. A yellow or orange precipitate of 2,4-dinitrophenylhydrazone indicates the presence of a carbonyl group. A silver mirror on the inner wall of the test tube indicates the presence of an aldehyde. A brick-red precipitate of cuprous oxide indicates the presence of a reducing sugar or an aldehyde.
* Carboxyl group test: This is a specific test for the detection of carboxylic acids in an organic compound. It involves reacting the compound with sodium bicarbonate, sodium carbonate, or litmus paper and observing the changes. A brisk effervescence of carbon dioxide gas indicates the presence of a carboxyl group. A change in the color of the litmus paper from blue to red indicates the presence of an acid.
* Amino group test: This is a specific test for the detection of amines in an organic compound. It involves reacting the compound with nitrous acid, Hinsberg's reagent, or ninhydrin and observing the changes. A gas evolution with a characteristic odor indicates the presence of a primary, secondary, or tertiary amine, depending on the nature of the gas. A formation of a salt or a quaternary ammonium salt indicates the presence of a primary or a secondary amine, respectively. A blue or purple coloration of the solution indicates the presence of a primary amine or an amino acid.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- This note implies that the instructor has the authority and flexibility to select the experiments that are most suitable for the course objectives, the available resources, and the students' interests and abilities.
- The note also implies that the instructor can modify or replace up to two of the experiments from the list provided, if they find them too difficult, too easy, too boring, or too irrelevant for the course.
- The note does not specify which experiments are mandatory or optional, nor does it provide any criteria or guidelines for choosing or changing the experiments. Therefore, the instructor has to use their own judgment and discretion to make the best decisions for the course.
- The note also does not indicate how the experiments will be evaluated or graded, nor how much weight they will have in the final assessment. Therefore, the instructor has to communicate these details clearly to the students before they start the experiments.
- The note may have some advantages and disadvantages for the instructor and the students. Some possible advantages are:
  - The instructor can tailor the experiments to the specific needs and goals of the course, and avoid wasting time and resources on experiments that are not relevant or useful.
  - The instructor can also adapt the experiments to the level and pace of the students, and provide them with appropriate challenges and feedback.
  - The students can have some variety and diversity in the experiments, and learn from different methods and perspectives.
  - The students can also have some input and choice in the experiments, and express their preferences and interests to the instructor.
- Some possible disadvantages are:
  - The instructor may have to spend more time and effort to select and prepare the experiments, and ensure that they are aligned with the course outcomes and standards.
  - The instructor may also have to deal with some uncertainty and inconsistency in the experiments, and adjust their plans and expectations accordingly.
  - The students may have some confusion and ambiguity about the experiments, and not know what to expect or how to prepare for them.
  - The students may also have some difficulty and frustration with the experiments, and not get enough support or guidance from the instructor.

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, content, activities, and assessments.
- Course outcomes are measurable, observable, and achievable within the scope and duration of the course.
- Course outcomes are written in clear and concise language, using action verbs that indicate the level of cognitive skills required.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course.
- Course outcomes are used to guide the course design, delivery, and evaluation.
- Course outcomes are reviewed and revised periodically to ensure their relevance and effectiveness.

Upon completion of the course the student should be able to:

- Identify and explain the main concepts, theories and methods of the discipline.
- Apply the relevant knowledge and skills to analyze and solve problems in the domain of the course.
- Communicate effectively and professionally using appropriate terminology and formats.
- Demonstrate critical thinking and creativity in approaching complex and novel situations.
- Collaborate with others and work independently as required by the course objectives and expectations.
- Reflect on their own learning process and outcomes and seek feedback for improvement.
- Adhere to the ethical and academic standards of the discipline and the institution.

Course Outcomes Bloom's

- Course outcomes are statements that describe what students should be able to do or demonstrate after completing a course.
- Bloom's taxonomy is a framework for classifying educational objectives into six levels of cognitive complexity: remember, understand, apply, analyze, evaluate, and create.
- Course outcomes should be aligned with Bloom's taxonomy to ensure that students achieve the desired level of learning and mastery of the course

content.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART) to facilitate assessment and feedback.
- Course outcomes should be written using action verbs that correspond to the appropriate level of Bloom's taxonomy, such as define, explain, compare, contrast, solve, design, etc.
- Course outcomes should be communicated to students at the beginning of the course and throughout the course to guide their learning and expectations.
- Course outcomes should be assessed using various methods and tools, such as quizzes, assignments, projects, presentations, portfolios, etc., that match the level of Bloom's taxonomy and the course content.
- Course outcomes should be reviewed and revised periodically based on the feedback from students, instructors, and other stakeholders to ensure their relevance and effectiveness.

Level

- A level is a tool or device that is used to measure or indicate whether a surface is horizontal (level) or vertical (plumb).
- A level consists of a horizontal tube or vial that contains a liquid (usually alcohol or water) and an air bubble. The position of the bubble indicates the alignment of the surface relative to the earth's gravity.
- A level can be used for various purposes, such as construction, carpentry, engineering, surveying, landscaping, photography, etc.
- There are different types of levels, such as spirit levels, laser levels, digital levels, water levels, etc. Each type has its own advantages and disadvantages depending on the application and accuracy required.
- A level can be calibrated by placing it on a flat surface and adjusting it until the bubble is centered in the vial. Then, the level is rotated 180 degrees and placed on the same surface. If the bubble is still centered, the level is accurate. If not, the level needs to be adjusted or replaced.

CO-1 Get an understanding of the use of different analytical instruments. K3

- Analytical instruments are devices that are used to measure, analyze, or identify the chemical, physical, or biological properties of a sample or a substance .
- Analytical instruments can be classified into different categories based on the principle or technique of analysis, such as spectroscopy, mass spectrometry, electrochemistry, thermal analysis, separation analysis, microscopy, and hybrid techniques .
- Analytical instruments have various applications in different fields, such as

pharmaceutical, food-processing, chemical, clinical, environmental, forensic, and life science research .

- Analytical instruments can help in establishing the composition, structure, purity, quality, and functionality of a sample or a substance .
- Analytical instruments can also help in detecting, identifying, and quantifying contaminants, impurities, or inclusions in a sample or a substance .
- Analytical instruments can be used in the laboratory or on the field, depending on the type, size, and portability of the instrument.
- Analytical instruments can be used for routine analysis, quality control, process monitoring, research and development, or troubleshooting .

CO2 Molecular Properties and Surface Tension

- Carbon dioxide (CO2) is a linear molecule with a bond length of 116.3 pm and a bond angle of 180 degrees.
- CO2 is a non-polar molecule with a dipole moment of zero and a symmetrical distribution of electrons.
- CO2 has a molar mass of 44.0095 g/mol and a critical point at 304.13 K and 7.38 MPa.
- CO2 is a gas at standard temperature and pressure, but it can be liquefied or solidified under certain conditions.
- The density and specific weight of CO2 depend on the temperature and pressure, and they can be calculated using online tools or tables.
- Surface tension is the energy or work required to increase the surface area of a liquid due to intermolecular forces.
- The surface tension of CO2 is influenced by the presence of other substances, such as water or monoethanolamine (MEA), which can act as solvents or surfactants .
- The surface tension of CO2 decreases with increasing temperature and CO2 loading (the amount of CO2 dissolved in the liquid) .
- The surface tension of CO2 can be measured using various methods, such as the pendant drop method, the Wilhelmy plate method, or the maximum bubble pressure method .
- The surface tension of CO2 is important for various applications, such as carbon capture and storage, enhanced oil recovery, or refrigeration .

Viscosity

- Viscosity is the property of a fluid that measures its resistance to flow.
- Viscosity depends on the internal friction between the molecules of the fluid, which is affected by temperature, pressure, and the nature of the fluid.

- The unit of viscosity is the pascal-second (Pa · s) in the SI system, or the poise (P) in the CGS system. One poise is equal to 0.1 Pa · s.
- The formula for viscosity is:

  ```
   = F / (A * dv/dy)
  ```

  where is the viscosity, F is the force applied, A is the area of the surface, dv/dy is the velocity gradient perpendicular to the direction of flow.
- Viscosity can be classified into two types: dynamic viscosity and kinematic viscosity.
- Dynamic viscosity is the ratio of the shear stress to the shear rate of a fluid, and it measures the force required to move a layer of fluid over another layer.
- Kinematic viscosity is the ratio of the dynamic viscosity to the density of a fluid, and it measures the resistance of a fluid to flow under gravity.
- Examples of fluids with high viscosity are honey, molasses, and glycerin. Examples of fluids with low viscosity are water, air, and gasoline.

K3 Surface

- A K3 surface is a type of algebraic surface that has some special properties .
- A K3 surface is smooth, proper, geometrically connected, and has trivial canonical bundle .
- A K3 surface has Kodaira dimension zero, which means it has no holomorphic forms of positive degree .
- A K3 surface is simply connected, which means it has no non-trivial loops .
- A K3 surface has 22 singular cohomology groups, which are related to its geometry and topology .
- A K3 surface can be defined over any field, but the most studied case is when the field is the complex numbers .
- A K3 surface can be constructed in various ways, such as by taking a quartic equation in projective space, or by taking a double cover of the projective plane branched along a sextic curve .
- A K3 surface has many applications and connections to other areas of mathematics and physics, such as smooth 4-manifolds, mirror symmetry, string theory, and moduli spaces .

CO-3 Measure the hardness and alkalinity of the water. K3

- Hardness and alkalinity are two parameters that indicate the quality of water.
- Hardness measures the concentration of divalent salts, mainly calcium and magnesium, that can form deposits or scale in pipes, boilers, and appliances.
- Alkalinity measures the capacity of water to neutralize acids, or the amount of bases present in water, mainly bicarbonate and carbonate.
- Hardness and alkalinity are related through common ions formed in aquatic systems. The counter-ions associated with the bicarbonate and carbonate fraction of alkalinity are the principal cations responsible for hardness (usually Ca ++ and Mg ++).
- Hardness and alkalinity are influenced by the geology of the area where the water is located and the dissolution of carbon dioxide from the atmosphere. The ions responsible for hardness and alkalinity originate from the dissolution of geological minerals into rain and groundwater.
- Hardness and alkalinity can be measured by titration methods using standard solutions of acids and bases and indicators that change color at certain pH levels.
- Hardness can be classified as temporary or permanent. Temporary hardness is caused by the presence of bicarbonate ions that can be removed by boiling the water. Permanent hardness is caused by the presence of other ions that cannot be removed by boiling the water.
- Alkalinity can be classified as total, carbonate, or bicarbonate alkalinity. Total alkalinity is the amount of acid required to lower the pH of water to 4.5. Carbonate alkalinity is the amount of acid required to lower the pH of water to 8.3. Bicarbonate alkalinity is the amount of acid required to lower the pH of water to 4.5 minus the amount of acid required to lower the pH of water to 8.3.
- Hardness and alkalinity are expressed in units of milligrams per liter (mg/L) as calcium carbonate (CaCO3). The conversion factor between mg/L of CaCO3 and mg/L of other ions is based on their equivalent weights.
- Hardness and alkalinity can affect the taste, odor, color, corrosion, and biological activity of water. Hard water can cause scaling, reduce the effectiveness of detergents, and increase the energy consumption of heating appliances. Alkaline water can cause corrosion, increase the solubility of metals, and reduce the disinfection efficiency of chlorine.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic compound with a hydroxyl group (-OH) attached to a benzene ring. It is also known as carbolic acid or phenic acid. Phenol has antiseptic and disinfectant properties and is used in the manufacture of plastics, dyes, explosives, and drugs.

There are several methods to prepare phenol from different starting materials. Some of the common methods are:

- **From haloarenes:** Haloarenes are aromatic compounds with one or more halogen atoms (-X) attached to a benzene ring. Examples of haloarenes are chlorobenzene, bromobenzene, and iodobenzene. When haloarenes are fused with sodium hydroxide (NaOH) at high temperature and pressure, they undergo nucleophilic aromatic substitution and form sodium phenoxide (NaOC6H5), which can be converted to phenol by acidification with dilute acid or carbon dioxide. The general reaction is:

  C6H5X + NaOH $\rightarrow$ NaOC6H5 + NaX

  NaOC6H5 + H+ $\rightarrow$ C6H5OH + Na+

- **From benzene sulphonic acid:** Benzene sulphonic acid is an aromatic compound with a sulphonic acid group (-SO3H) attached to a benzene ring. It can be prepared by the sulphonation of benzene with concentrated sulphuric acid (H2SO4) in the presence of a catalyst. When benzene sulphonic acid or its sodium salt is fused with sodium hydroxide at high temperature, it undergoes desulphonation and forms sodium phenoxide, which can be converted to phenol by acidification. The general reaction is:

  C6H6 + H2SO4 $\rightarrow$ C6H5SO3H + H2O

  C6H5SO3H + NaOH $\rightarrow$ NaOC6H5 + NaHSO4

  NaOC6H5 + H+ $\rightarrow$ C6H5OH + Na+

- **From cumene:** Cumene is an aromatic compound with an isopropyl group (-CH(CH3)2) attached to a benzene ring. It can be prepared by the alkylation of benzene with propene in the presence of a catalyst. When cumene is oxidized by air, it forms cumene hydroperoxide, which undergoes acid-catalyzed cleavage to form phenol and acetone. The general reaction is:

  C6H6 + CH2=CHCH3 $\rightarrow$ C6H5CH(CH3)2

  C6H5CH(CH3)2 + O2 $\rightarrow$ C6H5C(CH3)2OOH

  C6H5C(CH3)2OOH + H+ $\rightarrow$ C6H5OH + CH3COCH3

- **From diazonium salts:** Diazonium salts are organic compounds with a diazonium group ($-N_2^+$) attached to an aromatic ring. They can be prepared by the reaction of aromatic amines with nitrous acid (HNO_2) in cold acidic solution. When diazonium salts are hydrolyzed with water, they form phenol and nitrogen gas. The general reaction is:

  $C_6H_5NH_2 + HNO_2 \rightarrow C_6H_5N_2^+ + H_2O$

  $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2$

These are some of the fundamental concepts of the preparation of phenol. There are other methods as well, such as from coal tar, from aniline, and from phenylboronic acid. Phenol can also be synthesized by various reactions, such as electrophilic aromatic substitution, nucleophilic aromatic substitution, and oxidation. Phenol is an important intermediate in organic chemistry and has many applications in various fields.

Formaldehyde

- Formaldehyde is a naturally occurring organic compound with the formula CH_2O and structure H−CHO.
- Formaldehyde is a colorless, highly toxic, and flammable gas at room temperature.
- Formaldehyde is an intermediate in the oxidation (or combustion) of methane, as well as of other carbon compounds, e.g. in forest fires, automobile exhaust, and tobacco smoke.
- Formaldehyde is also produced in the atmosphere by the action of sunlight and oxygen on atmospheric methane and other hydrocarbons, and it becomes part of smog.
- Formaldehyde is used widely by industry to manufacture building materials and numerous household products, such as fertilizer, paper, plywood, resins, antiseptics, medicines, and cosmetics .
- Formaldehyde is also used as a food preservative and in embalming fluids.
- Formaldehyde can cause irritation of the skin, eyes, nose, and throat, as well as respiratory and neurological effects .
- Formaldehyde is classified as a human carcinogen by the International Agency for Research on Cancer and the US National Toxicology Program, based on evidence from studies of workers exposed to high levels of formaldehyde and from animal experiments .
- Formaldehyde exposure can be reduced by using products that contain low or no formaldehyde, ventilating indoor spaces, avoiding smoking, and following safety precautions when handling formaldehyde-containing products .

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa holders are eligible to apply for a work permit and travel authorization while in the United States.
- K3 visa is valid for two years and can be extended if the immigrant visa petition or adjustment of status application is still pending.
- K3 visa is not a direct path to permanent residence or citizenship. K3 visa holders must still complete the process of obtaining an immigrant visa or adjusting their status to become lawful permanent residents.
- K3 visa is intended to reduce the separation time between the U.S. citizen and their foreign spouse, but it is not always faster or easier than the regular immigrant visa process.

CO-5 Estimate the rate constant of reaction K3

- The rate constant of a reaction is a proportionality factor that relates the reaction rate to the concentrations of the reactants.
- The rate constant depends on the temperature, the presence of catalysts, and the nature of the reaction mechanism.
- The rate constant can have different units depending on the order of the reaction, which is the sum of the exponents of the concentration terms in the rate law.
- For a termolecular reaction, such as A + B + C → products, the order of the reaction is 3, and the rate law is r = k3[A][B][C], where k3 is the termolecular rate constant.
- The units of k3 for a termolecular reaction are L2/mol2 · s, which can be derived by dividing the units of the rate (mol/L · s) by the units of the concentrations (mol/L)3.
- To estimate the value of k3 for a given reaction, one can either use experimental data or theoretical calculations.
- Experimental data can be obtained by measuring the initial rates of the reaction at different concentrations of the reactants and then using a method of linearization or nonlinear regression to find the best fit for k3.
- Theoretical calculations can be based on the collision theory or the transition state theory, which both assume that the reaction occurs when the reactants collide with a certain orientation and energy.
- The collision theory estimates k3 by multiplying the collision frequency of the three reactants by the fraction of collisions that have enough energy to overcome the activation energy and the fraction of collisions that have the correct orientation.
- The transition state theory estimates k3 by dividing the equilibrium constant for the formation of the activated complex (the intermediate state

between the reactants and the products) by the product of the Planck constant and the partition function of the activated complex.
- Both the collision theory and the transition state theory require the knowledge of the activation energy, the molecular geometry, and the vibrational frequencies of the reactants and the activated complex.
- The value of k3 can vary significantly depending on the reaction conditions and the reaction mechanism. For example, the termolecular rate constant for the reaction H + O2 + M $\rightarrow$ HO2 + M, where M is a third body that absorbs the excess energy, is about 2.1 $\times$ 10-31 L2/mol2 $\cdot$ s at 298 K and 1 atm, but it increases to 6.0 $\times$ 10-30 L2/mol2 $\cdot$ s at 1000 K and 1 atm.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be chosen by the speaker, the writer, the audience, or the context.
- A topic can be influenced by the genre, the tone, the style, the purpose, and the audience of the communication.
- A topic can be developed by using various strategies, such as brainstorming, clustering, freewriting, outlining, researching, etc.
- A topic can be organized by using different patterns, such as chronological, spatial, topical, problem-solution, cause-effect, etc.
- A topic can be supported by using different types of evidence, such as facts, statistics, examples, anecdotes, quotations, etc.
- A topic can be evaluated by using different criteria, such as relevance, clarity, coherence, accuracy, completeness, etc.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.
- The topic can be used to guide the research and analysis of the text, such as by finding relevant sources, evaluating the evidence, and drawing conclusions.

- The topic can be used to communicate the purpose and message of the text, such as by using a thesis statement, a hook, a transition, or a call to action.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed explicitly or implicitly, depending on the context and the audience.
- A topic can be derived from a question, a problem, a curiosity, a personal interest, or a common knowledge.
- A topic can be related to other topics, forming a network of associations and connections.
- A topic can be developed and explored in various ways, such as by providing examples, evidence, arguments, opinions, facts, or perspectives.
- A topic can be organized and structured in different formats, such as by using headings, subheadings, bullet points, paragraphs, or outlines.
- A topic can be evaluated and revised based on the feedback, the purpose, and the audience of the communication.

The topic is

- The topic is a general term for the subject or theme of a conversation, text, or presentation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be explicit or implicit, depending on how clearly it is stated or implied by the speaker or writer.
- The topic can be broad or narrow, depending on how much detail or scope it covers.
- The topic can be related to different domains, such as science, history, art, or politics.
- The topic can be informative, persuasive, descriptive, or narrative, depending on the purpose or genre of the communication.
- The topic can be changed, refined, or expanded, depending on the feedback or interest of the audience or the development of the argument or story.

Page 30 of 40

- The topic of this page is **the structure and function of DNA**.

- DNA stands for **deoxyribonucleic acid**, and it is the molecule that carries the genetic information for all living organisms.
- DNA is composed of **nucleotides**, which are the building blocks of DNA. Each nucleotide consists of three parts: a **nitrogenous base**, a **sugar**, and a **phosphate group**.
- The nitrogenous bases are **adenine (A)**, **thymine (T)**, **cytosine (C)**, and **guanine (G)**. They form the **code** for the genetic information by pairing with each other in a specific way: A with T, and C with G. This is called **base pairing** or **complementary base pairing**.
- The sugar in DNA is **deoxyribose**, which has one less oxygen atom than the sugar in RNA (ribonucleic acid), which is **ribose**. The sugar and the phosphate group form the **backbone** of the DNA molecule, while the nitrogenous bases stick out from the backbone.
- DNA has a **double helix** structure, which means that it consists of two strands of nucleotides that twist around each other like a spiral staircase. The two strands are held together by **hydrogen bonds** between the complementary base pairs. The two strands are also **antiparallel**, which means that they run in opposite directions. One strand has a **5' end** (pronounced "five prime end"), which has a free phosphate group, and a **3' end** (pronounced "three prime end"), which has a free hydroxyl group. The other strand has the opposite orientation: a 5' end with a free hydroxyl group, and a 3' end with a free phosphate group.
- The double helix structure of DNA was discovered by **James Watson** and **Francis Crick** in 1953, based on the X-ray diffraction data of **Rosalind Franklin** and **Maurice Wilkins**. They published their findings in the journal **Nature**, and they shared the Nobel Prize in Physiology or Medicine in 1962 with Wilkins (Franklin died in 1958 and was not eligible for the prize).
- The function of DNA is to **store** and **transmit** the genetic information for the development, function, and reproduction of living organisms. The genetic information is encoded in the **sequence** of the nitrogenous bases along the DNA strands. The sequence of bases determines the sequence of **amino acids** in the proteins that are synthesized by the cells. Proteins are the main molecules that perform the various functions of life, such as catalyzing chemical reactions, transporting substances, providing structure, regulating gene expression, and defending against pathogens.
- The process of copying the genetic information from DNA to RNA is called **transcription**, and the process of translating the genetic information from RNA to protein is called **translation**. These processes are collectively known as **gene expression**. Gene expression is regulated by various factors, such as **transcription factors**, **enhancers**, **silencers**, **epigenetic modifications**, and **non-coding RNAs**. Gene expression determines the **phenotype** of an organism, which is the observable physical and biochemical characteristics of an organism.
- The process of passing the genetic information from one generation to the next is called **inheritance**, and the study of inheritance is called **genetics**.

The genetic information is organized into **chromosomes**, which are long, linear molecules of DNA that are tightly coiled and packaged with proteins called **histones**. Humans have 23 pairs of chromosomes, which are numbered from 1 to 22, and one pair of sex chromosomes, which are either XX (female) or XY (male). Each chromosome has a specific location on the chromosome called a **locus**, where a particular gene is located. A gene is a segment of DNA that codes for a specific protein or RNA molecule. Different versions of a gene are called **alleles**, and they may result in different phenotypes. The combination of alleles that an organism has for a particular gene is called its **genotype**. The inheritance of alleles follows certain patterns, such as **Mendelian inheritance**, **incomplete dominance**, **codominance**, **multiple alleles**, **polygenic inheritance**, and **sex-linked inheritance**. The inheritance of traits is also influenced by **environmental factors**, such as temperature, nutrition, and exposure to chemicals or radiation.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be chosen by the speaker, the writer, the listener, the reader, or a combination of them.
- A topic can be influenced by the context, the audience, the genre, the tone, and the goal of the communication.
- A topic can be developed by providing details, examples, evidence, arguments, opinions, or perspectives on it.
- A topic can be organized by using strategies such as classification, comparison, contrast, cause and effect, problem and solution, or chronological order.
- A topic can be evaluated by applying criteria such as relevance, accuracy, clarity, coherence, completeness, or originality.

Engineering Physics Lab

Engineering physics lab is a practical course that complements the theoretical knowledge of physics and its applications in engineering and technology. It involves experimental techniques and modern equipment to perform experiments related to topics such as optics, electronics, mechanics, thermodynamics, quantum physics, nanoscience, etc. Engineering physics lab also helps to develop skills such as data analysis, error estimation, report writing, and scientific communication.

Some of the common experiments that are performed in engineering physics lab are:

- Measurement of physical constants such as Planck's constant, speed of light, gravitational constant, etc.
- Verification of laws and principles such as Ohm's law, Kirchhoff's laws, Hooke's law, Snell's law, etc.
- Study of phenomena such as interference, diffraction, polarization, photoelectric effect, etc.
- Characterization of devices and materials such as diodes, transistors, LEDs, solar cells, resistors, capacitors, etc.
- Demonstration of concepts such as resonance, Doppler effect, standing waves, etc.

The syllabus of engineering physics lab may vary depending on the institution and the curriculum. However, some of the common topics that are covered in engineering physics lab are:

- Basic laboratory skills and safety
- Measurement and uncertainty
- Error analysis and propagation
- Graphical methods and curve fitting
- Statistical methods and data presentation
- Optics and optical instruments
- Electricity and magnetism
- Electronics and circuits
- Mechanics and dynamics
- Thermodynamics and heat transfer
- Quantum physics and atomic physics
- Nanoscience and nanotechnology

The assessment of engineering physics lab may include:

- Attendance and participation
- Pre-lab quizzes and assignments
- Lab reports and notebooks
- Oral presentations and posters
- Final exam and project

Engineering physics lab is an essential course for engineering, physics, computer science, and mathematics majors. It provides a hands-on experience of the physical principles and phenomena that underlie the modern technology and innovation. It also enhances the analytical, critical, and creative thinking skills of the students. Engineering physics lab is a challenging and rewarding course that prepares the students for their future careers in science and engineering.

Course Objectives:

- To provide an overview of the basic concepts and principles of artificial intelligence (AI).
- To introduce the main techniques and methods for solving AI problems, such as search, knowledge representation, reasoning, planning, learning, and natural language processing.
- To illustrate the applications and limitations of AI in various domains, such as games, robotics, computer vision, and natural language understanding.
- To develop the skills and abilities to design, implement, and evaluate AI systems using appropriate tools and frameworks.
- To foster the critical thinking and ethical awareness of the social and ethical implications of AI.

1. To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- Analytical instruments can be used for various purposes, such as quality control, research and development, environmental monitoring, forensic analysis, clinical diagnosis, etc.
- Analytical instruments can provide qualitative or quantitative information about the samples, such as their identity, composition, structure, concentration, purity, activity, etc.
- Analytical instruments can have different components, such as sources, detectors, transducers, amplifiers, filters, displays, etc., that work together to produce, manipulate, and interpret signals from the samples.
- Analytical instruments can have different characteristics, such as sensitivity, selectivity, accuracy, precision, resolution, range, speed, etc., that affect their performance and reliability.
- Analytical instruments can have different advantages and limitations, such as cost, availability, ease of use, maintenance, safety, etc., that influence their suitability and feasibility for different applications.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- Chloride content is the amount of chloride ions (Cl-) present in water. Chloride ions are derived from sources such as seawater intrusion, industrial effluents, agricultural runoff, and road salt. Chloride content can affect the taste, corrosion, and disinfection of water.

- Hardness is the measure of the concentration of calcium (Ca2+) and magnesium (Mg2+) ions in water. Hardness can cause scaling, staining, and reduced effectiveness of detergents and soap. Hardness can be expressed as calcium carbonate equivalent (mg/L as CaCO3) or as degrees of hardness (German, French, or English).
- Alkalinity is the measure of the capacity of water to neutralize acids. Alkalinity is mainly due to the presence of bicarbonate (HCO3-), carbonate (CO32-), and hydroxide (OH-) ions in water. Alkalinity can affect the pH, buffering, and corrosion of water.
- The analysis of chloride content, hardness, and alkalinity of water can be done by various methods, such as titration, colorimetry, spectrophotometry, ion chromatography, and electrochemical sensors. The choice of method depends on the accuracy, precision, sensitivity, and cost required for the analysis.
- The analysis of chloride content, hardness, and alkalinity of water can help to determine the quality, suitability, and treatment of water for various purposes, such as drinking, irrigation, industrial, and recreational uses. The analysis can also help to monitor the changes in water quality over time and space.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- pH is a measure of how acidic or basic a liquid is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. The pH scale ranges from 0 to 14, where 7 is neutral, lower than 7 is acidic, and higher than 7 is basic. The pH of a liquid can be measured using indicators, such as litmus paper, or pH meters, which are electronic devices that measure the voltage difference between two electrodes immersed in the liquid.
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is defined as the amount of force per unit length required to break or stretch the surface of a liquid. The surface tension of a liquid depends on the intermolecular forces, the temperature, and the presence of impurities or surfactants. The surface tension of a liquid can be measured using methods such as the drop weight method, the ring method, or the bubble pressure method.
- Viscosity is a measure of how resistant a liquid is to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. The shear stress is the force per unit area applied tangentially to the fluid, and the shear rate is the rate of change of velocity per unit distance in the direction perpendicular to the force. The viscosity of a liquid depends on the intermolecular forces, the temperature, and the pressure. The viscosity of a liquid can be measured using methods such as the capillary tube method, the rotating cylinder method, or the falling ball method.

4. To enable the students to understand about the preparation of different resins.

- Resins are natural or synthetic organic substances that are usually solid or semi-solid at room temperature and have a high molecular weight.
- Resins are widely used in various industries such as paints, adhesives, plastics, rubber, pharmaceuticals, cosmetics, etc.
- Resins can be classified into two main types: thermoplastic and thermosetting.
- Thermoplastic resins are resins that can be softened by heating and hardened by cooling repeatedly. They are usually prepared by polymerization or polycondensation of monomers or low molecular weight compounds. Examples of thermoplastic resins are polyethylene, polyvinyl chloride, polystyrene, nylon, etc.
- Thermosetting resins are resins that can be hardened by heating or by the action of a catalyst or a hardener. They cannot be softened by heating again. They are usually prepared by cross-linking of linear or branched polymers or by condensation of polyfunctional compounds. Examples of thermosetting resins are phenol-formaldehyde, urea-formaldehyde, epoxy, polyester, etc.
- The preparation of different resins involves the following steps:
  - Selection of raw materials: The raw materials for resin synthesis should be chosen based on the desired properties, availability, cost, and environmental impact of the resin. The raw materials can be natural or synthetic, monomeric or polymeric, aliphatic or aromatic, etc.
  - Reaction conditions: The reaction conditions for resin synthesis should be optimized to achieve the desired degree of polymerization, molecular weight distribution, functionality, branching, cross-linking, etc. The reaction conditions can include temperature, pressure, time, catalyst, solvent, etc.
  - Purification: The purification of resins involves the removal of unreacted raw materials, by-products, impurities, etc. The purification methods can include filtration, distillation, extraction, precipitation, etc.
  - Characterization: The characterization of resins involves the determination of the physical, chemical, and mechanical properties of the resin. The characterization methods can include spectroscopy, chromatography, thermal analysis, rheology, etc.
  - Application: The application of resins involves the processing of the resin into the desired shape, size, and form. The processing methods can include molding, extrusion, injection, coating, curing, etc.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

- Adipic acid is a dicarboxylic acid that is used as a raw material for the production of nylon-6,6, a synthetic polymer widely used in various applications.
- Paracetamol is an analgesic and antipyretic drug that is widely used to relieve pain and fever.
- Conventional route of synthesis of adipic acid involves the oxidation of cyclohexane or a mixture of cyclohexanol and cyclohexanone with nitric acid, which produces large amounts of nitrous oxide, a greenhouse gas, as a by-product.
- Conventional route of synthesis of paracetamol involves the acetylation of phenol with acetic anhydride, which produces large amounts of acetic acid, a corrosive and toxic substance, as a by-product.
- Green route of synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in a microemulsion, in the presence of a catalyst such as sodium tungstate, which produces water as the only by-product .
- Green route of synthesis of paracetamol involves the reaction of 4-aminophenol with acetic acid in water, in the presence of a catalyst such as sodium acetate, which produces water as the only by-product.
- Green route of synthesis of adipic acid and paracetamol are environmentally friendly, as they use renewable and biodegradable feedstocks, avoid harmful solvents and reagents, reduce waste generation and energy consumption, and produce high yields and purity of the products .

LIST OF EXPERIMENTS

- An experiment is a scientific procedure that aims to test a hypothesis, answer a question, or demonstrate a known fact.
- Experiments usually involve manipulating one or more variables, measuring their effects, and comparing them with a control group.
- Experiments can be classified into different types based on their purpose, design, and outcome.
- Some common types of experiments are:
  - **Observational experiments**: These are experiments where the researcher observes and records the natural behavior of the subjects without interfering with them. For example, observing the feeding habits of wild animals.
  - **Quasi-experiments**: These are experiments where the researcher cannot randomly assign the subjects to different groups, but uses existing groups or natural events as proxies. For example, comparing

the academic performance of students from different schools.

- **Randomized controlled experiments**: These are experiments where the researcher randomly assigns the subjects to different groups, and applies different treatments or interventions to each group. For example, testing the effectiveness of a new drug on patients.
- **Factorial experiments**: These are experiments where the researcher manipulates two or more independent variables, and measures their effects on one or more dependent variables. For example, studying the effects of temperature and humidity on plant growth.
- **Single-subject experiments**: These are experiments where the researcher applies a treatment or intervention to a single subject, and measures its effects over time. For example, measuring the blood pressure of a patient before and after a medication.
- **Repeated-measures experiments**: These are experiments where the researcher measures the same dependent variable multiple times on the same subjects, under different conditions or over time. For example, measuring the reaction time of drivers before and after drinking coffee.

1. **Calibration of Analytical Equipment and apparatus**

- Calibration is the process of evaluating and adjusting the precision and accuracy of measurement equipment.
- Calibration ensures that the instruments produce valid and reliable results that are consistent with the standards and specifications .
- Calibration is essential for maintaining the quality and safety of analytical products and processes.
- Calibration should be done according to a predefined schedule and procedure that are based on the manufacturer's recommendations, the frequency of use, the stability of the instrument, and the requirements of the application .
- Calibration should be performed by qualified and trained personnel using appropriate reference standards and methods .
- Calibration should be documented and recorded with the details of the instrument, the reference standard, the method, the date, the results, and the corrective actions if any .
- Calibration should be verified periodically by using check standards or quality control samples to monitor the performance and accuracy of the instrument .
- Calibration should be repeated or adjusted if the instrument is repaired, modified, relocated, or subjected to environmental changes that may affect its performance .

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is the measure of the concentration of calcium and magnesium ions in water.
- EDTA (Ethylenediaminetetraacetic acid) is a chelating agent that can form complexes with metal ions such as Ca2+ and Mg2+.
- EDTA titration is a method of determining the hardness of water by titrating a water sample with a standard solution of EDTA in the presence of a suitable indicator.
- The indicator used is Eriochrome Black T (EBT), which is a dye that changes color from wine red to blue when it binds to metal ions.
- The procedure for EDTA titration is as follows :
  - Take a sample volume of 20 ml (V ml) of water to be tested.
  - Dilute the sample to 40 ml by adding 20 ml of distilled water in an Erlenmeyer flask.
  - Add 1 ml of ammonia buffer to adjust the pH to 10 $\pm$ 0.1. This is to prevent the precipitation of metal hydroxides and to ensure the formation of stable EDTA-metal complexes.
  - Add 1 or 2 drops of EBT indicator to the sample. The sample will turn wine red due to the presence of Ca2+ and Mg2+ ions.
  - Titrate the sample with a standard solution of EDTA of known concentration (C mg/ml) from a burette with vigorous shaking until the color changes from wine red to blue. This indicates the end point of the titration.
  - Record the volume of EDTA solution used (B ml).
- The calculation of hardness of water by EDTA titration is based on the following equation :
  - C * B = V * H
  - Where C is the concentration of EDTA solution in mg/ml, B is the volume of EDTA solution used in ml, V is the volume of water sample taken in ml, and H is the hardness of water in mg/ml as CaCO3 (calcium carbonate).
  - The hardness of water can be expressed in different units, such as ppm (parts per million), mg/L (milligrams per liter), or dH (degrees of hardness). One ppm or mg/L is equivalent to one mg of CaCO3 per liter of water. One dH is equivalent to 17.8 ppm or mg/L of CaCO3.
  - The hardness of water can be classified as soft, moderately hard, hard, or very hard based on the following ranges:
    - Soft: 0 - 75 ppm or 0 - 4.2 dH
    - Moderately hard: 75 - 150 ppm or 4.2 - 8.4 dH
    - Hard: 150 - 300 ppm or 8.4 - 16.8 dH
    - Very hard: above 300 ppm or above 16.8 dH

3. **Determination of Alkalinity of water sample.**

- Alkalinity is the measure of the capacity of water to neutralize acids. It is mainly caused by the presence of bicarbonates, carbonates, and hydroxides of calcium, magnesium, sodium, and potassium in water.

- Alkalinity is important for water quality because it affects the pH, corrosion, and disinfection processes. High alkalinity can cause scaling and reduce the effectiveness of chlorine. Low alkalinity can cause corrosion and pH fluctuations.

- Alkalinity can be determined by titrating a water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a suitable indicator, such as phenolphthalein or methyl orange, to detect the endpoint of the titration.

- The endpoint of the titration depends on the type and concentration of the alkaline species in the water sample. There are three types of alkalinity: phenolphthalein alkalinity (P-alkalinity), total alkalinity (T-alkalinity), and hydroxide alkalinity (OH-alkalinity).

- Phenolphthalein alkalinity is the amount of acid required to lower the pH of the water sample to 8.3, which is the color change point of phenolphthalein indicator from pink to colorless. Phenolphthalein alkalinity measures the presence of hydroxides and half of the carbonates in the water sample.

- Total alkalinity is the amount of acid required to lower the pH of the water sample to 4.5, which is the color change point of methyl orange indicator from yellow to orange. Total alkalinity measures the presence of hydroxides, carbonates, and bicarbonates in the water sample.

- Hydroxide alkalinity is the amount of acid required to lower the pH of the water sample to 10.5, which is the point where all the hydroxides are neutralized. Hydroxide alkalinity measures the presence of hydroxides only in the water sample.

- The relationship between the three types of alkalinity can be expressed as follows:

  - P-alkalinity = OH-alkalinity + 1/2 CO3-alkalinity
  - T-alkalinity = OH-alkalinity + CO3-alkalinity + HCO3-alkalinity
  - OH-alkalinity = T-alkalinity - P-alkalinity - HCO3-alkalinity

- The procedure for determining the alkalinity of a water sample is as follows:

  - Collect a representative water sample in a clean glass bottle and measure its initial pH using a pH meter or pH paper.

  - Transfer a known volume of the water sample (usually 50 mL or 100 mL) to a conical flask and add 2-3 drops of phenolphthalein indicator. If the solution turns pink, it indicates the presence of P-alkalinity. Record the initial burette reading of the standard acid solution.

  - Titrate the water sample with the standard acid solution until the pink color disappears. Record the final burette reading and calculate the volume of acid used. This is the P-alkalinity titration.
  - Add 2-3 drops of methyl orange indicator to the same flask and continue the titration with the standard acid solution until the solution turns from yellow to orange. Record the final burette reading and calculate the volume of acid used. This is the T-alkalinity titration.
  - Calculate the P-alkalinity and T-alkalinity of the water sample in mg/L as CaCO3 using the following formula:
    * Alkalinity (mg/L as CaCO3) = (Volume of acid used in mL) x (Normality of acid) x (50,000) / (Volume of water sample in mL)
  - If the P-alkalinity is zero, it means that the water sample contains only bicarbonates and no hydroxides or carbonates. In this case, the T-alkalinity is equal to the HCO3-alkalinity.
  - If the P-alkalinity is equal to the T-alkalinity, it means that the water sample contains only hydroxides and no bicarbonates or carbonates. In this case, the OH-alkalinity is equal to the T-alkalinity.
  - If the P-alkalinity is less than the T-alkalinity, it means that the water sample contains a mixture of hydroxides, carbonates, and bicarbonates. In this case, the OH-alkalinity, CO3-alk

4. **Determination of pH by titrimetric method**

- Titrimetric method is a technique of quantitative analysis that involves measuring the volume of a solution of known concentration (titrant) that is required to react completely with a solution of unknown concentration (analyte) .
- pH is a measure of the acidity or basicity of a solution, defined as the negative logarithm of the hydrogen ion concentration .
- pH can be determined by titrimetric method by using a pH meter, which measures the cell potential of the solution in reference to a standard hydrogen electrode .
- A plot of the pH of an acidic (or basic) solution as a function of the amount of added base (or acid) is called a pH curve or a titration curve .
- The pH curve can be used to determine the endpoint or equivalence point of the titration, which is the point where the moles of titrant and analyte are equal .
- The pH curve can also be used to determine the pKa or pKb of the analyte, which are the negative logarithms of the acid or base dissociation constants,

respectively .

- The shape and features of the pH curve depend on the strength and concentration of the acid and base involved in the titration .
- For example, a titration of a strong acid with a strong base will have a pH curve that starts at a low pH, rises sharply near the equivalence point, and ends at a high pH .
- A titration of a weak acid with a strong base will have a pH curve that starts at a higher pH, has a buffer region where the pH changes slowly, and has an equivalence point above 7 .
- A titration of a weak base with a strong acid will have a pH curve that starts at a lower pH, has a buffer region where the pH changes slowly, and has an equivalence point below 7 .
- A titration of a polyprotic acid or base will have multiple equivalence points and buffer regions, corresponding to the different stages of dissociation .
- To perform a pH titration, the following steps are required :
  - Prepare the titrant solution of known concentration and standardize it using a primary standard or a secondary standard.
  - Prepare the analyte solution of unknown concentration and transfer a measured volume to a titration flask.
  - Calibrate the pH meter using buffer solutions of known pH values.
  - Immerse the pH electrode and a reference electrode in the analyte solution and record the initial pH reading.
  - Add the titrant solution gradually to the analyte solution, stirring constantly, and record the pH reading after each addition.
  - Plot the pH readings against the volume of titrant added and observe the shape of the pH curve.
  - Identify the endpoint or equivalence point of the titration from the pH curve, either by locating the point of inflection or by using an indicator that changes color at the desired pH range.
  - Calculate the concentration of the analyte solution using the stoichiometry of the titration reaction and the volume and concentration of the titrant solution at the endpoint or equivalence point.
  - Calculate the pKa or pKb of the analyte solution using the pH and concentration of the analyte solution at the half-equivalence point, where the moles of titrant and analyte are equal to half of their initial values.

5. Determination of surface tension of given liquid.

- Surface tension is the property of a liquid that makes its surface behave like a stretched elastic membrane.
- Surface tension is caused by the cohesive forces between the molecules of the liquid, which tend to minimize the surface area of the liquid.
- Surface tension can be measured by various methods, such as the capillary rise method, the drop weight method, the ring method, and the

stalagmometer method.

- The capillary rise method is based on the principle that a liquid will rise or fall in a narrow tube (capillary) depending on the relative magnitude of the adhesive forces between the liquid and the tube, and the cohesive forces within the liquid.
- The drop weight method is based on the principle that the weight of a drop of liquid that detaches from a vertical capillary tube is proportional to the surface tension of the liquid.
- The ring method is based on the principle that a horizontal ring of wire (du Nouy ring) attached to a balance will experience a downward force when it is dipped in a liquid and then pulled out, due to the liquid film that clings to the ring. The surface tension of the liquid can be calculated from the force required to break the film.
- The stalagmometer method is based on the principle that the number of drops of a liquid that fall from a capillary tube of known dimensions is inversely proportional to the surface tension of the liquid.

6. Determination of Viscosity of a given liquid by viscometer.

- Viscosity is a measure of the resistance of a fluid to flow or deform under an applied force.
- A viscometer is a device that measures the viscosity of a fluid by observing its flow through a narrow tube or a capillary.
- The flow rate of a fluid through a capillary depends on the pressure difference across the capillary, the length and radius of the capillary, and the viscosity of the fluid.
- The Hagen-Poiseuille equation relates these variables as follows:

$$Q = \frac{\pi r^4 \Delta P}{8\eta L}$$

where Q is the volumetric flow rate, r is the radius of the capillary, ΔP is the pressure difference, η is the viscosity, and L is the length of the capillary.

- To determine the viscosity of a given liquid by viscometer, the following steps are followed:
  - Fill the viscometer with the liquid to be tested and immerse it in a water bath at a constant temperature.
  - Allow the liquid to flow through the capillary under the influence of gravity and measure the time taken for a fixed volume of liquid to flow out of the capillary.
  - Repeat the experiment for different volumes of liquid and calculate the average flow rate.
  - Measure the pressure difference across the capillary by using a manometer or a pressure gauge.

  - Measure the length and radius of the capillary by using a meter scale and a vernier caliper.
  - Substitute the values of Q, r, L, and ΔP in the Hagen-Poiseuille equation and solve for η.
  - Report the viscosity of the liquid with appropriate units and significant figures.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator.

- Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron(II) sulfate and ammonium sulfate, with the formula (NH4)2Fe(SO4)2 · 6H2O.
- It is used as a primary standard for titrations involving oxidizing agents, such as potassium permanganate (KMnO4) or potassium dichromate (K2Cr2O7).
- The strength of a ferrous ammonium sulfate solution can be determined by titrating it with a standard solution of an oxidizing agent, using an external indicator to detect the end point.
- An external indicator is a substance that changes color when it comes in contact with the excess oxidizing agent, after the ferrous ammonium sulfate has been completely oxidized to ferric ammonium sulfate.
- One example of an external indicator is potassium ferricyanide (K3[Fe(CN)6]), which forms a blue precipitate of Prussian blue (Fe4[Fe(CN)6]3) when it reacts with ferric ions.
- The procedure for the determination of the strength of ferrous ammonium sulfate using external indicator is as follows:
  - Prepare a standard solution of an oxidizing agent, such as potassium permanganate or potassium dichromate, by dissolving a known mass of the salt in distilled water and standardizing it against a primary standard, such as sodium oxalate or sodium thiosulfate.
  - Pipette a known volume of the ferrous ammonium sulfate solution into a conical flask and add a few drops of dilute sulfuric acid to prevent the hydrolysis of the iron(II) ions.
  - Fill a burette with the standard solution of the oxidizing agent and note the initial reading.
  - Titrate the ferrous ammonium sulfate solution with the oxidizing agent, swirling the flask constantly, until the solution becomes pale green or yellow, indicating that most of the iron(II) ions have been oxidized to iron(III) ions.

- Place a white tile under the flask and add a drop of potassium ferricyanide solution on the tile near the mouth of the flask.
- Continue the titration, adding the oxidizing agent dropwise, until a blue spot appears on the tile, indicating that the end point has been reached. Note the final reading of the burette.
- Repeat the titration at least two more times and take the average of the concordant readings.
- Calculate the strength of the ferrous ammonium sulfate solution using the following formula:
    * Strength of ferrous ammonium sulfate = (Volume of oxidizing agent × Normality of oxidizing agent × Equivalent mass of ferrous ammonium sulfate) / Volume of ferrous ammonium sulfate

8. Determination of the strength of Potassium dichromate using internal indicator.

- Potassium dichromate is a primary standard substance that can be used to determine the strength of a reducing agent such as ferrous ammonium sulphate.
- The reaction between potassium dichromate and ferrous ammonium sulphate is as follows:

$$K_2Cr_2O_7{+}6FeSO_4{+}7H_2SO_4 \rightarrow Cr_2(SO_4)_3{+}3K_2SO_4{+}6Fe_2(SO_4)_3{+}7H_2O$$

- The end point of the titration can be detected by the change in colour of the solution from green to reddish-brown, due to the formation of ferric ions.
- This colour change is the internal indicator of the titration, and no external indicator is required.
- The strength of potassium dichromate can be calculated by using the following formula:

$$\text{Strength of potassium dichromate} = \frac{6 \times \text{Normality of ferrous ammonium sulphate} \times \text{Volume of ferrous}}{\text{Volume of potassium dichromate used}}$$

- The normality of ferrous ammonium sulphate can be determined by titrating it against a standard solution of potassium permanganate, using phenolphthalein as an external indicator.

- The reaction between potassium permanganate and ferrous ammonium sulphate is as follows:

$$KMnO_4 + 5FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$$

- The end point of the titration is marked by the appearance of a faint pink colour in the solution, due to the excess of potassium permanganate.
- The normality of ferrous ammonium sulphate can be calculated by using the following formula:

$$\text{Normality of ferrous ammonium sulphate} = \frac{\text{Normality of potassium permanganate} \times \text{Volume of potassi}}{5 \times \text{Volume of ferrous ammonium sulphate}}$$

- The normality of potassium permanganate can be assumed to be 0.1 N for the purpose of this experiment.

9. Determination of available chlorine in bleaching powder

- Bleaching powder is a compound of calcium hypochlorite [Ca(OCl)2] and calcium chloride [CaCl2] that is used as a disinfectant and a bleaching agent.
- Available chlorine is the amount of chlorine that can be liberated from bleaching powder when it reacts with dilute acids. It is expressed as a percentage of the mass of bleaching powder.
- The reaction of bleaching powder with dilute acid is as follows:

  Ca(OCl)Cl + H2SO4 → CaSO4 + H2O + Cl2

- The available chlorine can be determined by titrating the chlorine gas evolved with a standard solution of sodium thiosulfate (hypo) using starch as an indicator. The reaction is as follows:

  Cl2 + 2Na2S2O3 → 2NaCl + Na2S4O6

- The procedure for the determination of available chlorine in bleaching powder is as follows:
  - Weigh about 0.5 g of bleaching powder and transfer it to a 250 mL conical flask.
  - Add about 100 mL of distilled water and shake well to dissolve the bleaching powder.
  - Add 10 mL of dilute sulfuric acid and stopper the flask immediately.

- Allow the flask to stand for 10 minutes to complete the reaction and release the chlorine gas.
- Titrate the solution with 0.1 N sodium thiosulfate solution until the yellow color of chlorine fades.
- Add a few drops of starch solution and continue the titration until the blue color disappears.
- Record the volume of sodium thiosulfate solution used as V mL.
- Repeat the procedure with a blank sample of distilled water and acid and record the volume of sodium thiosulfate solution used as V0 mL.
- Calculate the percentage of available chlorine in the bleaching powder sample using the formula:

  % available chlorine = (V0 - V) x N x 35.45 / W

  where N is the normality of sodium thiosulfate solution, W is the weight of bleaching powder sample in grams, and 35.45 is the equivalent weight of chlorine.

10. Determination of chloride content in water sample.

- Chloride is one of the major anions found in natural waters. It originates from various sources, such as seawater intrusion, industrial effluents, agricultural runoff, and domestic sewage.
- Chloride can affect the taste, corrosion, and biological activity of water. Therefore, it is important to monitor and control its concentration in drinking water and wastewater.
- The most common method for determining chloride content in water sample is the **argentometric method**. This method is based on the precipitation of chloride ions as silver chloride by adding a known excess of silver nitrate solution and titrating the excess silver ions with a standard solution of sodium or potassium thiocyanate using ferric ammonium sulfate as an indicator.
- The following steps are involved in the argentometric method:
  1. Prepare a standard solution of sodium or potassium thiocyanate (0.1 N) by dissolving an accurately weighed amount of the salt in distilled water and standardizing it against a known solution of silver nitrate (0.1 N) using ferric ammonium sulfate as an indicator. The end point is marked by the appearance of a faint brown color due to the formation of ferric thiocyanate complex.
  2. Pipette a 50 mL aliquot of the water sample into a 250 mL conical flask and add 2 mL of nitric acid (6 N) to acidify the solution. This

prevents the precipitation of other anions, such as carbonate, sulfate, and phosphate, as silver salts.

3. Add a few drops of potassium chromate indicator, which forms a yellow precipitate of silver chromate with silver ions. The indicator also serves as a buffer to maintain the pH around 7.5, which is optimal for the precipitation of silver chloride.

4. Titrate the sample with silver nitrate solution (0.1 N) from a burette until a brick-red color appears due to the formation of silver chromate. Record the volume of silver nitrate solution used as V1.

5. Add 2 mL of ferric ammonium sulfate indicator, which forms a red complex with thiocyanate ions. Titrate the excess silver ions with sodium or potassium thiocyanate solution (0.1 N) from a burette until a faint brown color appears due to the formation of ferric thiocyanate complex. Record the volume of thiocyanate solution used as V2.

6. Repeat the procedure with a blank sample of distilled water and record the volumes of silver nitrate and thiocyanate solutions used as V1' and V2', respectively.

7. Calculate the chloride content in the water sample using the following formula:

   Cl- (mg/L) = [(V1 - V1') - (V2 - V2')] x N x 35.45 x 1000 / 50

   where N is the normality of the thiocyanate solution and 35.45 is the equivalent weight of chloride.

11. Preparation of Phenol Formaldehyde (PF) Resin

- Phenol formaldehyde resin is a type of thermosetting polymer that is obtained by the condensation reaction of phenol and formaldehyde.
- Phenol formaldehyde resin can be prepared by either acid-catalyzed or base-catalyzed method, depending on the desired properties and applications of the resin.
- The acid-catalyzed method produces novolac resin, which is a linear polymer that requires a curing agent (such as hexamethylenetetramine) to crosslink and harden.
- The base-catalyzed method produces resol resin, which is a branched polymer that can self-cure and harden without any additional agent.
- The general steps for the preparation of phenol formaldehyde resin are as follows :
  - Weigh the appropriate quantity of the raw materials (phenol and formaldehyde) before mixing them in a reaction vessel.

  - Add the catalyst (acid or base) to initiate the polymerization reaction. The molar ratio of formaldehyde to phenol determines the degree of crosslinking and the type of resin formed. A ratio of 1:1 produces novolac resin, while a ratio of more than 1:1 produces resol resin.
  - Control the temperature and pH of the reaction mixture to optimize the reaction rate and the molecular weight distribution of the resin. The temperature is usually kept between 60°C and 100°C, and the pH is adjusted according to the catalyst used.
  - Monitor the viscosity and the water content of the resin as the reaction proceeds. The viscosity increases as the polymer chains grow longer and more branched. The water content decreases as the water is removed by distillation or evaporation.
  - Stop the reaction when the desired properties of the resin are achieved. The resin can be isolated by filtration, precipitation, or solvent extraction. The resin can be further purified by washing, drying, and grinding.
  - The resin can be stored as a solid or a liquid, depending on its solubility and stability. The resin can be cured and hardened by applying heat, pressure, and/or a curing agent (for novolac resin).

- Some of the properties and applications of phenol formaldehyde resin are:
  - It has high mechanical strength, thermal stability, chemical resistance, and electrical insulation.
  - It is widely used as an adhesive, a coating, a molding material, a laminating material, and a wood composite material.
  - It is the main component of bakelite, the first synthetic plastic.

12. Preparation of Urea formaldehyde (UF) resin

- Urea formaldehyde (UF) resin is a thermosetting polymer that is widely used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- UF resin is synthesized by the condensation reaction of urea and formaldehyde in the presence of a base or an acid catalyst .
- The reaction can be carried out in one or more steps, depending on the desired properties and applications of the resin .
- The general steps for the preparation of UF resin are as follows :
  1. An aqueous solution of formaldehyde (usually 37% or more) and urea are mixed in a molar ratio of 2-3:1 at pH 6-11. This step is called the methylolation or amination step, where urea reacts with formaldehyde to form monomethylolurea and dimethylolurea .
  2. The mixture is heated to at least 80°C to promote the condensation reaction between the methylol groups of the urea derivatives. This step is called the condensation or resinification step, where water is

eliminated and methylene and methylene ether linkages are formed . The degree of condensation determines the molecular weight, viscosity, and solubility of the resin.

3. An acid (such as sulfuric acid, hydrochloric acid, or phosphoric acid) is added to lower the pH to 0.5-3.5. This step is called the hardening or curing step, where the resin becomes insoluble and infusible due to the formation of more methylene linkages . The curing time and temperature depend on the type and amount of the acid catalyst.

- The UF resin can be modified by adding other compounds, such as ammonia, melamine, phenol, or resorcinol, to improve its properties, such as water resistance, thermal stability, and mechanical strength .

13. Preparation of Adipic acid / Paracetamol

- Adipic acid is the most important dicarboxylic acid, mainly used as a precursor for the production of nylon.
- Adipic acid can be prepared from a mixture of cyclohexanone and cyclohexanol called KA oil, which is oxidized with nitric acid to give adipic acid, via a multistep pathway .
- The steps involved in the preparation of adipic acid are:
  - Oxidation of KA oil with nitric acid to form a mixture of nitro compounds and nitric acid esters.
  - Hydrolysis of the nitric acid esters to form carboxylic acids and nitrous acid.
  - Decomposition of nitrous acid to form nitrogen oxides and water.
  - Recycle of the nitrogen oxides to nitric acid by absorption in water.
  - Purification of adipic acid by crystallization from water.
- Paracetamol is an effective antipyretic and analgesic, widely used for the relief of pain and fever.
- Paracetamol can be prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of concentrated sulfuric acid as a catalyst.
- The steps involved in the preparation of paracetamol are:
  - Acetylation of p-aminophenol with acetic anhydride to form N-acetyl-p-aminophenol (paracetamol) and acetic acid.
  - Filtration of the crude product and washing with water to remove excess acetic acid.
  - Recrystallization of paracetamol from water or ethanol to obtain pure crystals .

14. Determination of Cell Conductance of a solution.

- Cell conductance is a measure of how well a solution can conduct electric current between two electrodes.

- Cell conductance depends on the type, concentration, and temperature of the solution, as well as the geometry and material of the electrodes and the cell.

- Cell conductance is measured in siemens (S) or mho (℧), which are the reciprocal units of resistance (ohm, Ω).
- Cell conductance can be calculated by applying a known voltage across the electrodes and measuring the resulting current, using the formula:

  `G = I / V`

  where G is the cell conductance, I is the current, and V is the voltage.
- Cell conductance can also be expressed as the product of the specific conductance (or conductivity) of the solution and the cell constant, using the formula:

  `G = k * K`

  where k is the specific conductance, and K is the cell constant.
- The specific conductance of a solution is a property that depends only on the type and concentration of the ions in the solution, and is measured in S/m or S/cm.
- The cell constant is a property that depends only on the geometry and material of the electrodes and the cell, and is measured in m^{-1} or cm^{-1}.
- The cell constant can be determined by measuring the cell conductance of a solution with a known specific conductance, using the formula:

  `K = G / k`

  where K is the cell constant, G is the cell conductance, and k is the specific conductance.
- Alternatively, the cell constant can be calculated from the dimensions of the electrodes and the cell, using the formula:

  `K = L / A`

  where K is the cell constant, L is the distance between the electrodes, and A is the effective area of the electrodes.
- To determine the cell conductance of a solution, the following steps are followed:
  - Prepare the solution of known concentration and temperature, and fill the cell with it.
  - Connect the cell to a conductance meter or a voltmeter and an ammeter.
  - Apply a known voltage across the electrodes and measure the resulting current, or measure the voltage and current directly using the conductance meter.

  - Calculate the cell conductance using the formula:

    ```
    G = I / V
    ```

  - If the cell constant is known, calculate the specific conductance of the solution using the formula:

    ```
    k = G / K
    ```

  - If the cell constant is unknown, measure the cell conductance of a standard solution with a known specific conductance, and calculate the cell constant using the formula:

    ```
    K = G / k
    ```

  - Repeat the measurements for different concentrations and temperatures of the solution, and plot the cell conductance versus the concentration and temperature.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of determination of rate constant of hydrolysis of esters.

15. Determination of Rate constant of hydrolysis of esters

- Esters are organic compounds that have the general formula RCOOR', where R and R' are alkyl or aryl groups.
- Esters can be hydrolyzed by water in the presence of an acid or a base catalyst, resulting in the formation of a carboxylic acid and an alcohol.
- The hydrolysis of esters is a first-order reaction, meaning that the rate of the reaction depends only on the concentration of the ester.
- The rate constant of the hydrolysis of esters can be determined by measuring the change in the concentration of the ester or the products over time, using various methods such as titration, spectrophotometry, or conductometry.
- One of the methods to determine the rate constant of the hydrolysis of esters is by titrating the hydrolyzed solution with a standard base solution, such as sodium hydroxide, and using an indicator, such as phenolphthalein, to detect the end point of the titration.
- The procedure for this method is as follows:
  - Prepare a known volume and concentration of the ester solution in a conical flask and add a few drops of the indicator.
  - Add a known volume and concentration of the acid catalyst, such as hydrochloric acid, to the flask and start a stopwatch.

  - Swirl the flask gently and periodically titrate a small aliquot of the solution with the standard base solution, using a burette, until the indicator changes color.
  - Record the volume of the base solution used and the time elapsed for each titration.
  - Repeat the titration at regular intervals until the reaction is complete.
  - Plot a graph of the volume of the base solution used versus the time elapsed.
  - The slope of the graph is equal to the rate constant of the hydrolysis of the ester, k, divided by the concentration of the acid catalyst, [H+].
  - The rate constant of the hydrolysis of the ester can be calculated by multiplying the slope by the concentration of the acid catalyst.

- The advantages of this method are that it is simple, inexpensive, and accurate. The disadvantages are that it requires frequent sampling, which may disturb the reaction, and that it may be affected by side reactions, such as the hydrolysis of the acid catalyst or the indicator.

16. Element detection and identification of functional groups in organic compounds.

- Organic compounds are composed of carbon, hydrogen, and other elements such as oxygen, nitrogen, sulfur, halogens, etc.
- The presence of these elements and the functional groups they form can be detected and identified by various chemical tests.
- Functional groups are specific arrangements of atoms or bonds within a molecule that determine its chemical properties and reactions.
- Some common functional groups are alcohols, aldehydes, ketones, carboxylic acids, esters, amines, amides, etc.

16.1 Element detection

- The detection of carbon and hydrogen in an organic compound can be done by heating it with copper oxide in a test tube. The carbon is oxidized to carbon dioxide, which turns lime water milky, and the hydrogen is oxidized to water, which condenses on the cooler part of the test tube.
- The detection of nitrogen in an organic compound can be done by the Lassaigne's test, which involves heating the compound with sodium metal in a fusion tube. The nitrogen forms sodium cyanide, which can be extracted with water and tested with ferrous sulfate and ferric chloride solutions. The formation of Prussian blue color indicates the presence of nitrogen.
- The detection of sulfur in an organic compound can be done by the Lassaigne's test as well, but the sodium sulfide formed can be extracted with water and tested with lead acetate solution. The formation of a black precipitate of lead sulfide indicates the presence of sulfur.

- The detection of halogens in an organic compound can be done by the Beilstein test, which involves heating the compound with a copper wire in a flame. The halogens form volatile copper halides, which impart a characteristic green color to the flame.

16.2 Functional group identification

- The identification of functional groups in an organic compound can be done by various chemical tests that involve adding specific reagents and observing the changes in color, odor, precipitate, etc.
- Some examples of functional group tests are:
  - Alcohols can be identified by the Lucas test, which involves adding Lucas reagent (a mixture of zinc chloride and concentrated hydrochloric acid) to the alcohol. The formation of a turbid solution indicates the presence of an alcohol. The rate of turbidity depends on the type of alcohol: tertiary alcohols react fastest, followed by secondary alcohols, and primary alcohols react very slowly or not at all.
  - Aldehydes can be identified by the Fehling's test, which involves adding Fehling's solution (a mixture of copper sulfate, sodium carbonate, and potassium sodium tartrate) to the aldehyde. The formation of a red precipitate of copper oxide indicates the presence of an aldehyde. Ketones do not react with Fehling's solution.
  - Carboxylic acids can be identified by the sodium bicarbonate test, which involves adding sodium bicarbonate solution to the carboxylic acid. The formation of bubbles of carbon dioxide gas indicates the presence of a carboxylic acid. Other organic acids, such as phenols, do not react with sodium bicarbonate.
  - Amines can be identified by the Hinsberg test, which involves adding Hinsberg reagent (benzenesulfonyl chloride) to the amine. The formation of a soluble or insoluble sulfonamide derivative indicates the presence of an amine. The solubility of the sulfonamide depends on the type of amine: primary amines form soluble sulfonamides, secondary amines form insoluble sulfonamides, and tertiary amines do not react with Hinsberg reagent.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- This note indicates that the instructor has the authority and flexibility to select the experiments that are most suitable for the course objectives and the available resources.
- The instructor can choose any 10 experiments from the list of experiments given in the syllabus or the lab manual, depending on the relevance, difficulty, duration, and learning outcomes of each experiment.

- The instructor can also modify or replace any two of the experiments from the list, if they find that the existing experiments are outdated, impractical, or insufficient for the course requirements.
- The instructor should inform the students about the selected and changed experiments at the beginning of the course or the lab session, and provide the necessary instructions, guidelines, and materials for conducting the experiments.
- The instructor should also explain the rationale and the objectives of the selected and changed experiments, and how they are related to the course topics and the learning outcomes.
- The instructor should ensure that the selected and changed experiments are aligned with the course curriculum and the assessment criteria, and that they are fair, feasible, and effective for the students' learning and evaluation.

Course Outcomes:

- A course outcome is a statement of what a student should be able to do or demonstrate by the end of the course.
- Course outcomes are usually derived from the course objectives, which are the broad goals of the course.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART).
- Course outcomes should align with the course content, activities, assessments, and evaluations.
- Course outcomes should also align with the program outcomes, which are the overarching goals of the program or degree.
- Course outcomes should be communicated to the students at the beginning of the course and throughout the course.
- Course outcomes should be reviewed and revised periodically to ensure their relevance and effectiveness.

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Some possible responses for the topic are:

Upon completion of the course the student should be able to:

- Apply the fundamental concepts and principles of computer science to solve problems.
- Design, implement, test, and debug programs using various programming languages and paradigms.
- Analyze the efficiency and correctness of algorithms and data structures.
- Use appropriate tools and techniques for software development and documentation.
- Work effectively in teams and communicate clearly and professionally.

- Demonstrate ethical and social responsibility in computing contexts.

Course Outcomes Bloom's

- Course outcomes are statements that describe what students should be able to do or demonstrate after completing a course.
- Bloom's taxonomy is a framework for classifying educational objectives into six levels of cognitive complexity: remember, understand, apply, analyze, evaluate, and create.
- Course outcomes can be aligned with Bloom's taxonomy to ensure that they cover a range of cognitive skills and learning domains.
- Some examples of course outcomes and their corresponding Bloom's levels are:
  - Remember: Identify the main components of a computer system.
  - Understand: Explain the difference between procedural and object-oriented programming.
  - Apply: Use appropriate data structures and algorithms to solve a given problem.
  - Analyze: Compare and contrast different sorting techniques in terms of their efficiency and complexity.
  - Evaluate: Critique the design and implementation of a software project.
  - Create: Develop a web application using HTML, CSS, and JavaScript.

Level

- A level is a measure of the amount or degree of something, such as height, quantity, quality, or intensity.
- Levels can be expressed in different units, such as meters, liters, degrees, or percentages.
- Levels can be compared, ordered, or classified according to their values, such as high, low, medium, or optimal.
- Levels can be used to describe the state or condition of a system, such as water level, sound level, blood sugar level, or difficulty level.
- Levels can be changed or adjusted by adding, removing, increasing, or decreasing the amount or degree of something, such as raising the water level, lowering the sound level, regulating the blood sugar level, or changing the difficulty level.

CO-1 Get an understanding of the use of different analytical instruments. K3

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.

- Different analytical instruments have different principles, applications, advantages, and limitations.
- Some examples of analytical instruments are:
  - Spectrophotometers: measure the amount of light absorbed or transmitted by a sample at different wavelengths.
  - Chromatographs: separate the components of a mixture based on their affinity to a stationary phase and a mobile phase.
  - Mass spectrometers: identify the molecular mass and structure of compounds by ionizing them and measuring their mass-to-charge ratio.
  - Electrochemical analyzers: measure the electrical potential or current generated by a chemical reaction or a redox process.
  - Microscopes: magnify the image of small objects or structures using visible light, electrons, or other forms of radiation.
  - Biosensors: detect the presence or concentration of a specific biomolecule or analyte using a biological recognition element and a transducer.
- To get an understanding of the use of different analytical instruments, one should:
  - Learn the basic theory and working principle of each instrument.
  - Know the types, components, and functions of each instrument.
  - Understand the sample preparation, calibration, and operation procedures of each instrument.
  - Be aware of the sources of error, uncertainty, and interference in each instrument.
  - Interpret the data and results obtained from each instrument.
  - Compare and contrast the performance and suitability of different instruments for different analytical tasks.

CO2 Molecular Properties and Surface Tension

- Carbon dioxide (CO2) is a linear molecule with a bond length of 116.3 pm and a bond angle of 180 degrees.
- CO2 is a non-polar molecule with a dipole moment of zero and a symmetrical distribution of electrons.
- CO2 has a molar mass of 44.0095 g/mol and a critical point at 304.25 K and 7.38 MPa.
- CO2 is a gas at standard temperature and pressure, but it can be liquefied or solidified under certain conditions.
- The density and specific weight of CO2 depend on the temperature and pressure, and can be calculated using online tools or tables.
- The surface tension of CO2 is the energy required to increase the surface area of the liquid phase due to intermolecular forces.

- The surface tension of CO_2 decreases with increasing temperature and pressure, as the intermolecular forces become weaker .
- The surface tension of CO_2 can be affected by the presence of solutes, such as monoethanolamine (MEA), which can act as surfactants and reduce the surface tension .
- The surface tension of CO_2 can be measured using various methods, such as the pendant drop method, the Wilhelmy plate method, or the capillary rise method .

Viscosity

- Viscosity is the property of a fluid that measures its resistance to flow.
- Viscosity depends on the internal friction between the molecules of the fluid, which is affected by factors such as temperature, pressure, and composition.
- Viscosity is expressed by the formula:

$$\eta = \frac{F}{A} \cdot \frac{1}{\frac{dv}{dy}}$$

where η is the viscosity, F is the force applied to the fluid, A is the area of contact, dv is the change in velocity, and dy is the change in distance between two layers of the fluid. - The SI unit of viscosity is the pascal-second ($\mathrm{Pa \cdot s}$), which is equivalent to one newton-second per square meter ($\mathrm{N \cdot s/m^2}$). Other common units are the poise (P), which is equal to 0.1 $\mathrm{Pa \cdot s}$, and the centipoise (cP), which is equal to 0.001 $\mathrm{Pa \cdot s}$. - Viscosity can be classified into two types: dynamic viscosity and kinematic viscosity. Dynamic viscosity is the ratio of the shear stress to the shear rate of a fluid, while kinematic viscosity is the ratio of the dynamic viscosity to the density of the fluid. - Viscosity can also be classified into two categories: Newtonian and non-Newtonian fluids. Newtonian fluids are fluids that have a constant viscosity regardless of the shear rate, such as water, air, and oil. Non-Newtonian fluids are fluids that have a variable viscosity depending on the shear rate, such as blood, ketchup, and honey.

K3

- K3 is a type of artificial intelligence that can generate natural language texts from structured data or queries.
- K3 uses a neural network model that is trained on a large corpus of text and data pairs, such as Wikipedia tables and summaries, or SQL queries and natural language answers.
- K3 can generate texts in different languages, domains, and styles, depending on the data source and the user's preferences.
- K3 can also answer questions, summarize information, and generate creative content, such as poems, stories, or jokes, from data or queries.

- K3 is designed to be fast, accurate, and engaging, and to provide useful and relevant information to the user.

CO-3 Measure the hardness and alkalinity of the water. K3

- Hardness and alkalinity are two parameters that indicate the water quality and the suitability of water for various purposes.
- Hardness measures the total amount (concentration) of divalent salts, mainly calcium and magnesium, that can form insoluble deposits or scale in pipes, boilers, and appliances .
- Alkalinity measures the total amount of bases, mainly bicarbonate and carbonate, that can neutralize acids and maintain the pH of water .
- Hardness and alkalinity are related through common ions formed in aquatic systems. The counter-ions associated with the bicarbonate and carbonate fraction of alkalinity are the principal cations responsible for hardness (usually Ca ++ and Mg ++) (Equations 3 and 4).
- Hardness and alkalinity often occur together, because some compounds can contribute both alkalinity and hardness ions.
- Hardness can be classified into two types: temporary and permanent. Temporary hardness, also known as carbonate hardness, is caused by the presence of bicarbonate and carbonate ions that can be removed by boiling or lime treatment. Permanent hardness, also known as non-carbonate hardness, is caused by the presence of other ions, such as sulfate, chloride, and nitrate, that cannot be removed by boiling or lime treatment .
- Alkalinity can be classified into three types: total, carbonate, and bicarbonate. Total alkalinity is the sum of all the bases that can react with acid. Carbonate alkalinity is the amount of bases that can react with strong acid. Bicarbonate alkalinity is the amount of bases that can react with weak acid.
- Hardness and alkalinity can be measured by titration methods using standard solutions of acid and indicators. The choice of acid and indicator depends on the type and range of hardness and alkalinity to be determined .

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic compound with a hydroxyl group (-OH) attached to a benzene ring. It is also known as carbolic acid or phenic acid. Phenol has antiseptic and disinfectant properties and is used in the manufacture of plastics, dyes, explosives, and drugs.

There are several methods to prepare phenol from different sources, such as haloarenes, benzene, cumene, diazonium salts, and coal tar. Some of the common methods are:

- **From haloarenes:** Haloarenes are aromatic compounds with a halogen atom (-X) attached to a benzene ring. Examples of haloarenes are chlorobenzene, bromobenzene, and iodobenzene. When haloarenes are fused with sodium hydroxide at high temperature and pressure, sodium phenoxide is produced. Sodium phenoxide is then acidified with dilute acid or carbon dioxide to give phenol. This reaction is known as the Dow process or the Fries process.

image

The mechanism of this reaction involves the nucleophilic substitution of the halogen atom by the hydroxide ion, followed by the elimination of water and the formation of a phenoxide ion. The phenoxide ion is then protonated by the acid or carbon dioxide to form phenol.

- **From benzene:** Benzene is an aromatic compound with a six-carbon ring and three double bonds. Benzene can be converted to phenol by two methods: by sulphonation and by Raschig's process.

  - **By sulphonation:** Benzene is treated with concentrated sulphuric acid to form benzene sulphonic acid. Benzene sulphonic acid is then fused with sodium hydroxide at high temperature to form sodium phenoxide. Sodium phenoxide is then acidified with dilute acid or carbon dioxide to give phenol.

    image

    The mechanism of this reaction involves the electrophilic substitution of a hydrogen atom by the sulphonic acid group, followed by the nucleophilic substitution of the sulphonic acid group by the hydroxide ion, and the protonation of the phenoxide ion.

  - **By Raschig's process:** Benzene is reacted with hydrogen chloride and air in the presence of a catalyst (such as copper chloride) at high temperature to form chlorobenzene. Chlorobenzene is then hydrolyzed with steam to form phenol and hydrochloric acid.

    image

    The mechanism of this reaction involves the electrophilic substitution of a hydrogen atom by the chlorine atom, followed by the nucleophilic substitution of the chlorine atom by the hydroxyl group.

- **From cumene:** Cumene is an aromatic compound with an isopropyl group (-CH(CH3)2) attached to a benzene ring. Cumene is oxidized by air in the presence of a catalyst (such as cobalt acetate) to form cumene hydroperoxide. Cumene hydroperoxide is then cleaved by heating with dilute acid to form phenol and acetone. This reaction is known as the cumene process or the Hock process.

  image

  The mechanism of this reaction involves the free radical oxidation of cumene to form cumene hydroperoxide, followed by the acid-catalyzed cleavage of the peroxide bond and the formation of phenol and acetone.

- **From diazonium salts:** Diazonium salts are organic compounds with a diazonium group (-N2+) attached to an aromatic ring. Examples of diazonium salts are benzenediazonium chloride, anilinium chloride, and nitrobenzenediazonium chloride. When diazonium salts are treated with water, phenol and nitrogen gas are produced. This reaction is

Formaldehyde

- Formaldehyde is a naturally occurring organic compound with the formula CH2O and structure H−CHO.
- Formaldehyde is a colorless, highly toxic, and flammable gas at room temperature.
- Formaldehyde is an intermediate in the oxidation (or combustion) of methane, as well as of other carbon compounds, e.g. in forest fires, automobile exhaust, and tobacco smoke.
- Formaldehyde is also produced in the atmosphere by the action of sunlight and oxygen on atmospheric methane and other hydrocarbons, and it becomes part of smog.
- Formaldehyde is used widely by industry to manufacture building materials and numerous household products, such as fertilizer, paper, plywood, resins, antiseptics, medicines, and cosmetics .
- Formaldehyde is also used as a food preservative and in embalming fluids.
- Formaldehyde can cause irritation of the skin, eyes, nose, and throat, as well as respiratory and neurological effects .
- Formaldehyde is classified as a human carcinogen by the International Agency for Research on Cancer and the US National Toxicology Program, based on evidence from studies of workers exposed to high levels of formaldehyde and from animal experiments .
- Formaldehyde exposure can be reduced by using products that contain low or no formaldehyde, ventilating indoor spaces, avoiding smoking, and following safety precautions when handling formaldehyde-containing products .

Urea formaldehyde resin

- Urea formaldehyde resin is a synthetic polymer made by reacting urea and formaldehyde in an aqueous solution.
- Urea formaldehyde resin is a thermosetting resin, meaning that it can be cured by heat and pressure to form a hard and durable material.
- Urea formaldehyde resin is widely used as an adhesive for wood products, such as particleboard, plywood, and medium-density fiberboard, as well as for laminates, coatings, and molding compounds.
- Urea formaldehyde resin is also used as a fertilizer, a slow-release nitrogen source for plants.

- Urea formaldehyde resin can release formaldehyde gas during its production, processing, and use, which can cause health effects similar to those of formaldehyde exposure .
- Urea formaldehyde resin can also emit other volatile organic compounds (VOCs), such as methanol, acetaldehyde, and ammonia, which can contribute to indoor air pollution and smog formation.
- Urea formaldehyde resin can be replaced by other resins that have lower or no formaldehyde emissions, such as phenol formaldehyde resin, melamine formaldehyde resin, or soy-based resin .

Adipic acid

- Adipic acid is an organic compound with the formula $C_6H_{10}O_4$ and structure $HOOC-(CH_2)_4-COOH$.
- Adipic acid is a white, crystalline solid that is slightly soluble in water and has a sour taste.
- Adipic acid is mainly used as a precursor for the production of nylon, a synthetic fiber used for clothing, carpets, and other applications.
- Adipic acid is also used as a food additive, a flavoring agent, a gelling agent, and a leavening agent.
- Adipic acid can be produced from various raw materials, such as cyclohexane, benzene, phenol, or glucose.
- Adipic acid can also be produced biologically by certain bacteria or fungi, which can convert glucose or other sugars into adipic acid.
- Adipic acid can cause irritation of the skin, eyes, and respiratory tract if inhaled, ingested, or contacted.
- Adipic acid can also cause metabolic acidosis, a condition where the blood becomes too acidic, if ingested in large amounts.
- Adipic acid can be degraded by microorganisms in the environment, and it is not considered to be a persistent or bioaccumulative pollutant.

Paracetamol

- Paracetamol is an organic compound with the formula $C_8H_9NO_2$ and structure $HOC_6H_4NHCOCH_3$.
- Paracetamol is a white, crystalline powder that is slightly soluble in water

K3

- K3 is a synthetic, artificially produced form of vitamin K that doesn't occur naturally .
- K3 is also known as menadione .
- K3 may be converted into K2, a natural form of vitamin K, in the liver.
- K3 is important for blood clotting and bone health, as it helps activate proteins that regulate these processes.
- K3 has demonstrated anticancer and antibacterial properties in test-tube studies, but it has also been associated with toxicity and adverse effects

in humans and animals .

- K3 is not approved for use as a dietary supplement or a food additive in the United States, Canada, and the European Union, due to its potential risks .
- K3 is sometimes used as a medication to treat vitamin K deficiency or bleeding disorders, but only under medical supervision and in low doses .
- K3 is also the name of a schedule form for the U.S. Internal Revenue Service, which reports the partner's share of income, deductions, credits, etc. from international sources.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction is a proportionality factor that relates the reaction rate to the concentrations of the reactants.
- The rate constant depends on the temperature, the activation energy, and the frequency factor of the reaction.
- The rate constant can have different units depending on the order of the reaction.
- For a termolecular reaction, where three molecules collide simultaneously, the rate constant is denoted by k3 and has the units of L2/mol2 · s.
- To estimate the rate constant of a termolecular reaction, one can use the Arrhenius equation: k3 = Ae^(-Ea/RT), where A is the frequency factor, Ea is the activation energy, R is the gas constant, and T is the temperature.
- Alternatively, one can use experimental data to plot the natural logarithm of the rate constant versus the inverse of the temperature, and obtain the slope and the intercept of the linear regression. The slope is equal to -Ea/R and the intercept is equal to ln A.
- If the reaction takes place in multiple steps, the overall rate constant can be expressed as a function of the individual rate constants of each step . For example, if the reaction is A + B + C -> D and it occurs in three steps, the overall rate constant k can be written as k = k2k1k3/(k1 + k2 + k3).

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of

The topic is

- The topic is a general term for the subject or theme of a conversation, text, or presentation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be broad or narrow, depending on the scope and depth of the information or discussion.
- The topic can be chosen by the speaker, the writer, the audience, or the context of the situation.

- The topic can be related to various fields of knowledge, such as science, history, art, literature, politics, etc.
- The topic can be informative, persuasive, argumentative, descriptive, narrative, or explanatory, depending on the purpose and mode of communication.
- The topic can be supported by evidence, examples, facts, opinions, arguments, or anecdotes, depending on the type and quality of information.
- The topic can be organized in different ways, such as by chronological order, spatial order, order of importance, cause and effect, problem and solution, comparison and contrast, etc.
- The topic can be revised or refined, based on the feedback, research, or analysis of the information or discussion.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn about. You have not greeted me, so I will not introduce myself further.

Some possible ways to continue the conversation are:

- Ask me to explain a concept or a term related to the topic.
- Ask me to give you an example or an exercise related to the topic.
- Ask me to summarize or review the main points of the topic.
- Ask me to test your knowledge or understanding of the topic.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed explicitly or implicitly, depending on the context and the audience.
- A topic can be chosen by the speaker or writer, or assigned by a teacher, editor, or moderator.
- A topic can be informative, persuasive, entertaining, or expressive, depending on the genre and mode of the communication.
- A topic can be related to other topics, such as subtopics, main topics, or background topics, depending on the structure and organization of the communication.
- A topic can be developed and supported by various types of evidence, such as facts, opinions, examples, statistics, anecdotes, or quotations, depending on the argument and the style of the communication.
- A topic can be analyzed and evaluated by various criteria, such as relevance, clarity, coherence, accuracy, validity, or originality, depending on the purpose and the standards of the communication.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have not specified the topic you want to write about.

Please enter the topic name after the colon:

The topic is:

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

- History
- Science
- Mathematics
- Literature
- Art
- Music
- Sports
- Technology
- Philosophy
- Politics

Please choose a topic or type your own.

Page 31 of 40

- The topic of this page is **the structure and function of the human eye**.
- The human eye is a complex organ that allows us to see the world around us. It consists of several parts, each with a specific role in vision.
- The main parts of the human eye are:
  - The **cornea**: a transparent layer that covers the front of the eye and refracts (bends) light rays entering the eye.
  - The **iris**: a circular muscle that controls the size of the pupil, the opening in the center of the eye that regulates the amount of light entering the eye.
  - The **lens**: a flexible structure behind the pupil that changes shape to focus light rays onto the retina, the light-sensitive layer at the back of the eye.
  - The **ciliary body**: a ring of muscles that attaches to the lens and helps adjust its shape and position.
  - The **aqueous humor**: a clear fluid that fills the space between the cornea and the lens and provides nourishment and pressure to the eye.
  - The **vitreous humor**: a clear gel that fills the space between the lens and the retina and helps maintain the shape of the eye.
  - The **retina**: a thin layer of cells that lines the back of the eye and converts light into electrical signals that are sent to the brain via the

optic nerve.

- The **macula**: a small area in the center of the retina that contains a high concentration of cone cells, which are responsible for color vision and sharp central vision.
- The **fovea**: a tiny pit in the center of the macula that contains only cone cells and provides the highest visual acuity (resolution).
- The **optic nerve**: a bundle of nerve fibers that carries the electrical signals from the retina to the brain, where they are processed and interpreted as images.
- The **optic disc**: a circular area where the optic nerve exits the eye and where there are no photoreceptor cells, creating a blind spot in the visual field.
- The **sclera**: a tough white layer that covers the outside of the eye and protects it from injury and infection.
- The **conjunctiva**: a thin membrane that covers the sclera and the inner surface of the eyelids and lubricates the eye with tears.
- The **eyelids**: movable folds of skin that cover the eye and protect it from dust, debris, and excessive light.
- The **eyelashes**: short hairs that grow on the edges of the eyelids and help prevent foreign particles from entering the eye.
- The **eyebrows**: curved rows of hairs above the eyes that help prevent sweat and rain from dripping into the eye.
- The **tear glands**: small glands that produce tears, which moisten, cleanse, and protect the eye from infection and irritation.
- The **tear ducts**: small tubes that drain the excess tears from the eye to the nose.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be chosen by the speaker, the writer, the listener, the reader, or a combination of them.
- A topic can be influenced by the context, the audience, the genre, the tone, and the goal of the communication.
- A topic can be developed by providing details, examples, evidence, arguments, or opinions that support or challenge it.
- A topic can be organized by using different strategies, such as chronological order, spatial order, comparison and contrast, cause and effect, problem and solution, or classification and division.
- A topic can be revised by adding, deleting, rearranging, or clarifying information to make it more clear, coherent, relevant, or persuasive.

ENGINEERING MATHEMATICS LAB KCS

- Engineering Mathematics Lab KCS is a course that aims to provide students with the skills and knowledge of applying mathematical concepts and methods to solve engineering problems.
- The course consists of lectures, labs, and recitations, where students learn and practice various topics such as linear algebra, differential equations, numerical methods, probability, and statistics.
- The course is taught by the faculty of the College of Engineering and Computer Science, and is based on engineering applications taken directly from core engineering courses.
- The course has the following objectives:
  - To enhance the students' mathematical proficiency and confidence in solving engineering problems.
  - To develop the students' ability to use mathematical software and tools for computation, simulation, and visualization.
  - To foster the students' understanding of the connections between mathematics and engineering disciplines.
  - To prepare the students for advanced courses in engineering and mathematics.
- The course has the following outcomes:
  - The students will be able to apply the knowledge of mathematics, science, engineering fundamentals, and engineering specialization to the solution of complex engineering problems.
  - The students will be able to identify, formulate, review research literature, and analyze complex engineering problems using principles of mathematics.
  - The students will be able to use appropriate techniques, skills, and modern engineering tools, including mathematical software, for modeling, simulation, and analysis of engineering systems.
  - The students will be able to communicate effectively the mathematical solutions and results of engineering problems using oral, written, and graphical modes.
- The course covers the following topics:
  - Linear algebra: matrix operations, systems of linear equations, determinants, eigenvalues and eigenvectors, linear transformations, orthogonality, and least squares.
  - Differential equations: first-order and higher-order equations, linear and nonlinear equations, homogeneous and nonhomogeneous equations, initial and boundary value problems, Laplace transforms, and series solutions.
  - Numerical methods: error analysis, root finding, interpolation, numerical differentiation and integration, numerical solutions of ordinary and partial differential equations, and optimization.
  - Probability and statistics: basic concepts of probability, random variables, probability distributions, sampling, estimation, hypothesis

testing, and regression.

- The course requires the students to complete the following assignments and assessments:
  - Lab exercises: the students have to perform various tasks using mathematical software such as MATLAB, Mathematica, or Python, and submit their codes and outputs .
  - Quizzes: the students have to answer multiple-choice or short-answer questions on the topics covered in the lectures and labs.
  - Midterm and final exams: the students have to solve problems on the topics covered in the course using analytical and numerical methods.

Course Objectives:

- To understand the basic concepts and principles of artificial intelligence (AI) and its applications.
- To learn the fundamental techniques and algorithms for solving AI problems, such as search, knowledge representation, reasoning, planning, learning, and natural language processing.
- To develop the skills and abilities to design, implement, and evaluate AI systems and agents using various programming languages and tools.
- To explore the ethical, social, and legal implications of AI and its impact on human society and the environment.

To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- Analytical instruments can be used for various purposes, such as quality control, research and development, environmental monitoring, forensic analysis, clinical diagnosis, etc.
- Analytical instruments can provide qualitative or quantitative information about the samples, such as their identity, composition, structure, concentration, purity, etc.
- Analytical instruments can have different components, such as sources, detectors, transducers, amplifiers, filters, displays, etc., that work together to produce, manipulate, or record signals from the samples.
- Analytical instruments can have different characteristics, such as sensitivity, selectivity, accuracy, precision, resolution, range, etc., that affect their performance and reliability.
- Analytical instruments can have different advantages and limitations, such

as speed, cost, complexity, portability, safety, etc., that determine their suitability and feasibility for different applications.

Analysis of Chloride Content, Hardness, and Alkalinity

- Chloride content is the amount of chloride ions (Cl^-) present in a water sample. Chloride ions are derived from sources such as seawater intrusion, industrial effluents, agricultural runoff, and domestic sewage. Chloride content is an important parameter for assessing the quality of drinking water, irrigation water, and industrial water.
- Hardness is the measure of the concentration of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions in a water sample. These ions are mainly derived from the dissolution of minerals such as limestone, dolomite, and gypsum. Hardness affects the properties of water such as taste, corrosiveness, scaling, and soap consumption. Hardness is expressed in terms of equivalent calcium carbonate ($CaCO_3$) concentration.
- Alkalinity is the measure of the capacity of a water sample to neutralize acids. Alkalinity is mainly due to the presence of bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), and hydroxide (OH^-) ions in water. Alkalinity is important for maintaining the pH and buffering capacity of water. Alkalinity is also expressed in terms of equivalent calcium carbonate ($CaCO_3$) concentration.

Methods of Analysis

- Chloride content can be determined by titrating a water sample with a standard solution of silver nitrate ($AgNO_3$) using potassium chromate (K_2CrO_4) as an indicator. The end point of the titration is marked by the formation of a reddish-brown precipitate of silver chromate (Ag_2CrO_4). The chloride content can be calculated from the volume and concentration of the silver nitrate solution used.
- Hardness can be determined by titrating a water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T (EBT) as an indicator. The end point of the titration is marked by the change of color of the indicator from wine red to blue. The hardness can be calculated from the volume and concentration of the EDTA solution used.
- Alkalinity can be determined by titrating a water sample with a standard solution of sulfuric acid (H_2SO_4) using phenolphthalein and methyl orange as indicators. The end point of the titration with phenolphthalein is marked by the change of color of the indicator from pink to colorless. The end point of the titration with methyl orange is marked by the change of color of the indicator from yellow to orange. The alkalinity can be calculated from the volume and concentration of the sulfuric acid solution

used.

Water

Water is a substance that is essential for life on Earth. It has the following properties and characteristics:

- Water is a liquid at standard temperature and pressure, but it can also exist as a solid (ice) or a gas (water vapor).
- Water is composed of two hydrogen atoms and one oxygen atom, forming the chemical formula H2O.
- Water is a polar molecule, meaning that it has a slight positive charge on one end and a slight negative charge on the other. This makes water able to form hydrogen bonds with other water molecules and other polar substances.
- Water has a high specific heat capacity, meaning that it can absorb or release a lot of heat without changing its temperature much. This helps regulate the temperature of the Earth and living organisms.
- Water has a high heat of vaporization, meaning that it takes a lot of energy to change water from liquid to gas. This allows water to evaporate from the surface of the Earth and form clouds, which then return water to the land as precipitation.
- Water has a high surface tension, meaning that it tends to stick together and form droplets or bubbles. This allows water to form capillary action, which helps water move through narrow spaces such as plant roots or blood vessels.
- Water is a universal solvent, meaning that it can dissolve many other substances. This makes water a medium for chemical reactions and transport of nutrients and wastes in living organisms.
- Water is transparent, meaning that it allows light to pass through it. This enables photosynthesis, the process by which plants use light to make food from water and carbon dioxide.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of how acidic or basic a solution is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. pH = -log[H+]
- The pH scale ranges from 0 to 14, with 7 being neutral, below 7 being acidic, and above 7 being basic. The lower the pH, the higher the acidity, and the higher the pH, the lower the acidity.

- pH can be measured using indicators, such as litmus paper, phenolphthalein, or universal indicator, which change color depending on the acidity or basicity of the solution. pH can also be measured using a pH meter, which is an electronic device that measures the voltage difference between a reference electrode and a pH-sensitive electrode immersed in the solution.
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is defined as the force per unit length required to stretch or break the surface of a liquid. Surface tension = F/L
- Surface tension causes liquids to form spherical droplets, to minimize their surface area. It also allows some objects, such as insects or needles, to float or rest on the surface of a liquid, if they do not break the surface.
- Surface tension can be measured using a force gauge, which measures the force required to pull a ring or a blade out of the surface of a liquid. Surface tension can also be measured using a stalagmometer, which measures the number of drops of a liquid that fall from a capillary tube under gravity.
- Viscosity is a measure of how resistant a fluid is to flow. It is defined as the ratio of the shear stress to the shear rate of a fluid. Viscosity = tau/gamma
- The shear stress is the force per unit area applied tangentially to the fluid, and the shear rate is the rate of change of velocity per unit distance in the direction perpendicular to the force. The higher the viscosity, the more force is required to make the fluid flow, and the slower the fluid flows.
- Viscosity can be measured using a viscometer, which is a device that measures the flow rate of a fluid through a tube or a gap under a known pressure difference or force. Viscosity can also be measured using a rheometer, which is a device that measures the relationship between the shear stress and the shear rate of a fluid under various conditions.

A liquid

- A liquid is one of the four states of matter, along with solid, gas, and plasma.
- A liquid is a fluid that has a definite volume but no fixed shape. It can flow and take the shape of its container.
- A liquid is composed of molecules that are loosely bound by intermolecular forces, such as hydrogen bonds or van der Waals forces.
- A liquid can undergo phase transitions to a solid (freezing) or a gas (evaporation or boiling) depending on the temperature and pressure.
- A liquid has a higher density than a gas, but a lower density than a solid. It can be compressed slightly, but not as much as a gas.
- A liquid can transmit sound, heat, and electricity, but its properties may vary depending on the type and concentration of the liquid.
- A liquid can exhibit various phenomena, such as surface tension, viscosity,

capillarity, and diffusion.

Preparation of Different Resins

Resins are organic compounds that are solid or semi-solid at room temperature and have a high molecular weight. Resins are widely used in various industries, such as paints, adhesives, plastics, rubber, and cosmetics.

There are two main types of resins: natural and synthetic. Natural resins are derived from plants or animals, such as rosin, shellac, amber, and lac. Synthetic resins are artificially produced by chemical reactions, such as phenol-formaldehyde, urea-formaldehyde, epoxy, and polyester.

The preparation of different resins depends on the type and properties of the resin, the raw materials, and the desired application. Some general steps involved in the preparation of resins are:

- Selection of raw materials: The raw materials for resin synthesis should be pure, stable, and compatible with each other. The raw materials may include monomers, solvents, catalysts, fillers, additives, and modifiers.
- Polymerization: The raw materials are reacted together to form long chains of molecules called polymers. Polymerization can be initiated by heat, light, or chemical agents. The degree of polymerization determines the molecular weight and viscosity of the resin.
- Curing: The polymer chains are cross-linked to form a three-dimensional network that gives the resin strength and durability. Curing can be done by heat, radiation, or chemical agents. The curing time and temperature affect the hardness and flexibility of the resin.
- Finishing: The cured resin is processed to obtain the final product. Finishing may include grinding, filtering, drying, molding, coating, or blending. The finishing process depends on the shape, size, and quality of the resin product.

Synthesis of Adipic Acid

- Adipic acid is a dicarboxylic acid with the formula C6H10O4. It is mainly used for the production of nylon, polyurethanes, and other polymers.
- The conventional synthesis of adipic acid involves the oxidation of cyclohexane or a mixture of cyclohexanol and cyclohexanone (KA oil) with nitric acid .
- This process generates large amounts of nitrous oxide (N2O), a greenhouse gas and an ozone-depleting agent, as well as nitric acid waste .
- Therefore, alternative routes for the synthesis of adipic acid from renewable and less harmful sources have been explored, such as glucose and its derivatives.

- Some of the methods for the synthesis of adipic acid from glucose and its derivatives are:
  - Oxidation of glucose to glucaric acid, followed by dehydration and hydrogenation to adipic acid.
  - Hydrogenolysis of levulinic acid, a product of glucose hydrolysis, to -valerolactone, followed by oxidation and ring-opening to adipic acid.
  - Oxidation of 2,5-dimethylfuran, a product of glucose dehydration, to 2,5-furandicarboxylic acid, followed by hydrogenation to adipic acid.
  - Oxidation of muconic acid, a product of glucose oxidation, to adipic acid.
- These methods require the development of efficient and selective catalysts, such as metal nanoparticles, metal oxides, and metal-organic frameworks, to achieve high yields and selectivities of adipic acid.

Acid and Paracetamol by Conventional and Green Route

- Paracetamol, also known as acetaminophen, is a common painkiller and antipyretic drug that is derived from phenol.
- The conventional route for the synthesis of paracetamol involves the following steps :
  - Nitration of phenol with nitric acid to form p-nitrophenol.
  - Reduction of p-nitrophenol with tin and hydrochloric acid to form p-aminophenol.
  - Acetylation of p-aminophenol with acetic anhydride to form paracetamol.
- The conventional route has some drawbacks, such as:
  - Low yield and selectivity of p-nitrophenol due to the formation of o-nitrophenol as a by-product.
  - Generation of toxic and corrosive wastes, such as nitric acid, tin chloride, and acetic acid.
  - Use of hazardous and expensive reagents, such as nitric acid, tin, and acetic anhydride.
- The green route for the synthesis of paracetamol involves the following steps :
  - Hydroxylation of phenol with hydrogen peroxide and a catalyst, such as iron(III) chloride, to form hydroquinone.
  - Amidation of hydroquinone with acetic acid and a catalyst, such as sodium acetate, under microwave irradiation to form paracetamol.
- The green route has some advantages, such as:
  - High yield and selectivity of paracetamol due to the direct conversion of phenol to paracetamol without intermediate steps.
  - Reduction of environmental impact and cost due to the use of benign and cheap reagents, such as hydrogen peroxide, acetic acid, and

sodium acetate.
- Enhancement of reaction rate and efficiency due to the use of microwave irradiation as a non-conventional energy source.

List of Experiments

- An experiment is a scientific procedure that is carried out to test a hypothesis, observe a phenomenon, or measure a variable.
- Experiments usually involve manipulating one or more independent variables and measuring their effects on one or more dependent variables.
- Experiments can be classified into different types based on their design, purpose, or outcome, such as:
  - Controlled experiments: These are experiments where all the variables except the independent variable are kept constant or controlled, and the effect of the independent variable on the dependent variable is measured.
  - Randomized experiments: These are experiments where the subjects or units are randomly assigned to different groups or treatments, and the outcomes are compared across the groups or treatments.
  - Natural experiments: These are experiments where the independent variable is not manipulated by the experimenter, but varies naturally due to some external factor, and the effect of the independent variable on the dependent variable is observed.
  - Quasi-experiments: These are experiments where the independent variable is not randomly assigned, but is determined by some preexisting condition, and the effect of the independent variable on the dependent variable is estimated.
  - Field experiments: These are experiments that are conducted in a natural setting, rather than in a laboratory or a controlled environment.
  - Laboratory experiments: These are experiments that are conducted in a controlled and artificial setting, such as a laboratory or a simulation.
  - Single-subject experiments: These are experiments where only one subject or unit is studied over time, and the effect of different treatments or conditions on the subject or unit is measured.
  - Factorial experiments: These are experiments where two or more independent variables are manipulated simultaneously, and the effects of each independent variable and their interactions on the dependent variable are measured.
  - Longitudinal experiments: These are experiments where the same subjects or units are followed over time, and the changes in the dependent variable are measured at different points in time.

- Cross-sectional experiments: These are experiments where different subjects or units are measured at the same point in time, and the differences in the dependent variable are compared across the subjects or units.

Calibration of Analytical Equipment and Apparatus

- Calibration is the process of evaluating and adjusting the precision and accuracy of measurement equipment.
- Calibration is important for ensuring the quality and reliability of analytical results, as well as the safety and performance of the equipment and apparatus.
- Calibration involves comparing the output of the equipment or apparatus with a known reference standard, and applying corrections if necessary.
- Calibration should be done according to standard operating procedures (SOPs) that specify the frequency, method, criteria, and documentation of calibration.
- The frequency of calibration depends on several factors, such as the usage, stability, and criticality of the equipment or apparatus.
- The method of calibration may vary depending on the type and function of the equipment or apparatus, but generally involves the following steps:
  - Selecting a suitable reference standard that has a higher accuracy and traceability than the equipment or apparatus to be calibrated.
  - Preparing the equipment or apparatus for calibration by ensuring that it is clean, stable, and in good working condition.
  - Performing the calibration by measuring the output of the equipment or apparatus at different input values, and comparing it with the reference standard.
  - Calculating the error, uncertainty, and correction factors for the equipment or apparatus based on the calibration results.
  - Adjusting the equipment or apparatus if the error exceeds the acceptable limits, or applying the correction factors to the measurements if the error is within the limits.
  - Documenting the calibration results and procedures, and issuing a calibration certificate or report.
- Calibration should be verified periodically by using check standards or quality control samples to ensure that the equipment or apparatus remains within the calibration limits.
- Calibration should be repeated or revised whenever there is a change in the equipment or apparatus, such as repair, modification, relocation, or damage.

Determination of Hardness of Water by EDTA Method

- Hardness of water is a measure of the concentration of calcium and magnesium ions in water.
- EDTA (Ethylenediaminetetraacetic acid) is a chelating agent that can form stable complexes with metal ions, such as Ca2+ and Mg2+.
- EDTA titration is a method of determining the hardness of water by titrating a water sample with a standard solution of EDTA, using a suitable indicator to detect the end point.
- The indicator used is Eriochrome Black T (EBT), which is a dye that changes color from wine red to blue when it binds to metal ions.
- The pH of the water sample is adjusted to 10 ± 0.1 by adding ammonia buffer, to ensure that all the metal ions are in the form of free ions and not precipitated as hydroxides.
- The procedure for the EDTA titration method is as follows:
  - Take a sample volume of 20 mL (V mL) of water and transfer it to a conical flask.
  - Dilute the sample to 40 mL by adding 20 mL of distilled water.
  - Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1.
  - Add 1 or 2 drops of EBT indicator solution. The solution will turn wine red due to the presence of metal ions.
  - Titrate the sample with a standard solution of EDTA of known concentration (C M) from a burette, with constant stirring, until the color changes from wine red to blue. This indicates that all the metal ions have been complexed by EDTA and the indicator is free.
  - Note the volume of EDTA used (V E mL).
- The calculation of hardness of water by EDTA titration is based on the following equation:
  - C M * V E = C H * V
  - Where C M is the concentration of EDTA in mg/L as CaCO3, V E is the volume of EDTA used in mL, C H is the concentration of hardness in water in mg/L as CaCO3, and V is the volume of water sample in mL.
  - The concentration of EDTA can be determined by standardizing it against a known solution of calcium carbonate or calcium chloride.
  - The concentration of hardness in water can be calculated by rearranging the equation as follows:
  - C H = (C M * V E) / V

- The result can be reported as total hardness or calcium hardness, depending on whether the water sample contains only calcium ions or both calcium and magnesium ions.

Determination of Alkalinity of Water Sample

- Alkalinity is a measure of the capacity of water to neutralize acids. It is mainly caused by the presence of bicarbonate, carbonate, and hydroxide ions in water.
- Alkalinity is important for aquatic life, water quality, and water treatment. It affects the pH, buffering capacity, and corrosion potential of water.
- Alkalinity can be determined by titrating a water sample with a standard acid solution, such as sulfuric acid, and measuring the volume of acid required to reach a specific pH endpoint.
- The endpoint pH depends on the type and concentration of alkalinity present in the water sample. For example, if the water sample contains only bicarbonate alkalinity, the endpoint pH is 4.5. If the water sample contains both bicarbonate and carbonate alkalinity, the endpoint pH is 8.3. If the water sample contains hydroxide alkalinity, the endpoint pH is 10.0.
- The titration curve of alkalinity shows the change in pH as acid is added to the water sample. The curve has different regions corresponding to different forms of alkalinity. The equivalence point is the point where the amount of acid added is equal to the amount of alkalinity in the water sample.
- The alkalinity of the water sample can be calculated from the volume and concentration of the acid solution and the endpoint pH using the following formula:

  Alkalinity (mg/L as CaCO3) = (Volume of acid (mL) x Normality of acid (eq/L) x 50,000) / Volume of water sample (mL)
- The alkalinity of the water sample can also be expressed in terms of milliequivalents per liter (meq/L) or millimoles per liter (mmol/L) by dividing the alkalinity in mg/L as CaCO3 by 50.

Determination of pH by titrimetric method

- Titrimetric method is a technique of quantitative analysis that involves measuring the volume of a solution of known concentration (titrant) that is required to react completely with a solution of unknown concentration (analyte).

- pH is a measure of the acidity or alkalinity of a solution, defined as the negative logarithm of the hydrogen ion concentration: pH = -log[H+].
- Determination of pH by titrimetric method involves using a pH meter to monitor the change in pH of the analyte solution as the titrant is added gradually.
- A pH meter consists of a glass electrode that responds to the hydrogen ion activity in the solution, and a reference electrode that provides a constant potential. The difference in potential between the two electrodes is proportional to the pH of the solution.
- A plot of the pH of the analyte solution versus the volume of the titrant added is called a titration curve. The titration curve can be used to determine the endpoint or equivalence point of the titration, where the analyte and the titrant are stoichiometrically equivalent.
- The shape and features of the titration curve depend on the type and strength of the acid and base involved in the titration. For example, a strong acid-strong base titration curve has a sharp rise in pH near the equivalence point, while a weak acid-strong base titration curve has a more gradual rise in pH and a buffer region before the equivalence point.

Determination of surface tension of given liquid

- Surface tension is a property of liquids that arises from the intermolecular forces between the molecules at the surface layer.
- Surface tension causes the liquid surface to behave like a stretched elastic membrane, minimizing its area and forming spherical drops.
- Surface tension can be measured by various methods, such as capillary rise method, drop weight method, ring method, etc.
- In this experiment, we will use the ring method, also known as the Du Nouy method, to determine the surface tension of a given liquid.
- The ring method involves suspending a thin metal ring from a torsion wire and dipping it into the liquid surface. The ring is then pulled out slowly by rotating the torsion wire until the liquid film breaks.
- The force required to break the liquid film is measured by a balance or a dynamometer attached to the torsion wire. The surface tension of the liquid is then calculated by using the following formula:

$$\gamma = \frac{F}{4\pi R}$$

where γ is the surface tension, F is the force, and R is the radius of the ring.

- The procedure of the experiment is as follows:
  - Clean the ring and the torsion wire with a suitable solvent and dry them.

  - Attach the ring to the torsion wire and suspend it from a stand. Adjust the height of the ring so that it is slightly above the liquid surface in a beaker.
  - Connect the other end of the torsion wire to a balance or a dynamometer and calibrate it to zero.
  - Lower the ring into the liquid surface and let it wet completely. Then raise it slowly by rotating the torsion wire until the liquid film breaks. Note the reading of the balance or the dynamometer at the moment of breakage. This is the force required to break the liquid film.
  - Repeat the experiment for at least five times and take the average value of the force.
  - Measure the diameter of the ring with a vernier caliper and calculate its radius.
  - Use the formula to calculate the surface tension of the liquid and record the result.
  - Repeat the experiment for different liquids and compare the results.

Determination of Viscosity of a given liquid by viscometer

- Viscosity is a measure of the resistance of a fluid to flow or deform under an applied force. It is also known as the internal friction of a fluid.
- A viscometer is a device that measures the viscosity of a fluid by observing its flow through a narrow tube or a capillary.
- There are different types of viscometers, such as Ostwald viscometer, Ubbelohde viscometer, Cannon-Fenske viscometer, etc. Each viscometer has a specific design and calibration for measuring the viscosity of different fluids.
- The basic principle of a viscometer is to measure the time taken by a known volume of fluid to flow through a capillary of known dimensions under a constant pressure difference. The viscosity of the fluid is then calculated using the following formula:

$$\eta = \frac{\pi r^4 \Delta P t}{8VL}$$

where η is the viscosity of the fluid, r is the radius of the capillary, ΔP is the pressure difference across the capillary, t is the time of flow, V is the volume of fluid, and L is the length of the capillary.

- The procedure for determining the viscosity of a given liquid by viscometer is as follows:
  - Clean the viscometer with a suitable solvent and dry it with air.
  - Fill the viscometer with the liquid to be tested and immerse it in a water bath at a constant temperature.

- Allow the liquid to reach the thermal equilibrium with the water bath.
- Open the valve at the upper end of the viscometer and allow the liquid to flow through the capillary until it reaches the lower mark on the viscometer.
- Close the valve and start a stopwatch.
- Open the valve again and record the time taken by the liquid to flow from the lower mark to the upper mark on the viscometer.
- Repeat the experiment at least three times and take the average time of flow.
- Calculate the viscosity of the liquid using the formula given above and the known parameters of the viscometer.
- Report the result with the appropriate units and the uncertainty.

Determination of the strength of Ferrous ammonium sulfate using external indicator

- Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron(II) sulfate and ammonium sulfate, with the formula (NH4)2Fe(SO4)2 · 6H2O.
- It is often used as a standard solution for titrations of reducing agents, such as oxalic acid, potassium permanganate, or potassium dichromate.
- The strength of a ferrous ammonium sulfate solution can be determined by titrating it with a standard solution of potassium permanganate, using an external indicator such as starch or potassium ferricyanide.
- The reaction between ferrous ammonium sulfate and potassium permanganate is:

(NH4)2Fe(SO4)2 + KMnO4 + H2SO4 -> K2SO4 + (NH4)2SO4 + Fe2(SO4)3 + MnSO4 + H2O

- The reaction is a redox reaction, in which iron(II) is oxidized to iron(III) and manganese(VII) is reduced to manganese(II).
- The end point of the titration is marked by the appearance of a faint pink color in the solution, due to the excess of potassium permanganate.
- The external indicator is used to detect the end point more accurately, as the color change is very faint and may be affected by the presence of other substances in the solution.
- Starch is used as an external indicator by adding it to the solution when the color becomes pale yellow. Starch forms a blue-black complex with iodine, which is produced by the reaction of potassium permanganate and potassium iodide. The end point is reached when the blue-black color disappears.
- Potassium ferricyanide is used as an external indicator by adding it to a separate test tube containing a small amount of the solution. Potassium

ferricyanide forms a blue precipitate with iron(II), which is present in the solution before the end point. The end point is reached when the blue precipitate no longer forms in the test tube.

- The strength of the ferrous ammonium sulfate solution can be calculated by using the following formula:

Strength of ferrous ammonium sulfate = (Volume of KMnO4 x Normality of KMnO4 x Equivalent weight of ferrous ammonium sulfate) / Volume of ferrous ammonium sulfate

- The equivalent weight of ferrous ammonium sulfate is 196 g/mol, as one mole of it reacts with one mole of potassium permanganate.

Determination of the strength of Potassium dichromate using internal indicator

- Potassium dichromate is a primary standard substance that can be used to determine the strength of an unknown solution of a reducing agent, such as ferrous ammonium sulphate, by titration.
- The titration is carried out in acidic medium, usually by adding dilute sulphuric acid to the solution.
- The internal indicator used for this titration is potassium ferricyanide, which forms a blue precipitate of Prussian blue with the excess of ferrous ions at the end point.
- The procedure of the titration is as follows:
  - Weigh accurately about 0.5 g of potassium dichromate and dissolve it in 100 mL of distilled water in a volumetric flask. This is the standard solution of potassium dichromate.
  - Pipette out 20 mL of the standard solution into a conical flask and add about 10 mL of dilute sulphuric acid.
  - Add a few drops of potassium ferricyanide solution as the internal indicator.
  - Titrate the solution with the unknown solution of ferrous ammonium sulphate from a burette until a faint blue colour appears, indicating the end point.
  - Note down the burette reading and repeat the titration until concordant results are obtained.
  - Calculate the strength of the unknown solution of ferrous ammonium sulphate using the following formula:

    * Strength of ferrous ammonium sulphate = (Strength of potassium dichromate x Volume of potassium dichromate x 10) / (Volume of ferrous ammonium sulphate x 3.92)
    * Where 3.92 is the molar ratio of potassium dichromate to ferrous ammonium sulphate in the reaction and 10 is the conversion factor from mL to L.

Determination of Available Chlorine in Bleaching Powder

- Bleaching powder is a white powder that contains calcium hypochlorite [Ca(OCl)2] and calcium chloride (CaCl2).
- Bleaching powder is used as a disinfectant and a bleaching agent.
- Available chlorine is the amount of chlorine that can be liberated from bleaching powder when it reacts with dilute acids.
- Available chlorine is expressed as a percentage of the mass of bleaching powder.
- The formula for calculating the percentage of available chlorine is:

  % available chlorine = (mass of chlorine liberated / mass of bleaching powder) x 100
- To determine the available chlorine in bleaching powder, the following steps are followed:
  1. Weigh a sample of bleaching powder and dissolve it in water.
  2. Transfer a measured volume of the solution to a flask and add excess of acetic acid and potassium iodide.
  3. The acetic acid reacts with the calcium hypochlorite and liberates chlorine gas.
  4. The chlorine gas reacts with the potassium iodide and liberates iodine.
  5. Titrate the iodine solution with sodium thiosulfate solution using starch as an indicator.
  6. The end point is the disappearance of the blue color of the starch-iodine complex.
  7. Calculate the mass of chlorine liberated using the volume and concentration of sodium thiosulfate solution.
  8. Calculate the percentage of available chlorine using the formula.

Determination of chloride content in water sample

- Chloride ions in water can be determined by various methods, such as argentometric method, potentiometric method, or ion chromatography method.
- Argentometric method is based on the precipitation and titration of chloride ions with silver nitrate solution using an indicator such as potassium chromate or potassium thiocyanate .
- Potentiometric method is based on the measurement of the potential of an electrochemical cell consisting of a chloride specific ion electrode and a reference electrode.
- Ion chromatography method is based on the separation and detection of chloride ions by passing the water sample through a column of ion exchange resin and a conductivity detector.

Procedure for chloride content determination in water by argentometric method (Mohr's method)

1. Take 50 mL of water sample (V mL) and dilute to 100 mL in a volumetric flask.
2. If the sample is highly coloured, add 3 mL of aluminium hydroxide and shake well, allow to settle, filter, wash and collect filtrate.
3. Bring the sample pH to 7-8 by adding acid or alkali .
4. Add 1 mL of indicator (potassium chromate) and mix well .
5. Titrate the solution against standard silver nitrate solution until a reddish brown precipitate of silver chromate is obtained .
6. Note the volume of silver nitrate solution used (V1 mL) .
7. Repeat the titration with a blank solution of distilled water and note the volume of silver nitrate solution used (V2 mL) .

Calculation of chloride content in water by argentometric method (Mohr's method)

- The reaction between chloride ions and silver nitrate is given by :

Cl- + Ag+ → AgCl(s)

- The reaction between chromate ions and silver nitrate is given by :

2 CrO42- + 2 Ag+ → Ag2CrO4(s)

- The concentration of chloride ions in the water sample can be calculated by the following formula :

C = (V1 - V2) x N x 35.45 / V

where,

C = concentration of chloride ions in mg/L

V1 = volume of silver nitrate solution used for the sample in mL

V2 = volume of silver nitrate solution used for the blank in mL

N = normality of silver nitrate solution

35.45 = equivalent weight of chloride ion

V = volume of water sample taken for titration in mL

- The percentage of chloride ions in the water sample can be calculated by the following formula:

% Cl = C x 100 / 1000

where,

% Cl = percentage of chloride ions in the water sample

C = concentration of chloride ions in mg/L

1000 = conversion factor from mg/L to g/L

Preparation of Phenol Formaldehyde (PF) Resin

- Phenol formaldehyde resin is a type of thermosetting polymer that is obtained by the condensation reaction of phenol and formaldehyde.
- Phenol and formaldehyde can react in different ratios to produce different types of PF resins, such as novolacs, resoles, and co-condensed resins.
- The reaction can be catalyzed by either acid or base, depending on the desired properties and applications of the resin.
- The general steps for the preparation of PF resin are as follows :
    1. Weigh the appropriate quantity of the raw materials (phenol and formaldehyde) before mixing them in a reaction vessel.
    2. Add the catalyst (acid or base) to the mixture and heat it under controlled conditions of temperature, time, and pressure.
    3. Monitor the reaction progress by measuring the viscosity, pH, and water content of the resin.
    4. Stop the reaction when the desired degree of polymerization and cross-linking is achieved.
    5. Cool and filter the resin to remove any impurities or unreacted materials.
    6. Store the resin in a suitable container or use it for further processing or application.
- Some examples of the preparation of specific types of PF resins are:

- Novolac resin: This is a linear and thermoplastic resin that is prepared by reacting phenol and formaldehyde in a molar ratio of 1:0.8 to 1:1.5, using an acid catalyst such as hydrochloric acid or sulfuric acid . The reaction is stopped before the resin becomes insoluble and infusible. Novolac resin can be cured by adding a cross-linking agent such as hexamethylenetetramine (HMTA) and heating it above 150°C.
- Resole resin: This is a branched and thermosetting resin that is prepared by reacting phenol and formaldehyde in a molar ratio of 1:1.5 to 1:3, using a base catalyst such as ammonia or sodium hydroxide . The reaction is continued until the resin becomes insoluble and infusible. Resole resin can be cured by heating it above 120°C without any additional cross-linking agent.
- Co-condensed resin: This is a modified resin that is prepared by reacting phenol and formaldehyde with another phenolic compound such as urea, melamine, or resorcinol . The reaction is carried out using either an acid or a base catalyst, depending on the type of comonomer. Co-condensed resin can have improved properties such as water resistance, flame retardance, and adhesion.

Preparation of Urea Formaldehyde (UF) Resin

- Urea formaldehyde (UF) resin is a thermosetting polymer that is widely used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- UF resin is synthesized by the condensation reaction of urea and formaldehyde in the presence of a base or an acid catalyst .
- The reaction can be carried out in one or more steps, depending on the desired properties and applications of the resin .
- The general steps for the preparation of UF resin are as follows:
  - Step 1: Preparation of methylolureas. An aqueous solution of formaldehyde (more than 50%) and urea are mixed in a molar ratio of 2-3:1 at pH 6-11. The mixture is heated to at least 80°C and a base catalyst such as ammonia or pyridine is added. The reaction produces various methylolureas, which are intermediates with one or more hydroxymethyl groups attached to the urea molecule.
  - Step 2: Condensation of methylolureas. The methylolureas undergo further condensation reactions to form linear or branched oligomers with methylene or methylene ether linkages. The reaction can be controlled by adjusting the pH, temperature, and catalyst concentration. An acid catalyst such as sulfuric acid or oxalic acid is added to lower the pH to 0.5-3.5. The reaction can be stopped at this stage to obtain a low molecular weight resin that can be used as an adhesive.

- Step 3: Curing of the resin. The resin can be cured by heating, pressure, or catalysts to form a cross-linked network with high molecular weight and high strength. The curing process releases formaldehyde as a by-product, which can cause health and environmental problems. Therefore, the resin should have a low formaldehyde content and a high degree of curing to minimize the emissions. The curing conditions depend on the type and amount of catalyst, the resin composition, and the substrate.

- The properties and applications of UF resin depend on the reaction conditions, the catalysts, and the additives used in the synthesis . Some of the advantages of UF resin are:
  - It is inexpensive and easy to produce.
  - It has high tensile strength, hardness, and heat resistance.
  - It has good adhesion to wood and other materials.
  - It can be modified with fillers, extenders, plasticizers, and other resins to improve its performance.
- Some of the disadvantages of UF resin are:
  - It is brittle and prone to cracking.
  - It has poor water resistance and durability.
  - It emits formaldehyde during curing and service, which can cause health and environmental hazards.
  - It is sensitive to pH and temperature changes.

Preparation of Adipic acid / Paracetamol

Adipic acid

- Adipic acid is the organic compound with the formula (CH2)4(COOH)2. It is the most important dicarboxylic acid, mainly used as a precursor for the production of nylon.
- Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .
- The steps involved in the oxidation of KA oil are:
  - Formation of nitrosonium ion (NO+) from nitric acid and sulfuric acid.
  - Nitrosation of cyclohexanone and cyclohexanol to form nitrosocyclohexanone and nitrosocyclohexanol, respectively.
  - Rearrangement of nitrosocyclohexanone and nitrosocyclohexanol to form 6-nitrosohexanone and 6-nitrosohexanol, respectively.
  - Oxidation of 6-nitrosohexanone and 6-nitrosohexanol to form 6-nitrohexanoic acid and 6-nitrohexanol, respectively.

  - Hydrolysis of 6-nitrohexanoic acid and 6-nitrohexanol to form adipic acid and nitrous acid, respectively.
  - Recycle of nitrous acid to nitric acid by oxidation with air.

Paracetamol

- Paracetamol is the common name for acetaminophen, an analgesic and antipyretic drug with the formula C8H9NO2. It is widely used for the relief of pain and fever.
- Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulfuric acid as catalyst.
- The reaction involved in the synthesis of paracetamol is:
  - Acetylation of p-aminophenol with acetic anhydride to form N-acetyl-p-aminophenol (paracetamol) and acetic acid.
  - The mechanism of the reaction is:
    * Protonation of the oxygen atom of acetic anhydride by sulfuric acid to form a good leaving group.
    * Nucleophilic attack of the nitrogen atom of p-aminophenol on the carbonyl carbon of acetic anhydride, forming a tetrahedral intermediate.
    * Elimination of the leaving group (acetic acid) and deprotonation of the nitrogen atom, forming paracetamol and regenerating sulfuric acid.
- Paracetamol is available as tablets, capsules, syrups, suspensions, and injections, with various additives and excipients.

Determination of Cell Conductance of a Solution

- Cell conductance of a solution is the reciprocal of its resistance, i.e., G = 1/R, where G is the conductance and R is the resistance.
- The resistance of a solution is measured by using a special type of cell known as a conductivity cell and a Wheatstone bridge.
- A conductivity cell consists of two electrodes of known area and distance, immersed in the solution whose conductance is to be measured.
- The cell constant, C, is defined as the ratio of the distance between the electrodes, l, to the area of cross-section of the electrodes, a, i.e., C = l/a. The unit of cell constant is m^{-1}.
- The cell constant can be determined by measuring the resistance of a cell filled with a solution of known conductivity, such as KCl (usually 0.0200 M).
- The conductivity of a solution, K, is defined as the conductance of a unit volume of the solution, i.e., K = G x V, where V is the volume of the solution. The unit of conductivity is S m^{-1} or ohm^{-1} m^{-1}.
- The conductivity of a solution depends on the concentration, the number

of charges, and the mobilities of the ions it contains.

- The conductivity of a solution can be calculated from the experimental resistance by using the formula: K = C/R, where C is the cell constant and R is the resistance.

Determination of Rate constant of hydrolysis of esters

- Esters are organic compounds that have the general formula RCOOR', where R and R' are alkyl or aryl groups.
- Esters can be hydrolyzed by water in the presence of an acid or a base to form a carboxylic acid and an alcohol.
- The hydrolysis of esters is a first-order reaction, which means that the rate of the reaction depends only on the concentration of the ester.
- The rate constant of the hydrolysis of esters can be determined by measuring the change in concentration of the ester or the products over time, using a suitable method such as titration, spectrophotometry, or conductometry.
- The rate constant of the hydrolysis of esters can be affected by factors such as temperature, pH, and catalysts.
- The rate constant of the hydrolysis of esters can be used to compare the reactivity of different esters, or to estimate the shelf life of ester-containing products.

Element detection and identification of functional groups in organic compounds

- Organic compounds are those that contain carbon atoms, usually bonded to hydrogen, oxygen, nitrogen, sulfur, or halogens.

- Functional groups are specific groups of atoms or bonds within organic molecules that are responsible for their chemical properties and reactions.

- Element detection is the process of identifying the presence of certain elements in an organic compound, such as carbon, hydrogen, nitrogen, sulfur, halogens, etc.

- Identification of functional groups is the process of determining the type and location of functional groups in an organic compound, such as alcohols, aldehydes, ketones, carboxylic acids, amines, etc.

- Element detection and identification of functional groups are important steps in the qualitative analysis of organic compounds, which helps to determine their structure and properties.

- Some common methods for element detection and identification of functional groups are:
  - Solubility test: This test involves dissolving the organic compound in different solvents, such as water, dilute acid, or dilute base, and observing the solubility or insolubility. This can give a clue about the polarity and acidity or basicity of the functional groups present in the compound.
  - Lassaigne's test: This test involves fusing the organic compound with sodium metal in a test tube, and then dissolving the fused mass in water. This produces sodium salts of the elements present in the compound, which can be detected by various chemical tests. For example, sodium cyanide (NaCN) can be detected by adding ferrous sulfate (FeSO4) and ferric chloride (FeCl3), which gives a Prussian blue precipitate of ferric ferrocyanide (Fe4[Fe(CN)6]3). This indicates the presence of nitrogen in the organic compound.
  - 2,4-dinitrophenylhydrazine (2,4-DNP) test: This test involves adding a small amount of the organic compound to 2,4-DNP reagent, which is a yellow solution of 2,4-dinitrophenylhydrazine in sulfuric acid. If the compound contains a carbonyl group (C=O), such as in aldehydes or ketones, it will react with 2,4-DNP and form a yellow, orange, or red precipitate of 2,4-dinitrophenylhydrazone. The melting point and the color of the precipitate can help to identify the specific carbonyl compound.
  - Tollen's test: This test involves adding a few drops of the organic compound to a freshly prepared solution of Tollen's reagent, which is a silver-ammonia complex ([Ag(NH3)2]+) in aqueous ammonia. If the compound contains an aldehyde group (RCHO), it will reduce the silver ions to metallic silver, which forms a shiny silver mirror on the inner surface of the test tube. This test can distinguish aldehydes from ketones, which do not react with Tollen's reagent.
  - Litmus test: This test involves dipping a strip of blue or red litmus paper in the organic compound or its aqueous solution. If the compound contains a carboxylic acid group (RCOOH), it will turn blue litmus paper red, indicating its acidic nature. If the compound contains an amine group (RNH2), it will turn red litmus paper blue, indicating its basic nature.
- These are some of the common methods for element detection and identification of functional groups in organic compounds. There are many other methods and tests that can be used for more specific and accurate analysis.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- This note implies that the instructor has the authority and flexibility to select the experiments that are most suitable for the course objectives, the available resources, and the students' interests and abilities.
- The note also implies that the instructor can modify or replace up to two experiments from the list provided above, if they find them too difficult, too easy, too boring, or too irrelevant for the course.
- The note does not specify which experiments are mandatory or optional, nor does it provide any criteria or guidelines for choosing or changing the experiments. Therefore, the instructor has to use their own judgment and discretion to make the best decisions for the course.
- The note also does not indicate how the experiments will be evaluated or graded, nor does it state how much weight they will have in the final assessment of the course. Therefore, the instructor has to communicate these details clearly to the students before they start the experiments.
- The note may have some advantages and disadvantages for the instructor and the students. Some possible advantages are:
  - The instructor can tailor the experiments to the specific needs and goals of the course, and avoid wasting time and resources on experiments that are not relevant or useful.
  - The instructor can adapt the experiments to the level and pace of the students, and provide them with appropriate challenges and feedback.
  - The instructor can introduce some variety and creativity to the experiments, and make them more engaging and enjoyable for the students.
  - The students can have some input and choice in the experiments they do, and express their preferences and interests to the instructor.
  - The students can learn from different types of experiments, and develop a range of skills and competencies.
- Some possible disadvantages are:
  - The instructor may have to spend more time and effort to select and design the experiments, and ensure that they are aligned with the course outcomes and standards.
  - The instructor may have to deal with some uncertainty and inconsistency in the experiments, and adjust their expectations and plans accordingly.
  - The instructor may have to face some challenges and difficulties in managing and supervising the experiments, and ensuring their quality and safety.
  - The students may have to cope with some ambiguity and variability in the experiments, and adapt their strategies and methods accordingly.

- The students may have to deal with some confusion and frustration in the experiments, and seek clarification and guidance from the instructor.

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, content, activities, and assessments.
- Course outcomes are measurable, observable, and achievable within the scope and duration of the course.
- Course outcomes are written in clear and concise language, using action verbs that indicate the level of cognitive skills required.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course.
- Course outcomes are used to guide the course design, delivery, and evaluation.
- Course outcomes are reviewed and revised periodically to ensure their relevance and effectiveness.

Upon completion of the course the student should be able to:

- Define the basic concepts and terminology of artificial intelligence, such as agents, environments, rationality, search, knowledge, inference, planning, learning, and natural language processing.
- Apply various search algorithms, such as uninformed search, informed search, local search, and adversarial search, to solve problems that can be modeled as state-space graphs or game trees.
- Represent and manipulate knowledge using propositional logic, first-order logic, and other formalisms, such as semantic networks, frames, and ontologies.
- Use inference methods, such as resolution, forward chaining, and backward chaining, to derive new facts from existing knowledge bases.
- Design and implement planning systems that can generate sequences of actions to achieve goals in dynamic and uncertain domains.
- Explain the principles and techniques of machine learning, such as supervised learning, unsupervised learning, reinforcement learning, and deep learning, and apply them to various tasks, such as classification, clustering, regression, and reinforcement learning.
- Analyze and evaluate the performance and limitations of different artificial intelligence methods and systems using appropriate metrics and benchmarks.

- Demonstrate the ability to use artificial intelligence tools and libraries, such as Python, NumPy, SciPy, scikit-learn, TensorFlow, PyTorch, and NLTK, to implement and test artificial intelligence algorithms and applications.
- Communicate and present artificial intelligence concepts, methods, results, and ethical issues effectively using oral, written, and visual means.

Course Outcomes Bloom's

- Course outcomes are statements that describe what students should be able to do or demonstrate after completing a course.
- Bloom's taxonomy is a framework for classifying educational objectives into six levels of cognitive complexity: remember, understand, apply, analyze, evaluate, and create.
- Course outcomes should be aligned with Bloom's taxonomy to ensure that students achieve the intended learning outcomes and develop higher-order thinking skills.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART).
- Course outcomes should be written using action verbs that correspond to the appropriate level of Bloom's taxonomy.
- Course outcomes should be assessed using appropriate methods and criteria that match the level of Bloom's taxonomy.
- Course outcomes should be communicated to students at the beginning of the course and throughout the course to guide their learning and self-assessment.

Level

- A level is a tool or device that is used to measure or indicate whether a surface is horizontal (level) or vertical (plumb).
- A level consists of a horizontal tube or vial that contains a liquid (usually alcohol or water) and an air bubble. The position of the bubble indicates the alignment of the surface relative to the earth's gravity.
- A level can be used for various purposes, such as construction, carpentry, engineering, surveying, landscaping, photography, etc.
- There are different types of levels, such as spirit levels, laser levels, digital levels, water levels, etc. Each type has its own advantages and disadvantages depending on the application and accuracy required.
- A level can be calibrated by placing it on a flat surface and adjusting it until the bubble is centered in the vial. Then, the level can be rotated 180 degrees and checked again. If the bubble is still centered, the level is accurate. If not, the level can be adjusted by turning the screws or knobs on the vial until the bubble is centered. This process can be repeated until

the level is calibrated.

CO-1 Get an understanding of the use of different analytical instruments. K3

- Analytical instruments are devices that are used to measure, analyze, and identify the chemical, physical, or biological properties of a sample or a substance.
- Analytical instruments can be classified into different categories based on the principle or technique of analysis, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, thermal analysis, microscopy, etc.
- Analytical instruments have various applications in different fields, such as pharmaceutical, food, environmental, clinical, forensic, life science, chemical, etc.
- Analytical instruments can help in determining the composition, structure, purity, quality, quantity, and function of a sample or a substance.
- Analytical instruments can also help in detecting, identifying, and quantifying the presence of contaminants, impurities, pollutants, pathogens, or other substances of interest in a sample or a substance.
- Analytical instruments can provide accurate, reliable, and reproducible results that can be used for research, development, testing, quality control, diagnosis, monitoring, or regulation purposes.

CO2 Molecular Properties and Surface Tension

- Carbon dioxide (CO2) is a linear molecule with a bond length of 116.3 pm and a bond angle of 180 degrees.
- CO2 is a non-polar molecule with a dipole moment of zero and a symmetrical distribution of electrons.
- CO2 has a molecular weight of 44.0095 g/mol and a molar volume of 22.26 L/mol at standard temperature and pressure.
- CO2 has a critical temperature of 304.13 K and a critical pressure of 7.38 MPa.
- CO2 has a triple point at 216.58 K and 0.517 MPa, where it can exist as a solid, liquid, and gas.
- CO2 has a sublimation point at 194.65 K and 1 atm, where it can transition directly from solid to gas.
- CO2 has a density of 1.977 kg/m3 at 0 °C and 1 atm, and 1.842 kg/m3 at 25 °C and 1 atm.
- CO2 has a specific heat capacity of 0.844 kJ/kg K at 25 °C and 1 atm, and 0.657 kJ/kg K at 100 °C and 1 atm.
- CO2 has a thermal conductivity of 0.0166 W/m K at 0 °C and 1 atm, and 0.0253 W/m K at 100 °C and 1 atm.

- CO_2 has a viscosity of 0.0148 mPa s at 0 °C and 1 atm, and 0.0152 mPa s at 25 °C and 1 atm.
- CO_2 has a surface tension of 0.0012 N/m at its triple point, and 0.0 N/m at its critical point.
- Surface tension is the energy, or work, required to increase the surface area of a liquid due to intermolecular forces.
- Surface tension depends on the nature of the liquid, the temperature, and the presence of solutes or surfactants.
- CO_2 has a low surface tension compared to water (0.072 N/m at 25 °C) because it has weaker intermolecular forces (London dispersion forces) than water (hydrogen bonds).
- CO_2 can form aqueous solutions with various solvents, such as monoethanolamine (MEA), which can affect its density and surface tension .
- The density and surface tension of CO_2-loaded aqueous MEA solutions depend on the amine mass ratio, the CO_2 loading, and the temperature .
- The density of CO_2-loaded aqueous MEA solutions decreases with increasing CO_2 loading and temperature, and increases with increasing amine mass ratio .
- The surface tension of CO_2-loaded aqueous MEA solutions decreases with increasing CO_2 loading and temperature, and increases with increasing amine mass ratio .
- The surface tension of CO_2-loaded aqueous MEA solutions is lower than that of pure water at the same temperature .

Viscosity

- Viscosity is the property of a fluid that measures its resistance to flow.

- Viscosity depends on the internal friction between the molecules of the fluid, which is affected by factors such as temperature, pressure, and composition.

- Viscosity is expressed by the formula:

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  = F / (A * dv / dx)
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  where is the viscosity, F is the force required to move a layer of fluid, A is the area of the layer, dv is the change in velocity, and dx is the distance between the layers.

- The SI unit of viscosity is the pascal-second (Pa · s), which is equivalent to the newton-second per square meter (N · s/m2). Another common unit is the poise (P), which is equal to 0.1 Pa · s.

- Viscosity can be classified into two types: dynamic viscosity and kinematic viscosity. Dynamic viscosity is the ratio of the shear stress to the shear rate of a fluid, while kinematic viscosity is the ratio of the dynamic viscosity to the density of the fluid.

- Viscosity can be measured by various methods, such as capillary viscometers, rotational viscometers, falling ball viscometers, and rheometers. These devices measure the flow rate, torque, or resistance of a fluid under different conditions.

- Viscosity is an important property in many applications, such as lubrication, transport, mixing, coating, and polymer processing. Viscosity affects the performance, efficiency, and safety of these processes.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa, which means it does not grant permanent residence or a green card. However, K3 visa holders can apply for adjustment of status to become permanent residents after their immigrant visa petition is approved.
- K3 visa is valid for two years and can be renewed if the immigrant visa petition or the adjustment of status application is still pending.
- K3 visa holders can work and study in the United States with the appropriate authorization from the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa holders can also travel outside the United States and re-enter with a valid visa and passport.
- To qualify for a K3 visa, the foreign spouse must be legally married to a U.S. citizen and have a pending Form I-130, Petition for Alien Relative, filed by the U.S. citizen spouse with the USCIS.
- The U.S. citizen spouse must also file a Form I-129F, Petition for Alien Fiancé(e), with the USCIS and request that the foreign spouse be issued a K3 visa.
- The foreign spouse must apply for a K3 visa at the U.S. consulate or embassy in their home country and undergo a medical examination and an interview.
- The foreign spouse must also provide evidence of their marriage to the U.S. citizen, such as a marriage certificate, photos, and letters.
- The foreign spouse must also demonstrate that they intend to enter the United States to reunite with their U.S. citizen spouse and not for any other purpose.

CO-3 Measure the hardness and alkalinity of the water. K3

- Hardness of water is the measure of the amount of calcium and magnesium salts dissolved in it. It is expressed in terms of equivalent calcium carbonate (CaCO3).
- Alkalinity of water is the measure of the capacity of water to neutralize acids. It is mainly due to the presence of bicarbonates, carbonates and hydroxides of calcium, magnesium and other metals. It is also expressed in terms of equivalent calcium carbonate (CaCO3).
- The hardness and alkalinity of water are important parameters for various applications, such as drinking water, industrial water, boiler water, irrigation water, etc.
- The hardness and alkalinity of water can be measured by titration methods using standard solutions of acids and bases.
- The hardness of water can be measured by two methods: total hardness and permanent hardness.
  - Total hardness is the sum of calcium and magnesium hardness. It can be measured by titrating a sample of water with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T as an indicator. The end point is marked by a color change from wine red to blue.
  - Permanent hardness is the hardness that remains after boiling the water. It is mainly due to the presence of chlorides and sulfates of calcium and magnesium. It can be measured by titrating a boiled sample of water with a standard solution of sodium carbonate using methyl orange as an indicator. The end point is marked by a color change from yellow to orange.
- The alkalinity of water can be measured by two methods: total alkalinity and phenolphthalein alkalinity.
  - Total alkalinity is the sum of all the bases that can react with acids. It can be measured by titrating a sample of water with a standard solution of sulfuric acid using phenolphthalein and methyl orange as indicators. The end point for phenolphthalein alkalinity is marked by a color change from pink to colorless. The end point for total alkalinity is marked by a color change from yellow to orange.
  - Phenolphthalein alkalinity is the alkalinity due to the presence of hydroxides and half of the carbonates. It can be measured by titrating a sample of water with a standard solution of sulfuric acid using phenolphthalein as an indicator. The end point is marked by a color change from pink to colorless.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic compound with a hydroxyl group (-OH) attached to a benzene ring. Phenol has the chemical formula C_6H_5OH and is also known as carbolic acid. Phenol is a weak acid and can form phenoxide ions by losing a proton from the hydroxyl group.

There are several methods to prepare phenol from different starting materials. Some of the common methods are:

- **From haloarenes:** Haloarenes are aromatic compounds with a halogen atom (-X) attached to a benzene ring. For example, chlorobenzene is a haloarene with a chlorine atom (-Cl) attached to a benzene ring. Haloarenes can be converted to phenols by fusing them with sodium hydroxide (NaOH) at high temperature and pressure. This reaction is called the Dow process. For example, chlorobenzene can be converted to phenol by the following reaction:

  $C_6H_5Cl + NaOH \rightarrow C_6H_5ONa + NaCl$

  $C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

  The sodium phenoxide (C_6H_5ONa) formed in the first step is hydrolyzed to phenol in the second step.

- **From benzene sulphonic acid:** Benzene sulphonic acid is an aromatic compound with a sulphonic acid group ($-SO_3H$) attached to a benzene ring. Benzene sulphonic acid can be prepared by the sulphonation of benzene with concentrated sulphuric acid (H_2SO_4). Benzene sulphonic acid can be converted to phenol by fusing it with sodium hydroxide (NaOH) at high temperature. This reaction is called the Raschig process. For example, benzene sulphonic acid can be converted to phenol by the following reaction:

  $C_6H_5SO_3H + NaOH \rightarrow C_6H_5ONa + NaHSO_4$

  $C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

  The sodium phenoxide (C_6H_5ONa) formed in the first step is hydrolyzed to phenol in the second step.

- **From cumene:** Cumene is an aromatic compound with an isopropyl group ($-CH(CH_3)_2$) attached to a benzene ring. Cumene can be prepared by the alkylation of benzene with propene. Cumene can be converted to phenol by the oxidation of cumene with air in the presence of a catalyst. This reaction is called the cumene process. For example, cumene can be converted to phenol by the following reaction:

  $C_6H_5CH(CH_3)_2 + O_2 \rightarrow C_6H_5OH + (CH_3)_2CO$

The phenol and acetone ($(CH_3)_2CO$) formed in this reaction can be separated by fractional distillation.

- **From diazonium salts:** Diazonium salts are compounds with a diazonium group ($-N_2^+$) attached to an aromatic ring. Diazonium salts can be prepared by the diazotization of aromatic amines with nitrous acid (HNO_2). Diazonium salts can be converted to phenols by the hydrolysis of diazonium salts with water. For example, benzenediazonium chloride can be converted to phenol by the following reaction:

  $C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + N_2 + HCl$

  The phenol and nitrogen gas (N_2) formed in this reaction can be separated by fractional distillation.

Formaldehyde and Urea-Formaldehyde Resin

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Formaldehyde is also a naturally occurring substance in the environment. It is produced by plants, animals, and humans as part of normal metabolism.
- Formaldehyde can cause health problems when inhaled or when it comes into contact with the skin, eyes, or mucous membranes. It can irritate the respiratory tract, cause allergic reactions, and increase the risk of some cancers.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer that is produced from urea and formaldehyde.
- UF is widely used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- UF has advantages such as low cost, high strength, water resistance, and fast curing.
- UF also has disadvantages such as high formaldehyde emission, poor weatherability, and low thermal stability.
- UF can be modified by adding melamine, polyvinyl alcohol, or adipic acid dihydrazide to reduce the free formaldehyde content and improve the performance of the resin.
- UF often shows the appearance of crystal domains, which are influenced by the synthesis conditions, the molecular structure, and the curing process of the resin .
- The crystallinity of UF can affect its mechanical, thermal, and optical properties .

Adipic Acid

- Adipic acid is a white, crystalline organic compound that has the formula $C_6H_{10}O_4$.
- Adipic acid is mainly used as a precursor for the production of nylon 6,6, a synthetic polymer that is widely used in textiles, carpets, plastics, and coatings.
- Adipic acid can also be used as a food additive, a flavoring agent, a plasticizer, a lubricant, a corrosion inhibitor, and a component of some pharmaceuticals.
- Adipic acid can be synthesized from various raw materials, such as cyclohexane, benzene, phenol, glucose, or biobased feedstocks.
- Adipic acid is a dicarboxylic acid, which means it has two carboxyl groups (-COOH) at each end of the molecule.
- Adipic acid can undergo various reactions, such as esterification, amidation, hydrogenation, oxidation, and polymerization.
- Adipic acid is a biodegradable and nontoxic substance, but it can cause skin irritation and eye damage if handled improperly.

Paracetamol

- Paracetamol, also known as acetaminophen, is a common over-the-counter medication that is used to treat pain and fever.
- Paracetamol belongs to a class of drugs called analgesics (pain relievers) and antipyretics (fever reducers).
- Paracetamol works by inhibiting the synthesis of prostaglandins, which are chemical messengers that mediate inflammation, pain, and fever in the body.
- Paracetamol is usually taken orally, but it can also be administered intravenously, rectally, or topically.
- Paracetamol is generally safe and effective when used as directed, but it can cause serious liver damage if taken in excess or in combination with alcohol or other drugs.
- Paracetamol can also cause allergic reactions, skin rashes, blood disorders, and kidney problems in some people.
- Paracetamol is metabolized by the liver and excreted by the kidneys.
- Paracetamol has the chemical formula $C_8H_9NO_2$ and the molecular weight of 151.16 g/mol.
- Paracetamol has a simple structure, consisting of a benzene ring with a hydroxyl group (-OH) and an amide group ($-NHCOCH_3$) attached to it.
- Paracetamol can be synthesized from phenol or p-aminophenol by various methods, such as acetylation, nitration, or reduction.

: https://pubmed.ncbi.nlm.nih.gov/336681

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa process is not completed within that time.
- K3 visa holders are eligible to apply for work authorization and travel permission while in the United States.
- K3 visa holders are also eligible to adjust their status to permanent residents once their immigrant visa petition is approved and a visa number is available.
- K3 visa is one of the options for spouses of U.S. citizens who wish to reunite with their partners in the United States, but it is not the only option. Depending on the circumstances, some spouses may prefer to apply for a CR1 or IR1 visa, which are immigrant visas that grant permanent residency upon arrival in the United States.

CO-5 Estimate the rate constant of reaction K3

- The rate constant of a reaction is a proportionality factor that relates the reaction rate to the concentrations of the reactants.
- The rate constant depends on the temperature, the activation energy, and the frequency factor of the reaction.
- The rate constant can have different units depending on the order of the reaction.
- For a termolecular reaction, such as A + B + C $\rightarrow$ products, the reaction rate is described by r = k3[A][B][C], where k3 is a termolecular rate constant.
- To estimate the rate constant of a termolecular reaction, one can use the Arrhenius equation: k3 = A3 exp(-Ea3/RT), where A3 is the frequency factor, Ea3 is the activation energy, R is the gas constant, and T is the temperature.
- Alternatively, one can use experimental data to plot the natural logarithm of the rate constant versus the inverse of the temperature, and obtain the slope and the intercept of the linear regression. The slope is equal to -Ea3/R and the intercept is equal to ln(A3).
- For example, a reaction takes place in three steps with an individual rate constant and activation energy, as given below :

Step	Rate constant	Activation energy
1	k1	Ea1 = 180 kJ/mol
2	k2	Ea2 = 80 kJ/mol
3	k3	Ea3 = 50 kJ/mol

- The overall rate constant k is given by k = k2k1k3.
- If the temperature is 300 K, the frequency factors are A1 = 10^13 s^-1, A2 = 10^9 L/mol s, and A3 = 10^6 $L^{2/mol}$2 s, then the rate constant of the third step can be estimated as:

k3 = A3 exp(-Ea3/RT) = 10^6 $L^{2/mol}$2 s * exp(-50000 J/mol / 8.314 J/mol K * 300 K) = 0.0036 $L^{2/mol}$2 s

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

- History
- Science
- Mathematics
- Literature
- Art
- Music
- Sports
- Politics
- Philosophy
- Psychology

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- Psychology

Please choose a topic or type your own.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

History

- The major events and periods of human history, such as ancient civilizations, world wars, revolutions, etc.
- The causes and effects of historical changes, such as political, social, economic, cultural, etc.
- The similarities and differences between different regions, cultures, and times in history.

Science

- The natural phenomena and laws that govern the physical and biological world, such as gravity, evolution, genetics, etc.
- The methods and tools of scientific inquiry and discovery, such as observation, experimentation, hypothesis, etc.
- The applications and implications of scientific knowledge and technology, such as medicine, engineering, environment, etc.

Mathematics

- The abstract concepts and structures that underlie the logic and reasoning of numbers, shapes, patterns, etc.
- The rules and operations that manipulate and transform mathematical objects, such as addition, multiplication, differentiation, etc.
- The problems and solutions that involve mathematical thinking and skills, such as algebra, geometry, calculus, etc.

Literature

- The written works and genres that express human thoughts, feelings, and experiences, such as novels, poems, essays, etc.
- The elements and techniques that create and analyze literary texts, such as plot, character, theme, style, etc.
- The contexts and influences that shape and interpret literary texts, such as history, culture, criticism, etc.

Art

- The visual and auditory forms and media that convey human creativity and expression, such as painting, music, sculpture, etc.
- The principles and methods that produce and evaluate artistic works, such as color, harmony, composition, etc.
- The functions and meanings that art serves and communicates, such as aesthetic, symbolic, social, etc.

Page 32 of 40

Topic: Introduction to Artificial Intelligence

- Artificial intelligence (AI) is the branch of computer science that deals with creating machines or software that can perform tasks that normally require human intelligence, such as reasoning, learning, decision making, natural language processing, computer vision, etc.
- AI can be classified into two main categories: weak AI and strong AI. Weak AI, also known as narrow AI, is the type of AI that is designed to perform a specific task or domain, such as playing chess, recognizing faces, or driving a car. Strong AI, also known as general AI or artificial general intelligence (AGI), is the type of AI that can perform any intellectual task that a human can, such as understanding natural language, solving complex problems, or exhibiting creativity and self-awareness.
- AI can also be classified into three main levels: artificial narrow intelligence (ANI), artificial general intelligence (AGI), and artificial superintelligence (ASI). ANI is the current state of AI, where machines can perform specific tasks at or above human level, but cannot generalize to other domains or tasks. AGI is the hypothetical future state of AI, where machines can perform any intellectual task that a human can, and can learn and improve themselves without human intervention. ASI is the hypothetical future state of AI, where machines can surpass human intelligence and capabilities in every aspect, and can potentially pose an existential threat to humanity.
- AI can be achieved by various methods and techniques, such as machine learning, deep learning, neural networks, natural language processing,

computer vision, expert systems, fuzzy logic, genetic algorithms, etc. Each method has its own advantages and limitations, and can be applied to different domains and tasks.
- AI has various applications and benefits for various fields and industries, such as healthcare, education, entertainment, business, security, etc. AI can help improve efficiency, accuracy, productivity, innovation, and quality of life for humans. However, AI also has various challenges and risks, such as ethical, social, legal, and technical issues, such as bias, privacy, accountability, transparency, security, etc. AI can also have negative impacts on human jobs, skills, values, and dignity. Therefore, AI should be developed and used responsibly and ethically, with respect to human rights and values.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for the main idea, the purpose, or the focus of the text, speech, or conversation.
- The topic can be used to organize, summarize, or evaluate the text, speech, or conversation.
- The topic can be related to other topics by using categories, subtopics, or keywords.

ENGINEERING MATHEMATICS LAB

- Engineering Mathematics Lab is a course that aims to provide students with practical skills and knowledge of various mathematical concepts and methods that are relevant to engineering problems and applications.
- The course covers topics such as linear algebra, differential equations, numerical methods, optimization, statistics, and probability.
- The course consists of lab sessions where students work on computer-based exercises and projects using software tools such as MATLAB, Python, or R.
- The course objectives are to:
  - Enhance the understanding of mathematical theory and its applications in engineering.
  - Develop the ability to formulate, analyze, and solve engineering problems using mathematical tools and techniques.
  - Improve the computational and programming skills of students.
  - Encourage the use of mathematical software for visualization, simulation, and experimentation.

- The course outcomes are to:
  - Demonstrate the knowledge of various mathematical concepts and methods and their relevance to engineering.
  - Apply the appropriate mathematical tools and techniques to model and solve engineering problems.
  - Use the mathematical software to perform numerical computations, data analysis, and graphical representation.
  - Communicate the results and findings of mathematical problems and projects effectively.

Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To develop the skills and techniques for designing, implementing, and evaluating AI systems and agents.
- To explore the ethical, social, and legal implications of AI and its impact on human society and the environment.
- To provide a foundation for further study and research in AI and related fields.

1. To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- Analytical instruments can be used for various purposes, such as quality control, research and development, environmental monitoring, forensic analysis, clinical diagnosis, etc.
- Analytical instruments can provide qualitative or quantitative information about the samples, such as their identity, composition, structure, concentration, purity, etc.
- Analytical instruments can have different components, such as sources, detectors, transducers, signal processors, data analyzers, etc.
- Analytical instruments can have different characteristics, such as sensitivity, selectivity, accuracy, precision, resolution, range, etc.
- Analytical instruments can have different advantages and limitations, such as speed, cost, reliability, complexity, etc.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- Chloride content is the amount of chloride ions (Cl^-) present in water. Chloride ions are derived from various sources such as rocks, soil, seawater, industrial effluents, and domestic sewage. Chloride content is an important parameter for assessing the quality of water for various purposes such as drinking, irrigation, and industrial use. Chloride content can be measured by titrating a known volume of water sample with a standard solution of silver nitrate ($AgNO_3$) using potassium chromate (K_2CrO_4) as an indicator. The end point of the titration is marked by the appearance of a reddish-brown precipitate of silver chromate (Ag_2CrO_4).

- Hardness of water is the measure of the concentration of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions in water. These ions are mainly derived from the dissolution of carbonate and bicarbonate minerals such as limestone, dolomite, and gypsum. Hardness of water affects the performance of soap and detergents, causes scaling and corrosion of pipes and boilers, and affects the taste and quality of water. Hardness of water can be classified into two types: temporary hardness and permanent hardness. Temporary hardness is caused by the presence of bicarbonate ions (HCO_3^-) and can be removed by boiling the water. Permanent hardness is caused by the presence of sulfate, chloride, and nitrate ions and cannot be removed by boiling. Hardness of water can be measured by titrating a known volume of water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T (EBT) as an indicator. The end point of the titration is marked by the change of color of the indicator from wine red to blue.

- Alkalinity of water is the measure of the capacity of water to neutralize acids. Alkalinity of water is mainly due to the presence of carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), and hydroxide (OH^-) ions in water. These ions are derived from the dissolution of carbonate and bicarbonate minerals, the hydrolysis of salts, and the photosynthesis of aquatic plants. Alkalinity of water affects the pH and buffering capacity of water, the solubility of metals and minerals, and the biological activity of water. Alkalinity of water can be measured by titrating a known volume of water sample with a standard solution of sulfuric acid (H_2SO_4) using phenolphthalein and methyl orange as indicators. The end point of the titration with phenolphthalein is marked by the disappearance of the pink color of the indicator, indicating the neutralization of hydroxide and carbonate ions. The end point of the titration with methyl orange is marked by the change of color of the indicator from yellow to orange, indicating the neutralization of bicarbonate ions. The difference between the two end points gives the phenolphthalein alkalinity (P-alkalinity) and the total alkalinity (T-alkalinity) of the water sample.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic you requested:

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- pH is a measure of how acidic or basic a liquid is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. The pH scale ranges from 0 to 14, where 7 is neutral, lower than 7 is acidic, and higher than 7 is basic. The pH of a liquid can be measured using indicators, such as litmus paper, or pH meters, which are electronic devices that measure the voltage difference between two electrodes immersed in the liquid.
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is expressed as the force per unit length that is needed to stretch or break the surface. Surface tension causes phenomena such as droplets, bubbles, and capillary action. The surface tension of a liquid depends on the intermolecular forces, the temperature, and the presence of surfactants, which are substances that reduce the surface tension. The surface tension of a liquid can be measured using methods such as the ring method, the drop weight method, or the pendant drop method, which involve measuring the force or weight of a ring, a drop, or a pendant of liquid, respectively.
- Viscosity is a measure of how resistant a liquid is to flow. It is defined as the ratio of the shear stress to the shear rate in a liquid. The shear stress is the force per unit area that is applied parallel to the surface of the liquid, and the shear rate is the rate of change of the velocity of the liquid along the direction of the shear stress. The viscosity of a liquid depends on the intermolecular forces, the temperature, and the pressure. The viscosity of a liquid can be measured using methods such as the capillary tube method, the falling ball method, or the rotational viscometer method, which involve measuring the flow rate, the terminal velocity, or the torque of a liquid, respectively.

4. To enable the students to understand about the preparation of different resins.

- Resins are natural or synthetic organic substances that are usually hard, transparent or translucent, and brittle solids or semi-solids.
- Resins are widely used in various industries, such as paints, varnishes, adhesives, plastics, rubber, and pharmaceuticals.
- Resins can be classified into two main types: thermoplastic and thermosetting.
- Thermoplastic resins are resins that can be softened by heating and hardened by cooling, without undergoing any chemical change. They can be repeatedly melted and reshaped. Examples of thermoplastic resins are

polyethylene, polypropylene, polystyrene, and nylon.

- Thermosetting resins are resins that undergo an irreversible chemical reaction when heated, forming a rigid and insoluble network. They cannot be remelted or reshaped. Examples of thermosetting resins are phenol-formaldehyde, urea-formaldehyde, epoxy, and polyester.
- The preparation of different resins depends on the type, source, and properties of the resin. Some general steps involved in the preparation of resins are:
  - Selection of raw materials: The raw materials for resins can be natural or synthetic, such as plant extracts, petroleum, coal, or natural gas. The raw materials should be pure, stable, and suitable for the desired resin.
  - Polymerization: Polymerization is the process of joining small molecules (monomers) to form large molecules (polymers). Polymerization can be initiated by heat, light, catalysts, or other agents. Polymerization can be classified into two types: addition and condensation.
    * Addition polymerization is the process of adding monomers to a growing polymer chain without the elimination of any by-product. Examples of addition polymerization are the formation of polyethylene from ethylene, and polystyrene from styrene.
    * Condensation polymerization is the process of joining monomers to form a polymer chain with the elimination of a small molecule, such as water, alcohol, or ammonia. Examples of condensation polymerization are the formation of phenol-formaldehyde from phenol and formaldehyde, and nylon from hexamethylenediamine and adipic acid.
  - Modification: Modification is the process of altering the properties of the resin by adding or removing certain functional groups, such as hydroxyl, carboxyl, amino, or epoxy groups. Modification can improve the solubility, viscosity, hardness, flexibility, adhesion, or compatibility of the resin. Examples of modification are the formation of alkyd resin from polyester and oil, and epoxy resin from bisphenol A and epichlorohydrin.
  - Blending: Blending is the process of mixing two or more resins to obtain a resin with the desired properties. Blending can enhance the strength, durability, resistance, or appearance of the resin. Examples of blending are the formation of acrylic resin from acrylic and styrene, and rubber from natural and synthetic rubber.
  - Curing: Curing is the process of hardening the resin by applying heat, pressure, or radiation. Curing can cross-link the polymer chains and form a rigid and insoluble network. Curing is essential for thermosetting resins, but optional for thermoplastic resins. Examples of curing are the formation of bakelite from phenol-formaldehyde and heat, and fiberglass from polyester and glass fibers.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

- Adipic acid is a dicarboxylic acid with the formula C6H10O4. It is mainly used as a precursor for the production of nylon.
- Paracetamol is an analgesic and antipyretic drug with the formula C8H9NO2. It is widely used for the relief of pain and fever.
- Conventional synthesis of organic compounds involves the use of harsh chemicals, high temperatures, and long reaction times. These methods often generate large amounts of waste and harmful by-products, which pose environmental and health hazards.
- Green synthesis of organic compounds aims to reduce or eliminate the use of hazardous substances, increase the efficiency and selectivity of the reactions, and minimize the waste and energy consumption. These methods often employ benign solvents, catalysts, and renewable resources, and use mild reaction conditions.
- The conventional synthesis of adipic acid involves the oxidation of cyclohexane or cyclohexanol with nitric acid, which produces large amounts of nitrous oxide, a greenhouse gas. The green synthesis of adipic acid can be achieved by the oxidation of cyclohexene or cyclohexanol with hydrogen peroxide, which produces only water as a by-product.
- The conventional synthesis of paracetamol involves the acetylation of phenol with acetic anhydride, which produces acetic acid as a by-product. The green synthesis of paracetamol can be achieved by the acetylation of phenol with acetic acid in the presence of a solid acid catalyst, such as zeolite, which can be reused and recycled.

LIST OF EXPERIMENTS

- An experiment is a scientific procedure that is carried out to test a hypothesis, observe a phenomenon, or measure a variable.
- Experiments usually involve manipulating one or more independent variables and measuring their effects on one or more dependent variables.
- Experiments can be classified into different types based on their design, purpose, or outcome. Some common types of experiments are:
  - Controlled experiments: These are experiments where all the variables except the independent variable are kept constant or controlled. The aim is to isolate the causal effect of the independent variable on the dependent variable. For example, testing the effect of different fertilizers on plant growth by using the same type of plant, soil, water, and sunlight in different pots.
  - Randomized experiments: These are experiments where the subjects or units are randomly assigned to different groups or treatments. The aim is to reduce the bias or confounding factors that may affect the

results. For example, testing the effect of a new drug on patients by randomly assigning them to either receive the drug or a placebo.
  - Natural experiments: These are experiments where the independent variable is not manipulated by the experimenter, but by some natural or historical event. The aim is to exploit the natural variation in the independent variable to study its effect on the dependent variable. For example, studying the effect of a volcanic eruption on the climate by comparing the temperature and precipitation before and after the eruption.
  - Quasi-experiments: These are experiments where the independent variable is manipulated by the experimenter, but the subjects or units are not randomly assigned to different groups or treatments. The aim is to approximate the causal effect of the independent variable on the dependent variable, but with less internal validity than randomized experiments. For example, testing the effect of a new curriculum on student achievement by using different schools or classes as the groups or treatments.
  - Field experiments: These are experiments that are conducted in a natural or real-world setting, rather than in a laboratory or controlled environment. The aim is to increase the external validity or generalizability of the results, but with less control over the variables and confounding factors. For example, testing the effect of a social media campaign on voter turnout by randomly sending different messages to potential voters.

1. **Calibration of Analytical Equipment and apparatus**

   - Calibration is the process of evaluating and adjusting the precision and accuracy of measurement equipment.
   - Calibration ensures that the instruments produce valid and reliable results that are consistent with the standards and specifications .
   - Calibration is essential for maintaining the quality and safety of analytical products and processes.
   - Calibration should be done according to a predefined schedule and procedure that are based on the manufacturer's recommendations, the frequency of use, the stability of the instrument, and the requirements of the application .
   - Calibration should be performed by qualified and trained personnel using appropriate reference standards and methods .
   - Calibration should be documented and recorded with details of the instrument, the reference standard, the method, the date, the result, and the acceptance criteria .
   - Calibration should be verified periodically by using check standards or quality control samples to monitor the performance and drift of the instrument .
   - Calibration should be repeated or adjusted if the instrument fails to meet

the acceptance criteria or shows signs of malfunction or damage .

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is the measure of the concentration of calcium and magnesium ions in water.
- EDTA (Ethylenediaminetetraacetic acid) is a chelating agent that can form stable complexes with metal ions, such as Ca2+ and Mg2+, by donating its six lone pairs of electrons.
- EDTA titration is a method of determining the hardness of water by adding a known amount of EDTA solution to a water sample until all the metal ions are complexed.
- The end point of the titration is detected by using an indicator, such as Eriochrome Black T (EBT), that changes color when the metal ions are bound by EDTA.
- EBT is a metallochromic indicator that forms a wine-red complex with Ca2+ and Mg2+ at a pH of 10 $\pm$ 0.1. When EDTA is added, it displaces EBT from the metal ions and forms a more stable complex. The free EBT then turns blue, indicating the end point of the titration.
- The procedure for the determination of hardness of water by EDTA titration is as follows :
  - Take a sample volume of 20 ml (V ml) of water in a conical flask.
  - Dilute the sample to 40 ml by adding 20 ml of distilled water.
  - Add 1 ml of ammonia buffer to adjust the pH to 10 $\pm$ 0.1.
  - Add 1 or 2 drops of EBT indicator solution. The solution should turn wine-red due to the presence of metal ions.
  - Titrate the sample with a standard EDTA solution of known concentration (C mol/L) from a burette, with constant stirring, until the color changes from wine-red to blue. Note the volume of EDTA used (B ml).
  - Repeat the titration with a blank solution of distilled water and note the volume of EDTA used (A ml).
  - Calculate the hardness of water using the following formula:

    Hardness of water (mg/L as CaCO3) = (B - A) x C x 1000 x 100 / V

    where B is the volume of EDTA used for the sample, A is the volume of EDTA used for the blank, C is the concentration of EDTA, and V is the volume of the sample.

- The advantages of EDTA titration method are:
  - It is simple, rapid, and accurate.
  - It can determine the total hardness of water in one titration.
  - It can be used for both clear and turbid water samples.
  - It can be modified to determine the individual hardness due to calcium or magnesium by using different indicators or masking agents.
- The disadvantages of EDTA titration method are:
  - It requires a pH adjustment to 10 ± 0.1, which may affect the accuracy of the titration.
  - It may give erroneous results if the water sample contains other metal ions, such as iron, copper, or zinc, that can also form complexes with EDTA.
  - It may not detect the hardness due to bicarbonates, which are weakly complexed by EDTA and may precipitate as carbonates at high pH.

3. Determination of Alkalinity of water sample.

- Alkalinity is the measure of the capacity of water to neutralize acids. It is mainly due to the presence of bicarbonates, carbonates and hydroxides of calcium, magnesium and other metals in water.
- Alkalinity is important for water quality because it affects the pH, corrosion, disinfection and biological processes in water.
- Alkalinity can be determined by titrating a water sample with a standard acid solution, usually sulfuric acid or hydrochloric acid, using a suitable indicator such as phenolphthalein or methyl orange.
- The titration curve of alkalinity shows the change of pH as the acid is added to the water sample. The curve has different regions depending on the types and concentrations of alkaline substances in water.
- The main types of alkalinity are:
  - Phenolphthalein alkalinity (P-alkalinity): It is the amount of acid required to lower the pH of water to 8.3, the endpoint of phenolphthalein indicator. It represents the sum of hydroxides and half of the carbonates in water.
  - Total alkalinity (T-alkalinity): It is the amount of acid required to lower the pH of water to 4.5, the endpoint of methyl orange indicator. It represents the sum of hydroxides, carbonates and bicarbonates in water.
  - Carbonate alkalinity (C-alkalinity): It is the difference between T-alkalinity and P-alkalinity. It represents the amount of carbonates in water.
  - Bicarbonate alkalinity (B-alkalinity): It is the difference between T-alkalinity and C-alkalinity. It represents the amount of bicarbonates in water.
- The procedure for determining the alkalinity of water sample is as follows:

- Take a 100 mL aliquot of the water sample in a conical flask and add a few drops of phenolphthalein indicator. If the color changes to pink, it indicates the presence of P-alkalinity. Record the initial burette reading of the standard acid solution.
- Titrate the water sample with the standard acid solution until the pink color disappears. Record the final burette reading and calculate the volume of acid used. This is the P-alkalinity in mL.
- Add a few drops of methyl orange indicator to the same flask and continue the titration with the standard acid solution until the color changes from yellow to orange. Record the final burette reading and calculate the volume of acid used. This is the T-alkalinity in mL.
- Calculate the C-alkalinity and B-alkalinity by subtracting the P-alkalinity from the T-alkalinity and the C-alkalinity from the T-alkalinity, respectively.
- Express the alkalinity in mg/L as CaCO3 by multiplying the volume of acid used by the normality of the acid and by a factor of 50. The factor of 50 is derived from the equivalent weight of CaCO3 (50 g/mol) and the volume of water sample (100 mL).
- Report the results as P-alkalinity, T-alkalinity, C-alkalinity and B-alkalinity in mg/L as CaCO3.

4. Determination of pH by titrimetric method

- Titrimetric method is a technique of quantitative analysis that involves measuring the volume of a solution of known concentration (titrant) that is required to react completely with a solution of unknown concentration (analyte) .
- pH is a measure of the acidity or alkalinity of a solution, defined as the negative logarithm of the hydrogen ion concentration: pH = -log[H3O+] .
- Determination of pH by titrimetric method involves using a pH meter to monitor the change in pH of the analyte solution as the titrant is added .
- A pH meter consists of a glass electrode that responds to the hydrogen ion activity in the solution, and a reference electrode that provides a constant potential . The difference in potential between the two electrodes is proportional to the pH of the solution .
- A plot of the pH of the analyte solution versus the volume of the titrant added is called a titration curve . The titration curve can be used to determine the endpoint or equivalence point of the titration, where the analyte and the titrant are stoichiometrically equivalent .
- The shape and features of the titration curve depend on the type and strength of the acid and base involved in the titration, as well as the buffer capacity of the solution .
- The titration curve can also be used to calculate the pKa or pKb of a weak acid or base, by finding the pH at the half-equivalence point, where the concentration of the weak acid or base is equal to the concentration of its conjugate base or acid .

- The titration curve can also be used to estimate the error in the pH measurement due to the titration method, by finding the difference between the pH at the endpoint and the pH at the equivalence point .

5. Determination of surface tension of given liquid.

- Surface tension is the property of a liquid that makes its surface behave like a stretched elastic membrane.
- Surface tension is caused by the cohesive forces between the molecules of the liquid, which tend to minimize the surface area of the liquid.
- Surface tension can be measured by various methods, such as the capillary rise method, the drop weight method, the ring method, and the stalagmometer method.
- In this experiment, we will use the stalagmometer method to determine the surface tension of a given liquid.
- A stalagmometer is a device that consists of a glass tube with a bulb at one end and a fine nozzle at the other end.
- The stalagmometer is filled with the liquid to be tested and then inverted over a graduated cylinder.
- The liquid drops from the nozzle and the number of drops required to fill a certain volume of the cylinder is counted.
- The surface tension of the liquid is proportional to the weight of each drop, which can be calculated from the density of the liquid and the volume of the cylinder.
- The formula for the surface tension of the liquid is:

$$\gamma = \frac{4}{3}\pi r^3 \rho g \frac{N}{V}$$

  where γ is the surface tension, r is the radius of the nozzle, ρ is the density of the liquid, g is the acceleration due to gravity, N is the number of drops, and V is the volume of the cylinder.

- To perform the experiment, we need the following apparatus:
  - A stalagmometer
  - A graduated cylinder
  - A balance
  - A thermometer
  - A stopwatch
  - A beaker
  - A liquid to be tested (e.g., water, alcohol, glycerin, etc.)

- The procedure of the experiment is as follows:
  - Clean and dry the stalagmometer and the graduated cylinder.
  - Fill the stalagmometer with the liquid to be tested up to the mark on the bulb.
  - Weigh the stalagmometer with the liquid and note the mass as M_1.
  - Invert the stalagmometer over the graduated cylinder and allow the liquid to drop from the nozzle.
  - Start the stopwatch when the first drop falls and stop it when the last drop falls.
  - Note the time taken as t and the volume of the liquid collected in the cylinder as V.
  - Count the number of drops that fell in the cylinder and note it as N.
  - Weigh the stalagmometer again and note the mass as M_2.
  - Calculate the mass of the liquid that dropped from the stalagmometer as $M = M_1 - M_2$.
  - Calculate the density of the liquid as $\rho = \frac{M}{V}$.
  - Measure the temperature of the liquid and note it as T.
  - Calculate the surface tension of the liquid using the formula given above.
  - Repeat the experiment for different liquids and compare the results.

6. Determination of Viscosity of a given liquid by viscometer.

- Viscosity is a measure of the resistance of a fluid to flow or deform under an applied force.
- A viscometer is an instrument that measures the viscosity of a fluid by observing its flow through a narrow tube or a capillary.
- The flow rate of a fluid through a capillary depends on the pressure difference across the capillary, the length and radius of the capillary, and the viscosity of the fluid.
- The Hagen-Poiseuille equation relates these parameters as follows:

$$Q = \frac{\pi r^4 \Delta P}{8\eta L}$$

where Q is the volumetric flow rate, r is the radius of the capillary, ΔP is the pressure difference, η is the viscosity, and L is the length of the capillary.

- To determine the viscosity of a given liquid by viscometer, the following steps are followed:
  - Fill the viscometer with the liquid to be tested and immerse it in a water bath at a constant temperature.
  - Allow the liquid to flow through the capillary under the action of gravity and measure the time taken for a fixed volume of liquid to flow out of the capillary.

- Repeat the experiment for different volumes of liquid and calculate the average flow rate.
- Measure the radius and length of the capillary using a micrometer and a meter scale.
- Measure the pressure difference across the capillary using a manometer or a pressure gauge.
- Substitute the values of Q, r, L, and ΔP in the Hagen-Poiseuille equation and solve for η.
- Report the viscosity of the liquid with appropriate units and significant figures.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator.

- Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron(II) sulfate and ammonium sulfate, with the formula (NH4)2Fe(SO4)2 · 6H2O.
- It is commonly used as a standard solution for titrating oxidizing agents, such as potassium permanganate or potassium dichromate, in redox titrations.
- The strength of a ferrous ammonium sulfate solution can be determined by titrating it with a standard solution of potassium permanganate, using an external indicator such as starch or potassium ferricyanide.
- The reaction between ferrous ammonium sulfate and potassium permanganate is as follows:

  10Fe^2+ + 2MnO4^- + 16H^+ -> 10Fe^3+ + 2Mn^2+ + 8H2O
- The end point of the titration is marked by the appearance of a faint pink color in the solution, due to the excess of potassium permanganate.
- The strength of the ferrous ammonium sulfate solution can be calculated by using the following formula:

  Strength of ferrous ammonium sulfate = (Volume of potassium permanganate x Normality of potassium permanganate x 392) / (Volume of ferrous ammonium sulfate x 1000)
- Where 392 is the equivalent weight of ferrous ammonium sulfate, and 1000 is the conversion factor from milliliters to liters.
- Alternatively, an external indicator such as starch or potassium ferricyanide can be used to detect the end point of the titration more accurately.
- Starch forms a blue-black complex with iodine, which is produced by the reaction of potassium permanganate and potassium iodide in acidic medium.

The end point is marked by the disappearance of the blue-black color, due to the reduction of iodine by ferrous ions.

- Potassium ferricyanide forms a blue precipitate of Prussian blue with ferrous ions, which is visible in the presence of a white background. The end point is marked by the disappearance of the blue precipitate, due to the oxidation of ferrous ions by potassium permanganate.

8. Determination of the strength of Potassium dichromate using internal indicator.

- Potassium dichromate is a primary standard substance that can be used to determine the strength of a reducing agent such as ferrous ammonium sulphate.
- The reaction between potassium dichromate and ferrous ammonium sulphate is as follows:

$$K_2Cr_2O_7+6FeSO_4+7H_2SO_4 \rightarrow Cr_2(SO_4)_3+3K_2SO_4+6Fe_2(SO_4)_3+7H_2O$$

- The end point of the titration can be detected by the change in color of the solution from green to reddish-brown, due to the formation of ferric ions.
- This color change is the internal indicator of the titration, and no external indicator is required.
- The strength of potassium dichromate can be calculated by using the following formula:

$$\text{Strength of } K_2Cr_2O_7 = \frac{6 \times \text{Normality of } FeSO_4 \times \text{Volume of } FeSO_4}{\text{Volume of } K_2Cr_2O_7}$$

- The procedure for the titration is as follows:
  - Prepare a standard solution of ferrous ammonium sulphate by dissolving 3.92 g of the salt in 250 mL of distilled water and adding 10 mL of dilute sulphuric acid.
  - Transfer 20 mL of this solution to a conical flask and add 10 mL of dilute sulphuric acid.
  - Pipette out 10 mL of potassium dichromate solution of unknown strength into a burette and note the initial reading.
  - Titrate the ferrous ammonium sulphate solution with potassium dichromate solution until the color changes from green to reddish-brown.

- Note the final reading of the burette and calculate the volume of potassium dichromate used.
- Repeat the titration for two more times and take the average volume of potassium dichromate used.
- Calculate the strength of potassium dichromate using the formula given above.

9. Determination of available chlorine in bleaching powder

- Bleaching powder is a compound of calcium hypochlorite [$Ca(OCl)2$] and calcium chloride [$CaCl2$] that is used as a disinfectant and a bleaching agent.
- Available chlorine is the amount of chlorine that can be liberated from bleaching powder when it reacts with dilute acids. It is expressed as a percentage of the mass of bleaching powder.
- The reaction of bleaching powder with dilute sulfuric acid is as follows:

  $Ca(OCl)Cl + H2SO4 \rightarrow CaSO4 + H2O + Cl2$

- The chlorine gas that is evolved can be estimated by titrating it with a standard solution of sodium thiosulfate (hypo) using starch as an indicator. The reaction is as follows:

  $Cl2 + 2Na2S2O3 \rightarrow 2NaCl + Na2S4O6$

- The procedure for determining the available chlorine in bleaching powder is as follows:
  - Weigh accurately about 0.5 g of bleaching powder and transfer it to a 250 mL conical flask.
  - Add about 100 mL of distilled water and shake well to dissolve the bleaching powder.
  - Add 10 mL of dilute sulfuric acid and stopper the flask immediately.
  - Allow the reaction to complete and then titrate the liberated chlorine with 0.1 N sodium thiosulfate solution using starch as an indicator.
  - Repeat the titration with a blank solution of water and acid without bleaching powder.
  - Calculate the percentage of available chlorine using the formula:

    % available chlorine = (B - S) x N x 35.46 / W

    where B is the volume of sodium thiosulfate for the blank, S is the volume of sodium thiosulfate for the sample, N is the normality of sodium thiosulfate, W is the weight of bleaching powder in grams, and 35.46 is the equivalent weight of chlorine.

10. Determination of chloride content in water sample.

- Chloride is one of the major anions found in natural waters. It originates from various sources, such as seawater intrusion, industrial effluents, agricultural runoff, and domestic sewage.
- Chloride can affect the taste, corrosion, and biological activity of water. Therefore, it is important to monitor and control its concentration in drinking water, wastewater, and environmental samples.
- The most common method for determining chloride content in water sample is the **argentometric method**. This method is based on the precipitation of chloride ions as silver chloride by adding a known excess of silver nitrate solution and titrating the excess silver ions with a standard solution of sodium or potassium thiocyanate using ferric ammonium sulfate as an indicator.
- The following steps describe the procedure of the argentometric method:
  1. Pipette a 50 mL aliquot of the water sample into a 250 mL conical flask and add 2 mL of 6 M nitric acid and a few drops of potassium chromate indicator. The indicator will form a yellow color in the solution.
  2. Titrate the solution with 0.02 M silver nitrate solution from a burette until a faint reddish-brown color appears at the end point. This indicates the formation of silver chromate precipitate. Record the volume of silver nitrate solution used as V1.
  3. Add 2 mL of 10% ammonium thiocyanate solution and a few drops of ferric ammonium sulfate indicator to the same flask. The indicator will form a red color in the solution.
  4. Titrate the solution with 0.1 M sodium or potassium thiocyanate solution from another burette until a permanent brown color appears at the end point. This indicates the formation of ferric thiocyanate complex. Record the volume of thiocyanate solution used as V2.
  5. Repeat the steps 1 to 4 for a blank solution containing only distilled water and the same reagents. Record the volumes of silver nitrate and thiocyanate solutions used as V1' and V2', respectively.
  6. Calculate the chloride content in the water sample using the following formula:

     `Cl- (mg/L) = (V1 - V1') x 0.02 x 35.45 x 1000 / 50 - (V2 - V2') x 0.1 x 35.45 x 1000 / 50`

     where 35.45 is the molar mass of chloride ion in g/mol.

11. Preparation of Phenol formaldehyde (PF) resin

- Phenol formaldehyde (PF) resin is a type of thermosetting resin that is obtained by the polymerization of phenol and formaldehyde.
- Phenol and formaldehyde react in a step-growth polymerization reaction that can be either acid- or base-catalyzed .
- The ratio of phenol to formaldehyde affects the structure and properties of the resin. A 1:1 ratio produces linear chains, while a higher ratio of formaldehyde produces cross-linked networks .
- The reaction can be carried out in one or two stages. In the one-stage process, phenol and formaldehyde are mixed with a catalyst and heated under reflux until the desired degree of polymerization is reached. In the two-stage process, a low-molecular-weight resin (called a resol) is first prepared by reacting phenol and formaldehyde with a base catalyst, and then cured by adding an acid catalyst and heating .
- The properties of PF resin depend on the type and amount of catalyst, the reaction temperature and time, the phenol to formaldehyde ratio, and the degree of curing. PF resin has high thermal stability, chemical resistance, hardness, and dimensional stability, but low flexibility and impact strength .
- PF resin is widely used in the production of wood-based composites, such as plywood, particleboard, and medium-density fiberboard (MDF), as well as in molding, laminating, coating, and adhesive applications .

12. Preparation of Urea formaldehyde (UF) resin

- Urea formaldehyde (UF) resin is a thermosetting polymer that is widely used as an adhesive and a coating material for wood products, such as plywood, particleboard, and fiberboard.
- UF resin is synthesized by the condensation reaction of urea and formaldehyde in an aqueous solution, usually in the presence of a catalyst or a base.
- The reaction can be divided into three main steps: methylolation, condensation, and curing.

Methylolation

- In this step, formaldehyde reacts with urea to form monomethylolurea (MMU) and dimethylolurea (DMU), which are the basic building blocks of UF resin.
- The reaction is reversible and depends on the molar ratio of formaldehyde to urea (F/U), the pH, and the temperature of the solution.
- A high F/U ratio (>2) and a high pH (>7) favor the formation of MMU and DMU, while a low F/U ratio (<2) and a low pH (<7) favor the reverse reaction.
- The reaction can be carried out at room temperature or at elevated temperatures (up to 80°C) to speed up the reaction rate.

- The reaction can be catalyzed by a base, such as ammonia or pyridine, which acts as a nucleophile and increases the reactivity of formaldehyde.
- The reaction can also be catalyzed by an acid, such as sulfuric acid or oxalic acid, which acts as a proton donor and increases the reactivity of urea.
- The methylolation step can be stopped by adding an acid to lower the pH and by cooling the solution.

Condensation

- In this step, MMU and DMU react with each other or with more urea and formaldehyde to form larger molecules, such as trimethylolurea (TMU), tetramethylolurea (TeMU), and polymeric chains with methylene (-CH2-) and methylene ether (-CH2-O-CH2-) linkages.
- The reaction is irreversible and depends on the F/U ratio, the pH, and the temperature of the solution.
- A high F/U ratio (>2) and a low pH (<5) favor the formation of methylene linkages, which are more stable and less prone to hydrolysis, while a low F/U ratio (<2) and a high pH (>7) favor the formation of methylene ether linkages, which are more flexible and more prone to hydrolysis.
- The reaction can be carried out at elevated temperatures (up to 100°C) to increase the reaction rate and the degree of polymerization.
- The reaction can be catalyzed by an acid, such as sulfuric acid or oxalic acid, which acts as a proton donor and increases the reactivity of the hydroxyl groups on MMU and DMU.
- The reaction can also be catalyzed by a base, such as ammonia or pyridine, which acts as a nucleophile and increases the reactivity of the carbonyl groups on urea and formaldehyde.
- The condensation step can be stopped by adding a base to raise the pH and by cooling the solution.

Curing

- In this step, the UF resin is applied to the wood substrate and heated to a high temperature (up to 200°C) to crosslink the polymer chains and form a hard and insoluble network.
- The curing process is irreversible and depends on the F/U ratio, the pH, and the temperature of the resin.
- A high F/U ratio (>2) and a low pH (<5) favor the formation of methylene linkages, which are more stable and less prone to hydrolysis, while a low F/U ratio (<2) and a high pH (>7) favor the formation of methylene ether linkages, which are more flexible and more prone to hydrolysis.
- The curing process can be accelerated by adding a hardener, such as ammonium chloride or ammonium sulfate, which acts as an acid catalyst and lowers the pH of the resin.
- The curing process can also be accelerated by adding a filler, such as wheat

flour or wood flour, which acts as a filler and reduces the amount of resin required.

- The curing process can be monitored by measuring the viscosity, the gel time, and the free formaldehyde content of the resin.

13. Preparation of Adipic acid / Paracetamol

- Adipic acid is the organic compound with the formula (CH2)4(COOH)2. It is the most important dicarboxylic acid, mainly used as a precursor for the production of nylon.
- Adipic acid can be prepared from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil . The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .
- The oxidation of KA oil involves the following steps :
  - Formation of nitrosonium ion (NO+) by the reaction of nitric acid and sulfuric acid.
  - Nitrosation of cyclohexanone and cyclohexanol to form nitrosocyclohexanone and nitrosocyclohexanol, respectively.
  - Tautomerization of nitrosocyclohexanone and nitrosocyclohexanol to form oximes.
  - Dehydration of oximes to form nitriles.
  - Hydrolysis of nitriles to form carboxylic acids.
  - Decarboxylation of cyclohexane-1,2-dicarboxylic acid to form adipic acid.
- The overall reaction can be written as :

  equation

- Paracetamol is the organic compound with the formula C8H9NO2. It is a widely used analgesic and antipyretic drug.
- Paracetamol can be prepared from phenol by the following steps :
  - Nitration of phenol to form 4-nitrophenol.
  - Reduction of 4-nitrophenol to form 4-aminophenol.
  - Acetylation of 4-aminophenol to form paracetamol.
- The overall reaction can be written as :

  equation

14. Determination of Cell Conductance of a solution.

- Cell conductance is a measure of how well a solution can conduct electric current between two electrodes.

- Cell conductance depends on the type, concentration and temperature of the solution, as well as the geometry and material of the electrodes and the cell.
- Cell conductance is usually expressed in siemens (S) or mho (℧), which are the reciprocal units of resistance (ohm, Ω).
- Cell conductance can be determined by applying a known alternating voltage across the electrodes and measuring the resulting alternating current.
- Cell conductance can be calculated by using Ohm's law: G = I / V, where G is the cell conductance, I is the current and V is the voltage.
- Cell conductance can also be expressed as the product of the specific conductance (or conductivity) of the solution and the cell constant: G = C, where is the specific conductance and C is the cell constant.
- The specific conductance of a solution is a property that depends only on the type and concentration of the ions in the solution, and is independent of the cell geometry and material.
- The cell constant is a factor that depends on the cell geometry and material, and is independent of the solution. It can be calculated by dividing the distance between the electrodes by the effective cross-sectional area of the electrodes: C = L / A, where L is the distance and A is the area.
- The cell constant can also be determined experimentally by measuring the cell conductance of a solution with a known specific conductance, and then using the formula: C = G / , where G is the cell conductance and is the known specific conductance.
- The cell conductance of a solution can be used to determine the concentration of an electrolyte in the solution, by using a calibration curve or a formula that relates the specific conductance to the concentration.

15. Determination of Rate constant of hydrolysis of esters.

- Hydrolysis of esters is a reaction in which an ester is converted into a carboxylic acid and an alcohol by the addition of water.
- The reaction is catalyzed by either an acid or a base.
- The rate of hydrolysis of esters depends on the nature of the ester, the concentration of the catalyst, the temperature, and the solvent.
- The rate constant of hydrolysis of esters can be determined by measuring the concentration of either the ester or the product (carboxylic acid or alcohol) as a function of time.
- One method to determine the rate constant of hydrolysis of esters is to use a conductometric titration, which measures the change in electrical conductivity of the solution as the reaction proceeds.
- The conductometric titration involves adding a known volume of a standard base (such as NaOH) to a known volume of the ester solution in the presence of an acid catalyst (such as HCl) and recording the conductivity

of the solution at regular intervals of time.

- The conductivity of the solution increases as the ester is hydrolyzed and the carboxylic acid is neutralized by the base, forming a salt and water.
- The conductivity of the solution reaches a maximum when the ester is completely hydrolyzed and the base is completely consumed.
- The rate constant of hydrolysis of esters can be calculated by plotting the conductivity versus time data and applying the integrated rate law for a first-order reaction.
- The integrated rate law for a first-order reaction is given by:

  `ln[A] = -kt + ln[A]0`

  where [A] is the concentration of the ester at time t, [A]0 is the initial concentration of the ester, k is the rate constant, and t is the time.
- The rate constant can be obtained by finding the slope of the linear plot of ln[A] versus t.

16. Element detection and identification of functional groups in organic compounds

- Element detection is the process of determining the presence and quantity of elements in an organic compound, such as carbon, hydrogen, nitrogen, oxygen, halogens, sulfur, and phosphorus.
- Identification of functional groups is the process of determining the type and location of the specific groups of atoms that are responsible for the characteristic chemical reactions of an organic compound, such as alcohols, aldehydes, ketones, carboxylic acids, amines, etc.
- Element detection and identification of functional groups are important for the analysis and synthesis of organic compounds, as they provide information about the structure, properties, and reactivity of the molecules.
- Some of the common methods for element detection and identification of functional groups are:
  - Solubility test: This test involves dissolving the organic compound in different solvents, such as water, dilute acid, or dilute base, and observing the solubility behavior. This test can help to identify some functional groups based on their polarity and acidity or basicity. For example, alcohols and phenols are soluble in water, carboxylic acids are soluble in dilute base, and amines are soluble in dilute acid.
  - Lassaigne's test: This test involves fusing the organic compound with sodium metal and extracting the sodium fusion extract with water. This test can help to detect the presence of nitrogen, sulfur, and halogens in the organic compound, as they form sodium cyanide, sodium

sulfide, and sodium halides, respectively, which can be detected by various chemical reactions.
  - 2,4-dinitrophenylhydrazine test: This test involves adding a small amount of the organic compound to 2,4-dinitrophenylhydrazine reagent and shaking well. This test can help to identify the presence of carbonyl groups (aldehydes and ketones) in the organic compound, as they form yellow, orange, or red precipitates of 2,4-dinitrophenylhydrazones, which have characteristic melting points.
  - Tollen's test: This test involves adding a few drops of the organic compound to a freshly prepared solution of ammoniacal silver nitrate and warming gently. This test can help to distinguish between aldehydes and ketones, as aldehydes reduce the silver ions to metallic silver, forming a silver mirror on the inner surface of the test tube, while ketones do not.
  - Fehling's test: This test involves adding a few drops of the organic compound to a mixture of Fehling's solution A (copper sulfate) and Fehling's solution B (sodium potassium tartrate) and heating. This test can help to distinguish between aldehydes and ketones, as aldehydes reduce the blue copper ions to red copper oxide, forming a red precipitate, while ketones do not.
  - Esterification test: This test involves adding a few drops of the organic compound to an alcohol and a few drops of concentrated sulfuric acid and heating. This test can help to identify the presence of carboxylic acids in the organic compound, as they react with alcohols to form esters, which have fruity odors.
  - Nitrous acid test: This test involves adding a few drops of the organic compound to a solution of sodium nitrite and dilute hydrochloric acid and observing the effervescence. This test can help to identify the presence of primary amines in the organic compound, as they react with nitrous acid to form nitrogen gas, water, and an alcohol.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- This note indicates that the instructor has the authority and flexibility to select the experiments that are most suitable for the course objectives and the available resources.
- The instructor can choose any 10 experiments from the list of experiments given in the syllabus or the lab manual, depending on the relevance, difficulty, duration, and learning outcomes of each experiment.
- The instructor can also modify or replace any two of the experiments from the list, as long as they are consistent with the course content and the learning objectives.
- The instructor should inform the students about the selected and changed experiments at the beginning of the semester or the lab session, and pro-

vide the necessary instructions, materials, and guidance for each experiment.
- The students should follow the instructor's directions and perform the experiments as per the schedule and the evaluation criteria.

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, which are derived from the program outcomes and the institutional mission and vision.
- Course outcomes are measurable, observable, and achievable within the duration and scope of the course.
- Course outcomes are written in terms of student learning, using verbs that indicate the level of cognitive skills required, such as analyze, apply, create, evaluate, etc.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course delivery, using various methods such as syllabus, rubrics, feedback, etc.
- Course outcomes are assessed using appropriate tools and methods, such as exams, assignments, projects, portfolios, etc.
- Course outcomes are used to evaluate the effectiveness of the course design, delivery, and improvement, as well as the student achievement and satisfaction.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply various artificial intelligence techniques and algorithms to solve problems, such as heuristic search, constraint satisfaction, logic, inference, probabilistic reasoning, decision making, reinforcement learning, neural networks, deep learning, natural language processing, computer vision, and robotics.
- Analyze the strengths and limitations of different artificial intelligence approaches and methods, and compare their performance and suitability for different domains and tasks.
- Design and implement artificial intelligence systems and applications using appropriate tools and frameworks, such as Python, TensorFlow, PyTorch, OpenAI Gym, NLTK, OpenCV, ROS, etc.
- Evaluate and test the functionality and quality of artificial intelligence systems and applications, and identify and debug errors and issues.
- Communicate and document the artificial intelligence solutions and results clearly and effectively, using appropriate terminology, notation, diagrams,

and formats.

Course Outcomes Bloom's

- Course outcomes are statements that describe what students should be able to do or demonstrate after completing a course.
- Bloom's taxonomy is a framework for classifying educational objectives into six levels of cognitive complexity: remember, understand, apply, analyze, evaluate, and create.
- Course outcomes can be aligned with Bloom's taxonomy to ensure that they cover a range of cognitive skills and learning domains.
- Some examples of course outcomes and their corresponding Bloom's levels are:
  - Remember: Identify the main components of a computer system.
  - Understand: Explain the difference between hardware and software.
  - Apply: Use a programming language to create a simple application.
  - Analyze: Compare and contrast different algorithms for sorting data.
  - Evaluate: Assess the strengths and weaknesses of various software development methodologies.
  - Create: Design and implement a software solution for a real-world problem.

Level

- A level is a measure of the amount or degree of something, such as height, quantity, quality, or intensity.
- A level can also refer to a position or rank in a hierarchy, such as a level of authority, education, or skill.
- A level can also mean a horizontal plane or surface that is perpendicular to the direction of gravity, such as the ground or the sea level.
- A level can also be a device or instrument that is used to determine if a surface is horizontal or vertical, such as a spirit level or a laser level.
- A level can also be a stage or phase in a process, such as a level of difficulty, achievement, or complexity.

CO-1 Get an understanding of the use of different analytical instruments. K3

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Different analytical instruments have different principles, applications, advantages, and limitations.
- Some examples of analytical instruments are:

- Spectrophotometers: measure the amount of light absorbed or transmitted by a sample at different wavelengths.
- Chromatographs: separate the components of a mixture based on their affinity to a stationary phase and a mobile phase.
- Mass spectrometers: identify the molecular mass and structure of compounds by ionizing them and measuring their mass-to-charge ratio.
- Electrochemical analyzers: measure the electrical potential or current generated by a chemical reaction or a redox process.
- Microscopes: magnify the image of small objects or structures using visible light, electrons, or other forms of radiation.
- Biosensors: detect the presence or concentration of a specific biomolecule or analyte using a biological recognition element and a transducer.

- To get an understanding of the use of different analytical instruments, one should:
  - Learn the basic theory and working principle of each instrument.
  - Know the types, components, and functions of each instrument.
  - Understand the sample preparation, calibration, and operation procedures of each instrument.
  - Recognize the sources of error, uncertainty, and interference in each instrument.
  - Interpret the data and results obtained from each instrument.
  - Compare and contrast the performance and suitability of different instruments for different purposes.

CO2 Molecular Properties and Surface Tension

- Carbon dioxide (CO2) is a linear molecule with a bond length of 116.3 pm and a bond angle of 180 degrees.
- CO2 is a non-polar molecule with a dipole moment of zero and a symmetrical distribution of electrons.
- CO2 has a molar mass of 44.0095 g/mol and a critical point at 304.25 K and 7.38 MPa.
- CO2 is a gas at standard temperature and pressure, but can be liquefied or solidified under certain conditions.
- The density and specific weight of CO2 vary with temperature and pressure, and can be calculated using online tools or tables.
- The surface tension of CO2 is the energy required to increase the surface area of the liquid phase due to intermolecular forces.
- The surface tension of CO2 depends on the temperature, pressure, and the presence of other substances in the solution .
- The surface tension of CO2 can be measured using various methods, such as the pendant drop method, the Wilhelmy plate method, or the capillary rise method.

- The surface tension of CO_2 is important for applications such as carbon capture and storage, enhanced oil recovery, and supercritical fluid extraction .

Viscosity, conductance of solution, chloride and iron content in the water

- Viscosity is a measure of the resistance of a fluid to deformation under shear stress. It is related to the internal friction of the fluid molecules. Viscosity depends on the temperature, pressure, and composition of the fluid.
- Conductance of solution is a measure of the ability of a solution to conduct electric current. It is proportional to the concentration and mobility of the ions in the solution. Conductance depends on the temperature, pressure, and type of electrolyte in the solution.
- Chloride and iron content in the water are indicators of the quality and purity of the water. Chloride is a common anion in natural waters and can affect the taste, corrosion, and conductivity of the water. Iron is a common metal in natural waters and can cause staining, discoloration, and taste problems in the water.
- Some important points to remember are:
  - Viscosity of water decreases with increasing temperature and pressure, and increases with increasing salinity and dissolved solids.
  - Conductance of water increases with increasing temperature and pressure, and increases with increasing concentration and mobility of ions. Chloride ions are one of the major contributors to the conductivity of seawater.
  - Chloride content in water can vary from less than 10 mg/L in freshwater to more than 19,000 mg/L in seawater. The World Health Organization (WHO) recommends a maximum chloride level of 250 mg/L for drinking water.
  - Iron content in water can vary from less than 0.01 mg/L to more than 50 mg/L depending on the source and treatment of the water. The WHO recommends a maximum iron level of 0.3 mg/L for drinking water.
  - Viscosity, conductance, chloride, and iron content can be measured by various methods, such as viscometers, conductivity meters, titration, colorimetry, spectrophotometry, etc .

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.

- K3 visa is a nonimmigrant visa, which means it does not grant permanent residence or a green card. However, K3 visa holders can apply for adjustment of status to become permanent residents after their immigrant visa petition is approved.
- K3 visa is valid for two years and can be renewed if the immigrant visa petition or the adjustment of status application is still pending.
- K3 visa holders can work and study in the United States with the appropriate authorization from the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa holders can also travel outside the United States and return with a valid visa and passport.
- To be eligible for a K3 visa, the foreign spouse must meet the following requirements:
  - Be legally married to a U.S. citizen.
  - Have a pending Form I-130, Petition for Alien Relative, filed by the U.S. citizen spouse with the USCIS.
  - Have a Form I-129F, Petition for Alien Fiancé(e), filed by the U.S. citizen spouse with the USCIS and approved by the National Visa Center (NVC).
  - Have a valid passport and a medical examination.
  - Have no criminal or immigration violations that would make them inadmissible to the United States.
  - Have sufficient financial support from the U.S. citizen spouse or a joint sponsor.
  - Have a valid relationship with the U.S. citizen spouse and intend to live together as spouses in the United States.

CO-3 Measure the hardness and alkalinity of the water. K3

- Hardness of water is the measure of the concentration of calcium and magnesium ions in water. It is expressed in terms of equivalent calcium carbonate (CaCO3).
- Alkalinity of water is the measure of the capacity of water to neutralize acids. It is mainly due to the presence of bicarbonates, carbonates and hydroxides of calcium, magnesium and other metals. It is expressed in terms of equivalent calcium carbonate (CaCO3) or milligrams per liter (mg/L).
- To measure the hardness of water, one of the methods is the EDTA titration method. EDTA (ethylenediaminetetraacetic acid) is a chelating agent that forms stable complexes with calcium and magnesium ions. The steps are as follows:
  - Take a known volume of water sample in a conical flask and add a few drops of Eriochrome Black T (EBT) indicator. The indicator forms a wine-red complex with the metal ions in water.
  - Titrate the sample with a standard solution of EDTA until the color changes from wine-red to blue. The end point indicates that all the

metal ions have been complexed by EDTA and the indicator is free.
  - Record the volume of EDTA used and calculate the hardness of water using the formula:
    Hardness (mg/L) = (Volume of EDTA x Normality of EDTA x 1000) / Volume of water sample
- To measure the alkalinity of water, one of the methods is the phenolphthalein and methyl orange titration method. Phenolphthalein and methyl orange are acid-base indicators that change color at different pH values. The steps are as follows:
  - Take a known volume of water sample in a conical flask and add a few drops of phenolphthalein indicator. If the water is alkaline, the indicator will turn pink. If the water is neutral or acidic, the indicator will remain colorless.
  - Titrate the sample with a standard solution of sulfuric acid until the color changes from pink to colorless. The end point indicates that all the hydroxides and half of the carbonates have been neutralized by the acid. Record the volume of acid used and calculate the phenolphthalein alkalinity using the formula:
    Phenolphthalein alkalinity (mg/L) = (Volume of acid x Normality of acid x 1000 x 50) / Volume of water sample
  - Add a few drops of methyl orange indicator to the same sample and continue the titration with the acid until the color changes from yellow to orange-red. The end point indicates that all the carbonates and bicarbonates have been neutralized by the acid. Record the total volume of acid used and calculate the total alkalinity using the formula:
    Total alkalinity (mg/L) = (Total volume of acid x Normality of acid x 1000 x 50) / Volume of water sample
- The hardness and alkalinity of water are important parameters for water quality and treatment. Hard water can cause scaling and corrosion of pipes and equipment, reduce the efficiency of detergents and soap, and affect the taste and appearance of water. Alkaline water can also cause scaling and corrosion, as well as affect the pH and buffering capacity of water. Therefore, it is necessary to measure and control the hardness and alkalinity of water for various domestic, industrial and environmental purposes.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic compound that has a hydroxyl group (-OH) attached to a benzene ring. Phenol is also known as carbolic acid or phenic acid. Phenol has many applications in industry, medicine, and organic synthesis.

There are several methods to prepare phenol from different starting materials. Some of the common methods are:

- From haloarenes: Haloarenes are aromatic compounds that have one or more halogen atoms (-X) attached to a benzene ring. Examples of haloarenes are chlorobenzene, bromobenzene, and iodobenzene. Haloarenes can be converted to phenol by heating them with sodium hydroxide (NaOH) at high temperature and pressure. This is called the Dow process. The reaction involves the replacement of the halogen atom by a hydroxyl group. For example, chlorobenzene can be converted to phenol as follows :

$C_6H_5Cl + NaOH \rightarrow C_6H_5ONa + NaCl$

$C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

- From benzene: Benzene is the simplest aromatic compound that has a six-carbon ring with alternating single and double bonds. Benzene can be converted to phenol by first sulfonating it with concentrated sulfuric acid (H_2SO_4) to form benzene sulfonic acid. Then, the benzene sulfonic acid is heated with sodium hydroxide to form sodium phenoxide, which is acidified with dilute acid or carbon dioxide to form phenol. This is called the Raschig process. The reactions are as follows :

  $C_6H_6 + H_2SO_4 \rightarrow C_6H_5SO_3H + H_2O$

  $C_6H_5SO_3H + NaOH \rightarrow C_6H_5ONa + NaHSO_4$

  $C_6H_5ONa + H_2O \rightarrow C_6H_5OH + NaOH$

- From cumene: Cumene is an organic compound that has a benzene ring attached to an isopropyl group ($-CH(CH_3)_2$). Cumene can be converted to phenol by first oxidizing it with air to form cumene hydroperoxide, which is then cleaved with dilute acid to form phenol and acetone. This is called the cumene process. The reactions are as follows :

  $C_6H_5CH(CH_3)_2 + O_2 \rightarrow C_6H_5C(CH_3)_2OOH$

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea formaldehyde (UF) is a nontransparent thermosetting resin or polymer that is produced from urea and formaldehyde. It is used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- UF resin often shows the appearance of crystal domains, which are influenced by the synthesis conditions, such as pH, temperature, and molar ratio .
- Adipic acid is a dicarboxylic acid that is mainly used as a precursor for the production of nylon. It can also be used as a modifier for UF resin to reduce the free formaldehyde content.

- Paracetamol, also known as acetaminophen, is a medication used to treat pain and fever. It is one of the most commonly used drugs in the world. It works by inhibiting the synthesis of prostaglandins, which are involved in inflammation and pain sensation.

K3

- K3 is a type of artificial intelligence that can generate natural language texts from structured data or queries.
- K3 uses a neural network model that is trained on a large corpus of text and data pairs, such as Wikipedia tables and summaries, or SQL queries and natural language answers.
- K3 can generate texts in different languages, domains, and styles, depending on the input data and the user's preferences.
- K3 can also answer questions, summarize information, generate reports, write essays, and create other types of texts from data.
- K3 is a powerful and versatile tool that can help users access, understand, and communicate data in natural language.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction, K3, is a measure of how fast the reaction proceeds at a given temperature and concentration of reactants.
- The rate constant of a reaction can be estimated by using the Arrhenius equation, which relates K3 to the activation energy, Ea, and the frequency factor, A, of the reaction.
- The Arrhenius equation is: $K3 = A * \exp(-Ea / RT)$, where R is the gas constant and T is the absolute temperature.
- To estimate K3, one needs to know the values of A and Ea, which can be obtained from experimental data or theoretical calculations.
- Alternatively, one can use the linearized form of the Arrhenius equation, which is: $\ln(K3) = \ln(A) - Ea / RT$, and plot $\ln(K3)$ versus $1 / T$ to obtain a straight line with slope $-Ea / R$ and intercept $\ln(A)$.
- By using the slope and intercept of the line, one can calculate the values of Ea and A, and then use them to estimate K3 at any desired temperature.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for the main idea or the most important information that the text, speech, or conversation conveys.
- The topic can be used to organize, summarize, or categorize the text, speech, or conversation.

- The topic can also be used to generate questions, arguments, or opinions about the text, speech, or conversation.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- A topic can be broad or narrow, depending on the scope and purpose of the communication.
- A topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- A topic can be chosen by the speaker or writer, or assigned by a teacher, editor, or audience.
- A topic can be influenced by the context, the genre, the audience, and the purpose of the communication.
- A topic can be developed and supported by using various types of information, such as facts, opinions, examples, statistics, anecdotes, or quotations.
- A topic can be organized and structured by using different methods, such as comparison and contrast, cause and effect, problem and solution, or chronological order.
- A topic can be evaluated and revised by considering the clarity, relevance, accuracy, completeness, and coherence of the communication.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn about. You have chosen the topic:

What is the difference between a virus and a bacterium?

Here are some points to help you understand the topic:

- A virus is a microscopic particle that consists of genetic material (DNA or RNA) enclosed in a protein coat. A bacterium is a single-celled organism that has a cell membrane, a cell wall, and a cytoplasm with various organelles.
- A virus cannot reproduce on its own. It needs to infect a host cell and use its machinery to make copies of itself. A bacterium can reproduce by binary fission, which is a process of splitting into two identical cells.
- A virus is not considered a living thing because it does not have all the characteristics of life, such as metabolism, growth, and response to stimuli. A bacterium is considered a living thing because it has all the characteristics of life.
- A virus can only infect specific types of cells, depending on the receptors on its surface and the host cell. A bacterium can infect a wide range of cells, depending on the nutrients and conditions available.

- A virus can cause diseases such as the common cold, influenza, AIDS, and COVID-19. A bacterium can cause diseases such as tuberculosis, strep throat, cholera, and anthrax.
- A virus can be prevented or treated by vaccines, antiviral drugs, or the immune system of the host. A bacterium can be prevented or treated by antibiotics, disinfectants, or the immune system of the host.

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.
- The topic can be used to guide the reader's or listener's attention and interest, such as by using hooks, transitions, examples, or questions.
- The topic can be used to communicate the purpose and goal of the text, such as by using a thesis statement, a claim, an argument, or a conclusion.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss.

Some possible topics are:

- History
- Science
- Literature
- Art
- Geography
- Mathematics
- Philosophy
- Music
- Sports
- Politics

Please choose a topic or type your own.

Page 33 of 40

Topic: Introduction to Artificial Intelligence

- Artificial intelligence (AI) is the branch of computer science that deals with creating machines or software that can perform tasks that normally require human intelligence, such as reasoning, learning, planning, decision

making, natural language processing, computer vision, speech recognition, etc.
- AI can be classified into two main categories: weak AI and strong AI. Weak AI, also known as narrow AI, is the type of AI that is designed to perform a specific task or domain, such as playing chess, driving a car, or recognizing faces. Strong AI, also known as general AI or artificial general intelligence (AGI), is the type of AI that can perform any intellectual task that a human can, such as understanding natural language, solving complex problems, or exhibiting creativity and self-awareness.
- AI can also be classified into three main levels: artificial narrow intelligence (ANI), artificial general intelligence (AGI), and artificial superintelligence (ASI). ANI is the current state of AI, where machines can perform specific tasks better than humans, but lack general intelligence or common sense. AGI is the hypothetical future state of AI, where machines can match or surpass human intelligence in any domain. ASI is the hypothetical future state of AI, where machines can exceed human intelligence in every possible way, and potentially pose an existential threat to humanity.
- AI can be achieved using various methods, techniques, and tools, such as logic, search, knowledge representation, machine learning, neural networks, deep learning, natural language processing, computer vision, speech recognition, etc. Each of these methods has its own advantages, limitations, and applications. AI systems can also combine multiple methods to achieve better performance or functionality.

The topic is

- The topic is a general term for the subject or theme of a conversation, text, or presentation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be explicit or implicit, depending on how clearly it is stated or implied by the speaker or writer.
- The topic can be broad or narrow, depending on how much information or detail it covers or excludes.
- The topic can be related to different domains, such as science, history, art, politics, etc.
- The topic can be chosen by the speaker or writer, or assigned by someone else, such as a teacher, a boss, or a client.
- The topic can be influenced by various factors, such as the purpose, audience, context, and genre of the communication.
- The topic can be developed or changed throughout the communication, depending on the feedback, questions, or arguments of the listener or reader.

ENGINEERING GRAPHICS LAB KCS

- Engineering graphics is the discipline of creating and interpreting graphical representations of engineering objects, such as mechanical parts, structures, circuits, etc.
- Engineering graphics lab is a practical course that teaches students how to use various tools and techniques to create and analyze engineering drawings, such as sketching, orthographic projection, isometric projection, dimensioning, scaling, etc.
- Engineering graphics lab also helps students to develop spatial visualization skills, which are essential for understanding and designing three-dimensional objects and systems.
- Engineering graphics lab is divided into several modules, each covering a specific topic or skill related to engineering graphics. Some of the modules are:
  - Introduction to engineering graphics: This module introduces the basic concepts and terminology of engineering graphics, such as point, line, plane, angle, etc. It also covers the types and formats of engineering drawings, such as multiview, pictorial, sectional, etc.
  - Classical geometry: This module covers the construction and properties of various geometric figures, such as triangles, quadrilaterals, circles, polygons, etc. It also teaches how to use compass, ruler, protractor, and other instruments to draw accurate geometric shapes.
  - Projection of points, lines, and planes: This module covers the principles and methods of orthographic projection, which is a technique to represent three-dimensional objects on a two-dimensional plane. It also teaches how to project points, lines, and planes in different quadrants and positions, and how to find their true lengths, inclinations, and traces.
  - Projection of solids: This module covers the projection of various solids, such as prisms, pyramids, cylinders, cones, spheres, etc. It also teaches how to find their true shapes, dimensions, and sections, and how to draw their development and intersection with other solids.
  - Isometric projection: This module covers the isometric projection, which is a technique to represent three-dimensional objects in a pictorial form. It also teaches how to draw isometric views of various solids and their combinations, and how to convert orthographic views to isometric views and vice versa.
  - Dimensioning and scaling: This module covers the rules and conventions of dimensioning and scaling, which are techniques to indicate the size and proportion of engineering objects on a drawing. It also teaches how to use different types of dimensions, such as linear, angular, radial, etc., and how to use different types of scales, such as plain, diagonal, vernier, etc.

- Engineering graphics lab requires students to perform various exercises and assignments based on the topics covered in each module. The exercises and assignments are designed to test the students' understanding and application of the concepts and skills learned in the lab.
- Engineering graphics lab also requires students to use various software tools, such as AutoCAD, SolidWorks, etc., to create and manipulate engineering drawings on a computer. The software tools help the students to enhance their productivity and accuracy, and to explore more complex and realistic engineering problems.

Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To develop the skills and techniques for designing, implementing, and evaluating AI systems and agents.
- To explore the ethical, social, and legal implications of AI and its impact on human society.
- To provide a foundation for further study and research in AI and related fields.

To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are devices that are used to measure, analyze, or monitor physical, chemical, or biological properties of samples.
- Analytical instruments can be classified into different categories based on their principles, applications, or techniques, such as spectroscopy, chromatography, mass spectrometry, electrochemistry, microscopy, etc.
- The fundamental concepts of analytical instruments include the following:
  - The basic components of an analytical instrument, such as the source, the sample, the detector, and the signal processor.
  - The types and characteristics of signals, such as analog, digital, noise, drift, etc.
  - The methods of signal processing, such as amplification, filtering, modulation, demodulation, etc.
  - The principles and applications of calibration, validation, and quality control of analytical instruments.
  - The factors affecting the performance, accuracy, precision, sensitivity, selectivity, and resolution of analytical instruments.
  - The advantages and limitations of different analytical techniques and instruments.

Analysis of Chloride Content, Hardness, and Alkalinity

- Chloride content is a measure of the amount of chloride ions (Cl^-) present in a water sample. Chloride ions are derived from sources such as seawater intrusion, industrial effluents, agricultural runoff, and road salt. Chloride content affects the taste, corrosion, and disinfection of water.
- Hardness is a measure of the amount of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions present in a water sample. Hardness is caused by the dissolution of minerals such as limestone, dolomite, and gypsum. Hardness affects the formation of scale, soap lather, and water heating efficiency.
- Alkalinity is a measure of the capacity of a water sample to neutralize acids. Alkalinity is mainly due to the presence of bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), and hydroxide (OH^-) ions. Alkalinity affects the pH, buffering, and corrosion of water.

Methods of Analysis

- Chloride content can be determined by titrating a water sample with a standard solution of silver nitrate ($AgNO_3$) using potassium chromate (K_2CrO_4) as an indicator. The end point is marked by the formation of a reddish-brown precipitate of silver chromate (Ag_2CrO_4).
- Hardness can be determined by titrating a water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T (EBT) as an indicator. The end point is marked by a color change from wine red to blue.
- Alkalinity can be determined by titrating a water sample with a standard solution of sulfuric acid (H_2SO_4) using phenolphthalein and methyl orange as indicators. The end point for phenolphthalein alkalinity is marked by a color change from pink to colorless. The end point for total alkalinity is marked by a color change from yellow to orange.

Applications and Significance

- Chloride content is important for assessing the suitability of water for drinking, irrigation, and industrial purposes. High chloride content can cause health problems, such as hypertension and dehydration, and can damage crops, pipes, and equipment.
- Hardness is important for assessing the quality of water for domestic, industrial, and agricultural purposes. High hardness can cause problems, such as scaling, reduced soap lather, increased water heating costs, and reduced plant growth.
- Alkalinity is important for maintaining the stability of water quality and preventing corrosion and acidification. High alkalinity can cause problems, such as scaling, reduced disinfection efficiency, and increased water consumption. Low alkalinity can cause problems, such as corrosion, metal

leaching, and pH fluctuations.

Water

Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states. It is one of the most plentiful and essential of compounds. A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.

Some of the main points about water are:

- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen. Water molecules cling to each other because of a force called hydrogen bonding. It's the reason why water can do amazing things.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid. Liquid water is the most common form of water on Earth. Gas water is water vapor, which is invisible and forms clouds. Solid water is ice, which can take various shapes such as snowflakes, hail, or glaciers.
- Water is the liquid that makes life on Earth possible. As water cycles from the air to the land to the sea and back again, water shapes our planet — and nearly every aspect of our lives. All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H2O. It is a transparent, tasteless, odorless, and nearly colorless chemical substance, and it is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).
- Water has many physical and chemical properties that make it unique and vital for life. Some of these properties are: high specific heat capacity, high heat of vaporization, high surface tension, high dielectric constant, and universal solvent.
- Water is also essential for human activities such as agriculture, industry, transportation, and recreation. Water is used for drinking, cooking, washing, irrigation, power generation, and many other purposes. Water also has cultural and spiritual significance for many people around the world.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of how acidic or basic a solution is. It is defined as the negative logarithm of the hydrogen ion concentration in moles per liter. pH = -log[H+]

- The pH scale ranges from 0 to 14, with 7 being neutral, lower than 7 being acidic, and higher than 7 being basic. The pH of pure water is 7 at 25°C.
- pH can be measured using indicators, such as litmus paper, phenolphthalein, or universal indicator, which change color depending on the acidity or basicity of the solution. pH can also be measured using a pH meter, which is an electronic device that measures the voltage difference between a reference electrode and a pH-sensitive electrode.
- Surface tension is a measure of how strongly the molecules of a liquid are attracted to each other at the surface. It is defined as the force per unit length required to stretch or break the surface of a liquid. Surface tension = F/L
- Surface tension causes liquids to form spherical droplets, to minimize their surface area. It also allows some insects, such as water striders, to walk on the surface of water without sinking.
- Surface tension can be measured using a force gauge, which measures the force required to pull a ring or a needle out of the surface of a liquid. Surface tension can also be measured using a stalagmometer, which measures the number of drops of a liquid that can be formed from a given volume.
- Viscosity is a measure of how resistant a fluid is to flow. It is defined as the ratio of the shear stress to the shear rate of a fluid. Viscosity = /
- The viscosity of a fluid depends on its temperature, pressure, and composition. Generally, liquids have higher viscosity than gases, and viscosity decreases with increasing temperature and pressure.
- Viscosity can be measured using a viscometer, which is a device that measures the time required for a fluid to flow through a narrow tube or a rotating spindle. Viscosity can also be measured using a rheometer, which is a device that measures the relationship between the shear stress and the shear rate of a fluid under different conditions.

A liquid

- A liquid is one of the four states of matter, along with solid, gas, and plasma.
- A liquid is a fluid that can flow and take the shape of its container, but has a constant volume.
- A liquid has a definite surface, unlike a gas, but can change its shape, unlike a solid.
- A liquid is composed of molecules or atoms that are loosely bound together by intermolecular forces, such as hydrogen bonds or van der Waals forces.
- A liquid can undergo phase transitions to a solid (freezing) or a gas (evaporation or boiling) depending on the temperature and pressure.
- A liquid can also form mixtures with other liquids (solutions) or solids (suspensions or colloids).
- A liquid has some properties that depend on its composition, such as density, viscosity, surface tension, and refractive index.

- A liquid has some properties that depend on its shape, such as pressure, buoyancy, and capillarity.

Preparation of Different Resins

Resins are natural or synthetic organic substances that are usually solid or semi-solid, and have a high molecular weight. Resins are widely used in various industries, such as paints, adhesives, plastics, rubber, and pharmaceuticals.

There are different types of resins, such as thermoplastic resins, thermosetting resins, and elastomeric resins. Each type of resin has different properties and applications, and requires different methods of preparation.

Thermoplastic Resins

Thermoplastic resins are resins that can be melted and reshaped by heating, and solidified by cooling. They are usually prepared by polymerization or polycondensation of monomers, such as ethylene, propylene, styrene, vinyl chloride, etc. Some examples of thermoplastic resins are polyethylene, polypropylene, polystyrene, polyvinyl chloride, etc.

The general steps for preparing thermoplastic resins are:

- Monomer selection: The monomers are chosen based on the desired properties and applications of the resin.
- Polymerization or polycondensation: The monomers are reacted with each other or with other substances, such as catalysts, initiators, or cross-linking agents, to form long chains of molecules called polymers. The reaction can be carried out in bulk, in solution, in suspension, or in emulsion, depending on the type of monomer and the desired characteristics of the resin.
- Purification: The polymer is separated from the unreacted monomers, catalysts, solvents, and other impurities by filtration, distillation, extraction, or precipitation.
- Drying: The polymer is dried to remove any moisture that may affect its properties or processing.
- Pelletizing or granulating: The polymer is cut or extruded into small pieces or pellets, which are easier to handle, store, and transport.

Thermosetting Resins

Thermosetting resins are resins that can be cured or hardened by heating, and cannot be melted or reshaped by further heating. They are usually prepared by polyaddition or polycondensation of monomers, such as phenol, formaldehyde, urea, melamine, etc. Some examples of thermosetting resins are phenolic resins, urea-formaldehyde resins, melamine-formaldehyde resins, etc.

The general steps for preparing thermosetting resins are:

- Monomer selection: The monomers are chosen based on the desired properties and applications of the resin.
- Polyaddition or polycondensation: The monomers are reacted with each other or with other substances, such as catalysts, hardeners, or fillers, to form a resin that is still liquid or viscous at room temperature. The reaction can be carried out in bulk, in solution, or in suspension, depending on the type of monomer and the desired characteristics of the resin.
- Molding or casting: The resin is poured or injected into a mold or a cavity, where it is heated and cured to form a solid product. The curing process can be accelerated by using heat, pressure, or radiation.
- Demolding or trimming: The product is removed from the mold or the cavity, and any excess resin or flash is trimmed or cut off.

Elastomeric Resins

Elastomeric resins are resins that have elastic properties, and can be stretched and recovered without permanent deformation. They are usually prepared by copolymerization of monomers, such as butadiene, isoprene, styrene, acrylonitrile, etc. Some examples of elastomeric resins are natural rubber, synthetic rubber, silicone rubber, etc.

The general steps for preparing elastomeric resins are:

- Monomer selection: The monomers are chosen based on the desired properties and applications of the resin.
- Copolymerization: The monomers are reacted with each other or with other substances, such as catalysts, initiators, or modifiers, to form a copolymer that has both elastic and plastic segments. The reaction can be carried out in bulk, in solution, in suspension, or in emulsion, depending on the type of monomer and the desired characteristics of the resin.
- Purification: The copolymer is separated from the unreacted monomers, catalysts, solvents, and other impurities by filtration, distillation, extraction, or precipitation.
- Drying: The copolymer is dried to remove any moisture that may affect its properties or processing.
- Vulcanization or cross-linking: The copolymer is treated with heat, pressure, or chemicals, such as sulfur, peroxides, or metal oxides, to

Synthesis of Adipic Acid

- Adipic acid is a dicarboxylic acid with the formula C6H10O4. It is mainly used for the production of nylon, polyurethanes, and other polymers.
- The conventional method for the synthesis of adipic acid is the oxidation of a mixture of cyclohexanone and cyclohexanol (KA oil) with nitric acid .

- The oxidation of KA oil involves several steps, such as:
  - The conversion of cyclohexanol to cyclohexanone, releasing nitrous acid.
  - The formation of a nitroso compound from cyclohexanone and nitrous acid.
  - The oxidation of the nitroso compound to a nitro compound with nitric acid.
  - The hydrolysis of the nitro compound to adipic acid and nitrous oxide.
- The main drawbacks of this method are the low yield, the high energy consumption, and the environmental impact of nitrous oxide, which is a greenhouse gas and an ozone-depleting agent .
- Alternative methods for the synthesis of adipic acid from renewable sources, such as glucose and its derivatives, have been explored using various catalysts and oxidants.
- Some examples of these methods are:
  - The oxidation of glucose to glucaric acid, followed by the dehydration and hydrogenation of glucaric acid to adipic acid.
  - The oxidation of sorbitol to gluconic acid, followed by the dehydration and hydrogenation of gluconic acid to adipic acid.
  - The oxidation of muconic acid, which can be derived from glucose or lignin, to adipic acid using hydrogen peroxide or molecular oxygen.
- These methods have the advantages of using renewable feedstocks, avoiding nitrous oxide emissions, and achieving higher yields and selectivities.
- However, these methods also face some challenges, such as the high cost and low stability of the catalysts, the low solubility and reactivity of the substrates, and the separation and purification of the products.

Acid and Paracet

- Paracet is a brand name of acetaminophen, also known as paracetamol, a widely used analgesic and antipyretic drug.
- Paracet works by inhibiting the synthesis of prostaglandins, which are mediators of pain and inflammation, in the central nervous system .
- Paracet does not have anti-inflammatory effects in the peripheral tissues, unlike aspirin and other non-steroidal anti-inflammatory drugs (NSAIDs) .
- Paracet is used for the treatment of mild to moderate pain and fever, such as headaches, menstrual cramps, toothaches, backaches, osteoarthritis, or cold and flu symptoms .
- Paracet is generally safe and well-tolerated when taken at the recommended doses, but it can cause serious liver damage if overdosed or taken with alcohol .
- Paracet overdose can also cause metabolic acidosis, a condition where the blood becomes too acidic, which can affect the function of vital organs.

- Paracet overdose requires immediate medical attention and treatment with an antidote called N-acetylcysteine, which can prevent or reduce liver damage if given early enough .
- Paracet can also cause some side effects, such as nausea, vomiting, rash, itching, sore throat, or bleeding problems, especially if taken with other drugs that affect the liver or blood clotting.
- Paracet should be used with caution in people with liver or kidney disease, alcohol dependence, or blood disorders, and in pregnant or breastfeeding women .
- Paracet should not be taken with other products that contain acetaminophen, such as cold and cough medicines, to avoid exceeding the maximum daily dose .
- Paracet should be taken as directed by the doctor or pharmacist, and the label should be read carefully before use .